Covariance-Aware Policy Optimization

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Covariance-Aware Policy Optimization (CAPO) is a GRPO-style, variance-aware trajectory reweighting framework for Reinforcement Learning with Verifiable Rewards (RLVR). CAPO starts from a generalized Bayesian working model for per-trajectory policy-gradient statistics and replaces token-independence assumptions with a principled treatment of within-trajectory covariance. With a Jeffreys prior on the global scale, the posterior-precision rule recovers ΔL normalization as the correctly specified independence special case. Beyond independence, CAPO's theory shows that structured token dependence induces an *induced variance law* $v(L) = \mathrm{Var}(g \mid L)$ that may grow linearly (summable/short-range dependence) or superlinearly (persistent/long-range dependence), and that the statistically efficient GRPO group reweighting is inverse-variance.

To connect theory to practice, we express CAPO as a *modular plug-in algorithm*: the policy update requires only an estimator of $v(L)$ (up to scale), which is converted into within-group precision weights. We develop and analyze four plug-ins of increasing structure: **L-CAPO**, the length-only/independence plug-in (including ΔL as the fixed-β=1 case); **LV-CAPO**, an empirical-Bayes plug-in that jointly estimates a length exponent and a structured dependence shape; **CAPO-Q**, a quadratic variance-law plug-in motivated by compound-symmetry/random-intercept covariance; and **CAPO-HAC**, a long-run-variance plug-in based on lag-window (Newey–West) estimation. We provide closed-form induced-variance derivations, grouped (prompt-wise) EB objectives and gradients, robustness guarantees for plug-in inverse-variance weighting, and diagnostics for when CAPO reduces variance without introducing bias. Across synthetic covariance benchmarks and CountDown RLVR experiments, CAPO variants stabilize training and improve efficiency relative to ΔL and GRPO/DAPO/Dr. GRPO normalizations.

Keywords: Reinforcement Learning with Verifiable Rewards (RLVR); Bayesian Methods; Variance Reduction; Policy Optimization; Large Language Models; Sequence Modeling; Stochastic Optimization; Robust Training; Statistical Inference.

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Contributions

Overview. We address gradient aggregation for Reinforcement Learning with Verifiable Rewards (RLVR) under highly variable response lengths and realistic token dependencies. Our approach, *Covariance-Aware Policy Optimization (CAPO)*, unifies prior length-based heuristics under a Bayesian perspective and introduces a simple, plug-in rule that yields unbiased, lower-variance updates without relying on token-independence assumptions.

Key contributions.

- **Unified analysis.** We place existing length-normalization rules (including ΔL) on a coherent Bayesian footing via scale-invariant reasoning, clarifying the invariances and assumptions behind them.
- **Vector-valued optimality under common covariance shape.** We show that when per-trajectory covariances share a common "shape" up to a scalar factor, the inverse-variance rule remains optimal for a broad class of matrix risk functionals (Proposition 3.2), justifying scalar proxy modeling in high-dimensional policy gradients.
- **Robustness to variance misestimation.** We prove that plug-in inverse-variance weights degrade variance by at most a quadratic factor in the multiplicative error of the variance estimates (Proposition 3.7), providing a formal stability guarantee under model misspecification.
- **CAPO aggregator.** We generalize beyond independence by estimating the *induced variance law* $v(L) = \mathrm{Var}(g \mid L)$ under structured token dependence and using the corresponding inverse-variance (precision) weights, optionally combined with sublinear length dampening ($\beta < 1$) to hedge variance-estimation noise.
- **Estimator-family view (submission-ready plug-ins).** We express CAPO as a modular GRPO-style reweighting rule that requires only a plug-in estimate of $v(L)$ (up to a common scale). This yields a coherent family of instantiations: **L-CAPO** (length-only/independence, including ΔL as the fixed-β=1 case), **LV-CAPO** (empirical Bayes with structured dependence), **CAPO-Q** (quadratic variance law motivated by compound symmetry / random-intercept covariance), and **CAPO-HAC** (lag-window/Newey–West long-run variance estimation).
- **Theory to practice.** We show that trajectory-level gradient variance grows approximately linearly with length under broad dependency classes, motivating CAPO's form and its guidance for choosing $\beta < 1$.
- **Simple implementation.** CAPO is a drop-in change at the level of group-based advantage reweighting: plug-ins produce per-trajectory precisions (or equivalently induced-variance estimates), and the main update normalizes these within each prompt group to obtain trajectory weights.
- **Diagnostics and robustness.** We provide practical checks (effective sample size, weight entropy, CV ratios, residual autocorrelation) to tune hyperparameters and detect covariance misfit.

- **Controlled synthetic covariance benchmark.** We provide a base-model-free simulator that generates length-heterogeneous rollout statistics under configurable within-trajectory covariance regimes and use it to compare GRPO, ΔL, and CAPO variants directly at the level of advantage magnitudes and signal allocation across lengths.
- **Empirical protocol.** We specify a minimal, controlled evaluation suite (§5) that compares CAPO against GRPO/DAPO/Dr. GRPO and ΔL under matched tokens. We report both end accuracy and stability diagnostics, and include targeted ablations (sublinear exponent and covariance window size) to isolate covariance-aware effects.

Takeaway. CAPO offers a principled, covariance-aware alternative to length-only normalization: it preserves unbiasedness, reduces variance, and delivers stable, efficient RLVR training with minimal engineering effort.

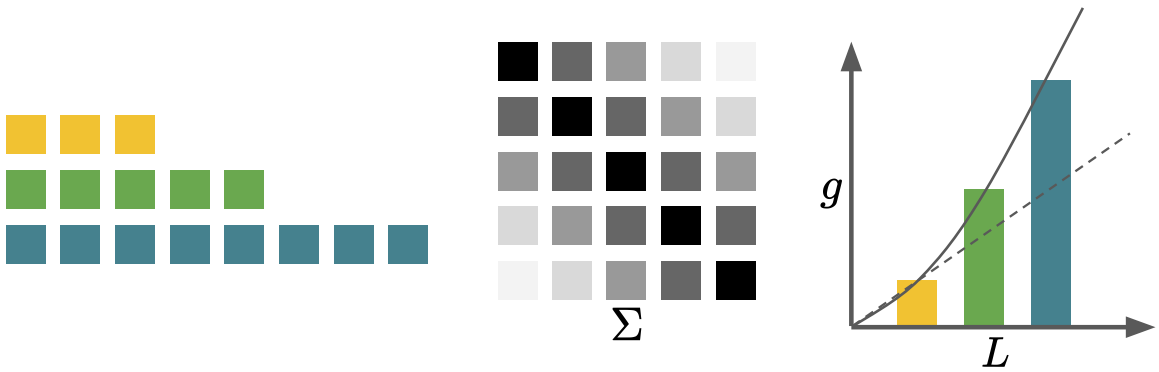

Figure 1 Schematic overview of CAPO. (a) Within-trajectory token dependence induces nontrivial covariance structure in trajectory gradients, which can amplify gradient variance as a function of response length. (b) CAPO reweights trajectories by learned posterior precisions (length and covariance-aware) to obtain an unbiased, lower-variance aggregate update.

0.1 Bayesian graphical model for CAPO

We summarize the **LV-CAPO** (empirical Bayes) instantiation of CAPO from §3 and §3.8 as a hierarchical Bayesian graphical model. Other CAPO variants correspond to alternative plug-in estimators of the induced variance law $v(L)$ (§3.10) and plug into the same grouped training loop (§4).

Generative story. At a given outer optimization step (suppressing the step index t for clarity):

1. **Hyperparameters.** Draw global parameters

$$\mu \sim p(\mu) \propto 1, \quad \sigma^2 \sim \text{Inv-Gamma}(a_0, b_0), \quad \beta \sim \pi_\beta, \quad \xi \sim \pi_\xi,$$

   where β controls the length–variance law and ξ parameterizes the dependence shape $s(L;\xi)$ (e.g., $\xi = (\rho, k, \eta)$ under the stretched-geometric k-band model of Definition 3.28).

2. **Trajectory-level structure.**

   For each trajectory $i = 1, \ldots, G$:

(a) The trajectory length $L_i \in \mathbb{N}$ is observed (exogenous, determined by prompts and stopping rules).

(b) Conditional on ξ and L_i, the token-level increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ form a second–order stationary process with auto-covariance $\gamma(h;\xi)$. Under the k-banded stretched-geometric family,

$$\gamma(0) = \tau^2, \qquad \gamma(h) = \tau^2 \rho^{|h|^\eta} \mathbf{1}\{|h| \le k\},$$

so that the induced variance of the scalarized trajectory gradient $g_i = \sum_{\tau=1}^{L_i} Y_{i,\tau}$ is

$$\mathrm{Var}(g_i \mid L_i, \xi) = \sigma^2 L_i^\beta s(L_i;\xi), \qquad s(L_i;\xi) > 0$$

as in Lemma 3.13 and equation 16.

(c) The scalarized trajectory-gradient statistic $g_i \in \mathbb{R}$ obeys the heteroscedastic Normal working model

$$g_i \mid \mu, \sigma^2, \beta, \xi, L_i \sim \mathcal{N}\big(\mu,\ v_i(\vartheta)\big), \qquad v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i;\xi),$$

where $\vartheta = (\beta, \xi)$.

Empirical Bayes layer and CAPO aggregator. Given observed $\{(g_i, L_i, \{Y_{i,\tau}\})\}_{i=1}^G$, CAPO performs Empirical Bayes (type-II ML) estimation of $\vartheta = (\beta, \xi)$ by maximizing the marginal objective $\ell(\beta, \xi)$ in Theorem 3.22. This induces precision factors

$$\omega_i(\beta,\xi) = \frac{1}{L_i^\beta s(L_i;\xi)}, \qquad w_i(\beta,\xi) = \frac{\omega_i(\beta,\xi)}{\sum_{j=1}^G \omega_j(\beta,\xi)},$$

and the CAPO aggregator

$$m(\beta,\xi) = \sum_{i=1}^G w_i(\beta,\xi)\, g_i,$$

which is the GLS/Bayes estimator of μ for fixed ϑ (Proposition 3.21). The weights $w_i(\beta,\xi)$ are then fed into the CAPO update as trajectory weights for the policy/teacher loss.

Plate diagram. Figure 2 shows the corresponding Bayesian plate model, highlighting the global hyperparameters $(\mu, \sigma^2, \beta, \xi)$, the trajectory-level variables (L_i, g_i), and the token-level increments $Y_{i,\tau}$ used to estimate the dependence shape via $s(L;\xi)$. The deterministic nodes $v_i(\vartheta)$, $\omega_i(\beta,\xi)$, and $m(\beta,\xi)$ summarize the variance law, precision weights, and CAPO aggregator, respectively.

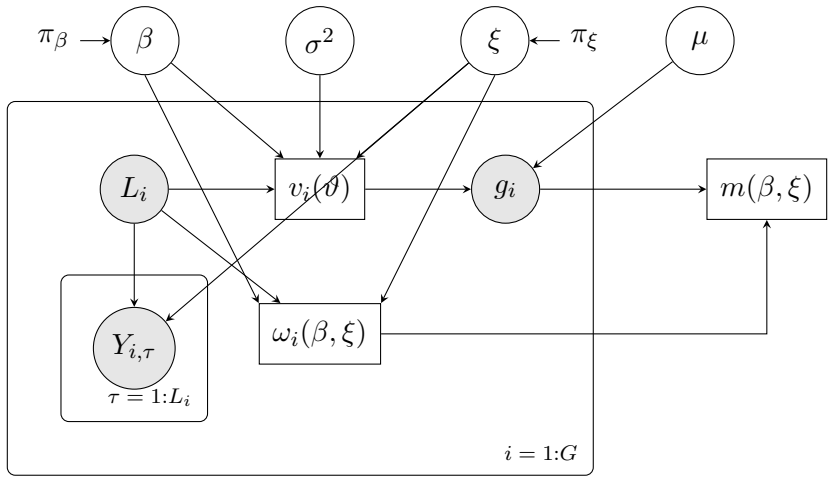

Figure 2 CAPO Model: Horizontal layout with enlarged plates and unified labeling.

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1 Introduction

Recent work has rapidly advanced the mathematical and logical reasoning abilities of large language models (LLMs) by leveraging *reinforcement learning with verifiable rewards* (RLVR), wherein a policy produces chain-of-thought (CoT) trajectories and receives rule-based, deterministic feedback on final-answer correctness (**????**). A distinctive challenge in RLVR is that response trajectories vary widely in length, often from tens to thousands of tokens and typically increasing as training progresses. So updates computed from a single batch can differ dramatically in both magnitude and variance (**??**). Consequently, training stability and sample efficiency hinge on how we aggregate per-trajectory gradients under this length variability.

A common remedy is *length-dependent* normalization. GRPO normalizes at the sample level by implicitly or explicitly dividing by trajectory length, DAPO normalizes at the batch level by the total length, and Dr. GRPO removes length terms to preserve an unbiased estimator (**???**). While such strategies can curb instabilities from long CoTs, their statistical properties differ: length-based factors may introduce bias in the policy-gradient direction, whereas length-agnostic updates can suffer high variance—both of which impede convergence (**????**). In particular, estimators with higher gradient variance are more prone to deviating from the optimal optimization direction, resulting in slow or unstable learning.

Related work and connections. CAPO is positioned at the intersection of three literatures.

First, it relates to modern RLHF/RLVR optimization pipelines for LLMs, which typically instantiate a policy-gradient method (most often a PPO-type update (**?**)) equipped with reward modeling, preference modeling, or verifiable rewards (**???**). The specific normalization rules compared in this paper (ΔL, GRPO-, DAPO-, and Dr. GRPO-type variants) can be interpreted as different choices of *trajectory-level reweighting* and *baseline* in a policy-gradient estimator. Our contribution is to place these heuristics into a single estimation-theoretic framework and to isolate a design axis that is largely implicit in prior work: how *within-trajectory dependence* and heteroscedasticity induced by variable-length CoT should enter the weighting rule.

Second, CAPO has close ties to classical weighted least squares and generalized least squares. The MVU weights in Lemma 3.6 are the Gauss–Markov/BLUE solution for estimating a common mean under heteroscedastic noise, and the dependence-aware extensions can be viewed as (structured) GLS with autocorrelated errors (**?**). This connection motivates both the functional form of the optimal weights and the emphasis on correctly modeling how variance grows with length and dependence.

Third, our empirical-Bayes (EB) construction in §3.8 follows a long tradition of EB modeling for structured variance and shrinkage, in which hyperparameters are fitted by marginal likelihood and then plugged into a decision rule (**?**). In the present setting, EB serves a pragmatic purpose: it yields closed-form weight expressions with stable large-L behavior while avoiding the computational overhead of full Bayesian integration over dependence parameters.

This paper proposes *Covariance-Aware Policy Optimization (CAPO)*, a principled, variance-

aware gradient aggregation framework for RLVR. We begin by casting per-step loss/gradient aggregation as a classical *minimum-variance unbiased estimation* (MVUE) problem over heterogeneous trajectory lengths, in the same spirit as **?**. We then introduce a generalized Bayesian working model for trajectory-level gradient statistics, place a Jeffreys prior on the scale parameter to obtain an invariant baseline, and show that the resulting posterior-precision aggregator recovers ΔL normalization as a special case (**??**). Finally, we extend beyond token-independence by modeling realistic within-trajectory dependencies via simple covariance structures (e.g., banded or geometrically decaying kernels). This yields a plug-in weighting rule that down-weights trajectories by both an empirically estimated variance and a sublinear length factor:

$$\bar{g} = \sum_{i=1}^{G} \tilde{w}_i g_i, \qquad \tilde{w}_i \propto \frac{L_i^{-\alpha}}{\widehat{v}_i}, \qquad \alpha \in (0,1), \tag{1}$$

where g_i is a scalarized trajectory-gradient statistic (§3.2), L_i is its length, and $\widehat{v}_i$ is a plug-in estimate of the induced variance law $v(L) = \mathrm{Var}(g \mid L)$ under a chosen covariance surrogate (§3.10). In particular, the same CAPO update can be instantiated with multiple variance-law plug-ins: **L-CAPO** (length-only/independence, including ΔL), **LV-CAPO** (length + structured dependence via empirical Bayes), **CAPO-Q** (quadratic/compound-symmetry variance law), and **CAPO-HAC** (lag-window long-run variance estimation).

Our analysis shows that, under broad dependence classes, the trajectory-level variance grows *approximately linearly* with length whenever within-trajectory correlations are summable (short-range dependence). For a simple but illustrative case with geometrically decaying correlations $\mathrm{Cov}(Z_t, Z_s) = \sigma^2 \rho^{|t-s|}$ (with $|\rho| < 1$), one obtains

$$\mathrm{Var}\left(\sum_{t=1}^{L} Z_t\right) = \sigma^2 \sum_{k=-(L-1)}^{L-1} (L - |k|)\, \rho^{|k|} = \Theta(L), \qquad |\rho| < 1, \tag{2}$$

so that naive length-proportional weighting can be unstable. In finite samples, additionally, variance estimation noise increases with L, which motivates the sublinear dampening in equation 1. We further study the *coefficient of variation* (CV) of the aggregated estimator under different normalization schemes and show that CAPO reduces CV relative to length-agnostic updates while avoiding the length-induced bias of sample- or batch-normalization.

Empirically, we evaluate CAPO on a focused RLVR testbed: a representative task (Count-Down), a single open-weight backbone model (Qwen2.5–1.5B–Instruct), two context-length regimes (2048 and 8192), and a compact set of aggregation baselines that isolate the effect of aggregation under matched tokens and identical decoding/clipping (GRPO Norm, DAPO Norm, Dr. GRPO Norm, ΔL, and CAPO). In addition to end performance, we report *stability diagnostics* (monotonicity of held-out accuracy, step-to-step drift, CV/kurtosis of trajectory weights, and length-decile stratification), together with a targeted ablation over the covariance bandwidth used by CAPO's empirical-Bayes estimator. Finally, to isolate the advantage transformation in a setting where the length–variance law is known by construction, we include a base-model-free synthetic covariance benchmark that compares CAPO, GRPO, and ΔL under controlled dependence regimes (§5.8 and Appendix D.1.2).

Summary of contributions. (i) A unified estimation-theoretic and Bayesian analysis of RLVR gradient aggregation that derives ΔL from MVUE/Jeffreys principles and clarifies invari-

ances and bias–variance trade-offs (**??**); (ii) *CAPO*, a covariance-aware generalization with closed-form inverse-variance weights and rigorous robustness guarantees for plug-in variance estimation; (iii) a modular *variance-law plug-in* perspective that yields multiple submission-ready instantiations—**L-CAPO** (length-only/independence), **LV-CAPO** (empirical Bayes with structured dependence), **CAPO-Q** (quadratic/compound symmetry), and **CAPO-HAC** (lag-window long-run variance); (iv) dependence-aware theory linking token-level autocovariance to trajectory-level variance growth, yielding principled corrections beyond length-only normalization; (v) an empirical protocol with stability diagnostics and targeted ablations that isolate covariance-aware weighting effects under matched compute; and (vi) a controlled synthetic covariance benchmark that compares GRPO, ΔL, and CAPO advantage constructions under known dependence regimes, making length amplification and signal allocation effects explicit.

Organization. Section 2 formalizes the trajectory aggregation problem and clarifies the role of the scaling constraint in equation 6. Section 3 develops the MVU and Bayesian foundations, derives dependence-aware variance laws, and establishes the EB estimator and its asymptotics. Section 4 provides implementable CAPO variants and complexity considerations. Section 5 describes the RLVR evaluation protocol and stability diagnostics. Appendices supply full proofs and illustrative covariance examples.

2 Problem Setup

Goal. Given a batch of trajectories with heterogeneous lengths and within-trajectory token dependence, we seek an unbiased linear aggregate of trajectory-level gradients that is statistically efficient. We cast the aggregation problem as estimation of a common mean under heteroscedastic noise whose variance scales with trajectory length and dependence structure, setting up a Bayesian treatment that will later recover and generalize the ΔL family of weights (**?**).

2.1 Setup and Notation

We consider an update step that aggregates a collection of $G \geq 1$ rollouts (trajectories) indexed by $i \in \{1, \ldots, G\}$. Trajectory i has a (random) token length $L_i \in \mathbb{N}$ and yields a sequence of token-level gradient contributions $\{\boldsymbol{Z}_{i,t}\}_{t=1}^{L_i} \subset \mathbb{R}^p$, where p is the parameter dimension. The (un-normalized) trajectory gradient is the token-sum

$$\boldsymbol{g}_i \;:=\; \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}. \tag{3}$$

Throughout, we regard the policy-gradient signal as a common mean vector $\boldsymbol{\mu} := \mathbb{E}[\boldsymbol{g}_i]$ (under the current policy and sampling scheme), and we study linear aggregators of the form equation 6. The scaling constant $M > 0$ in equation 6 is fixed by the training objective (e.g., M may correspond to a number of prompt groups in a batch, or to a normalization convention in the policy-gradient estimator).

To quantify *dispersion* of a vector-valued gradient, we introduce a scalar functional $\Psi : \mathbb{S}_+^p \to (0, \infty)$ on positive-semidefinite matrices (e.g., $\Psi(\boldsymbol{\Sigma}) = \mathrm{tr}(\boldsymbol{\Sigma})$ or $\Psi(\boldsymbol{\Sigma}) = \boldsymbol{u}^\top \boldsymbol{\Sigma} \boldsymbol{u}$ for a fixed

Symbol	Meaning
G	number of trajectories aggregated in the update
L_i	token length of trajectory i
p	parameter dimension; $\boldsymbol{Z}_{i,t} \in \mathbb{R}^p$
$\boldsymbol{Z}_{i,t}$	token-level gradient contribution at position t in trajectory i
$\boldsymbol{g}_i$	trajectory-level gradient $\sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}$
$\boldsymbol{\mu}$	common mean gradient, $\mathbb{E}[\boldsymbol{g}_i] = \boldsymbol{\mu}$
$\widehat{\boldsymbol{g}}$	aggregated gradient $\sum_{i=1}^{G} x_i \boldsymbol{g}_i$
x_i	trajectory weights; unbiasedness requires $\sum_i x_i = 1/M$
M	fixed scaling constant in equation 6
$\Psi(\cdot)$	scalar functional on covariances used to define v_i
v_i	proxy dispersion $\Psi(\mathrm{Cov}(\boldsymbol{g}_i))$
$\boldsymbol{\Gamma}_i(k)$	token-level lag-k autocovariance, equation 7
$s(L;\xi)$	dependence shape factor linking autocovariance to v_i
β	length exponent in the variance law $v_i \propto L_i^{\beta}$

Table 1 Core notation used throughout the paper.

direction $\boldsymbol{u}$) and define the proxy variance

$$v_i \;:=\; \Psi(\mathrm{Cov}(\boldsymbol{g}_i))\,. \tag{4}$$

The role of CAPO is to estimate how v_i depends on length and within-trajectory dependence, and to use (an estimate of) v_i^{-1} as the statistically efficient weighting rule.

Assumption 2.1 (Cross-trajectory uncorrelatedness)**.** *For the trajectories aggregated in a single update, we assume* $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) = \boldsymbol{0}$ *for all* $i \neq j$*. This assumption is exact when trajectories are sampled independently and is a standard working approximation when any remaining dependence is negligible compared to within-trajectory autocorrelation.*

Assumption 2.2 (Second-order stationarity within trajectories)**.** *For each trajectory* i*, conditional on its length* L_i*, the token-level process* $\{\boldsymbol{Z}_{i,t}\}_{t=1}^{L_i}$ *is second-order stationary in the sense that* $\mathbb{E}[\boldsymbol{Z}_{i,t}]$ *does not depend on* t *and the lag-*k *autocovariance* $\mathrm{Cov}(\boldsymbol{Z}_{i,t}, \boldsymbol{Z}_{i,t-k})$ *depends on* k *but not on* t *whenever both indices are in* $\{1, \ldots, L_i\}$*.*

Assumptions 2.1–2.2 are sufficient for the variance decompositions used in §3 and the appendices. We discuss several dependence models for $\boldsymbol{\Gamma}_i(k)$ and their implications for the length-variance law in §2.4 and §3.6.

2.2 Observed Entities and Target Quantity

We observe G trajectories indexed by $i \in \{1, \ldots, G\}$. Trajectory i has length $L_i \in \mathbb{N}$ tokens and produces the un-normalized trajectory-level gradient.

$$\boldsymbol{g}_i \;=\; \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t} \;\in\; \mathbb{R}^p, \qquad \mathbb{E}[\,\boldsymbol{g}_i\,] \;=\; \boldsymbol{\mu} \;\in\; \mathbb{R}^p, \tag{5}$$

where $\boldsymbol{Z}_{i,t}$ is the token-level score contribution (e.g., advantage-weighted policy-score increment). In GRPO-style training, trajectories are sampled in *prompt groups*; within a group

(fixed prompt) the learning signal is common, while across groups the location may shift. Our EB development therefore treats group-specific means as nuisance parameters integrated out (Section 3.8.1).

2.3 A Statistically Efficient, Linear, Unbiased Estimator

Training aggregates the per-trajectory gradients linearly,

$$\widehat{\boldsymbol{g}} \;=\; \sum_{i=1}^{G} x_i\, \boldsymbol{g}_i, \qquad \sum_{i=1}^{G} x_i \;=\; \frac{1}{M}, \tag{6}$$

for a fixed scaling constant $M > 0$. Under equation 5, any choice of weights satisfying the normalization in equation 6 yields an unbiased estimate, $\mathbb{E}[\widehat{\boldsymbol{g}}] = \boldsymbol{\mu}/M$. The design problem is to choose $\{x_i\}$ to minimize the dispersion of $\widehat{\boldsymbol{g}}$ subject to unbiasedness.

2.4 Non-iid Tokens

Token contributions within a trajectory are dependent. Define the auto-covariance function,

$$\boldsymbol{\Gamma}_i(k) \;:=\; \mathrm{Cov}\Big(\boldsymbol{Z}_{i,t},\, \boldsymbol{Z}_{i,t-k}\Big) \in \mathbb{R}^{p\times p}, \qquad k \in \mathbb{Z}. \tag{7}$$

Then the trajectory covariance, under stationarity condition, consult Appendix C, is

$$\mathrm{Var}(\boldsymbol{g}_i) \;=\; \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k). \tag{8}$$

For aggregation we summarize $\mathrm{Var}(\boldsymbol{g}_i)$ by a positive scalar functional

$$v_i \;:=\; \Psi\Big(\mathrm{Var}(\boldsymbol{g}_i)\Big) \;\in\; (0,\infty), \qquad \text{e.g., } \Psi(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A}) \text{ or } \Psi(\boldsymbol{A}) = \mathbb{E}_{\boldsymbol{u}}[\boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}], \tag{9}$$

where $\boldsymbol{u}$ is a unit vector (fixed or randomized). If $\{v_i\}$ were known, the minimum-variance unbiased (MVU) linear weights are

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^{G} v_j^{-1}}. \tag{10}$$

For a detailed proof along with the assumptions consult Appendix A. Hence the core statistical task is to model/estimate v_i as a function of length and dependence.

3 Method

Roadmap. We begin with a length-only model that recovers ΔL weighting method; we then present a heteroscedastic Gaussian model with flexible priors, derive the posterior-precision aggregator, identify Jeffreys' baseline (α=1), and show how correctness-linked variance reduction yields α<1; dependence corrections are incorporated via $s(L;\xi)$.

3.1 Length-Only Benchmark and the ΔL Family

Under an i.i.d. token model with $\mathrm{Var}(\boldsymbol{Z}_{i,t}) = \boldsymbol{\Sigma}_0$ and $\boldsymbol{\Gamma}_i(k) = \mathbf{0}$ for $k \neq 0$,

$$\mathrm{Var}(\boldsymbol{g}_i) = L_i\, \boldsymbol{\Sigma}_0 \quad \Longrightarrow \quad v_i \propto L_i. \tag{11}$$

Substituting into equation 10 yields the inverse-length rule

$$x_i \;\propto\; L_i^{-1}, \tag{12}$$

which is precisely the ΔL normalization of **?** with exponent $\alpha = 1$. More generally, following **?** we consider the one-parameter family

$$x_i \;\propto\; L_i^{-\beta}, \qquad \beta \in [0, 2], \tag{13}$$

interpolating between constant weighting ($\beta = 0$) and inverse-length weighting ($\beta = 1$), as a practical control of variance induced by heterogeneous lengths.

A two-parameter normalization family. Length-only normalization can be viewed as choosing a (possibly biased) *global* step scale and a (possibly variance-reducing) *relative* within-batch reweighting. To decouple these effects, we introduce a two-parameter family

$$x_i(\alpha, \kappa) \;:=\; \frac{1}{M_\kappa} \frac{L_i^{-\alpha}}{\sum_{j=1}^{G} L_j^{-\alpha}}, \qquad M_\kappa \;:=\; \Big(\sum_{j=1}^{G} L_j^{-\alpha} \Big)^{\kappa}, \tag{14}$$

where $\alpha \in [0, 1]$ controls the relative suppression of long responses and $\kappa \in \mathbb{R}$ controls the overall step magnitude. The standard ΔL estimator corresponds to $\kappa = 0$ (purely relative reweighting with a fixed M), while $\kappa = 1$ recovers a DAPO-style batch-length normalization where the step magnitude shrinks with the batch effective length. Section 3.8 shows that CAPO preserves unbiasedness by enforcing the linear constraint $\sum_i x_i = 1/M$ while learning the variance structure; empirically, we therefore treat (α, κ) as a diagnostic lens rather than as free hyperparameters for CAPO.

3.2 Bayesian Setup under a General Prior

We project each trajectory-level gradient onto a fixed direction to obtain a scalar summary. Let $\boldsymbol{u}$ be a unit vector and define

$$g_i \;=\; \boldsymbol{u}^\top \boldsymbol{g}_i, \qquad \mathbb{E}[g_i] = \mu, \qquad \mathrm{Var}(g_i) = v_i. \tag{15}$$

Vector-valued gradients and common-shape covariances. Our Bayesian and empirical-Bayes derivations proceed via a scalar projection equation 15. This is not merely a notational convenience: in a broad and practically relevant regime, the resulting weights are also optimal for *vector-valued* aggregation objectives.

Assumption 3.1 (Common-shape covariance)**.** *There exists a fixed positive-semidefinite matrix $\boldsymbol{\Sigma}_0 \in \mathbb{S}_+^p$ and positive scalars v_i such that, for every trajectory i,*

$$\mathrm{Cov}(\boldsymbol{g}_i) = v_i\, \boldsymbol{\Sigma}_0.$$

Proposition 3.2 (Inverse-variance weights are optimal under common shape)**.** *Assume cross-trajectory uncorrelatedness (Assumption 2.1) and the common-shape model in Assumption 3.1. Let $\Psi : \mathbb{S}_+^p \to (0, \infty)$ be any positive-homogeneous functional, i.e., $\Psi(c\boldsymbol{A}) = c\,\Psi(\boldsymbol{A})$ for all $c \geq 0$ and $\boldsymbol{A} \in \mathbb{S}_+^p$. Then the unique minimizer of $\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}}))$ over weights satisfying $\sum_{i=1}^G x_i = 1/M$ is given by the inverse-variance rule $x_i^\star \propto v_i^{-1}$ (with normalization fixed by $\sum_i x_i = 1/M$), i.e., the same weights as in Lemma 3.6.*

Proof. Under cross-trajectory uncorrelatedness,

$$\mathrm{Cov}(\widehat{\boldsymbol{g}}) = \sum_{i=1}^G x_i^2\,\mathrm{Cov}(\boldsymbol{g}_i) = \sum_{i=1}^G x_i^2\, v_i\, \boldsymbol{\Sigma}_0 = \left(\sum_{i=1}^G x_i^2 v_i\right) \boldsymbol{\Sigma}_0.$$

By positive homogeneity of Ψ,

$$\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}})) = \left(\sum_{i=1}^G x_i^2 v_i\right) \Psi(\boldsymbol{\Sigma}_0).$$

Since $\Psi(\boldsymbol{\Sigma}_0) > 0$ is constant in (x_i), minimizing $\Psi(\mathrm{Cov}(\widehat{\boldsymbol{g}}))$ is equivalent to minimizing $\sum_i x_i^2 v_i$ under $\sum_i x_i = 1/M$, whose unique minimizer is the inverse-variance rule in Lemma 3.6. □

Remark 3.3 (Beyond common shape). When the covariance matrices $\mathrm{Cov}(\boldsymbol{g}_i)$ are not approximately proportional to a common shape, the optimal weights can depend on the chosen scalarization objective Ψ (e.g., a particular direction $\boldsymbol{u}$ or a trace objective). CAPO focuses on the setting where a *single* scalar proxy v_i provides a stable and implementable approximation.

Conditional on the variance parameters $\vartheta := (\sigma^2, \beta, \xi)$,

$$g_i \,\Big|\, \mu, \vartheta \;\sim\; \mathcal{N}\Big(\mu,\; v_i(\vartheta)\Big), \qquad v_i(\vartheta) \;=\; \sigma^2\, L_i^\beta\, s(L_i;\, \xi), \tag{16}$$

where $\sigma^2 > 0$ is a global scale, $\beta \in [0, 2]$ controls length scaling, and $s(L_i; \xi) > 0$ is a shape factor encoding within-trajectory dependence. This modular treatment allows us to plug in different estimates for the shape factor neatly.

General priors. We adopt a flat prior on the mean and a flexible, proper prior on the variance and shape parameters:

$$p(\mu) \;\propto\; 1, \qquad \sigma^2 \sim \text{Inv-Gamma}(a_0, b_0), \qquad \beta \sim \pi_\beta, \qquad \xi \sim \pi_\xi. \tag{17}$$

Jeffreys' prior for scale is recovered in the limit $a_0, b_0 \to 0$. The prior π_β can be chosen agnostic (centered at 1 with moderate variance) or informative to reflect anticipated sub/superlinear growth. The prior π_ξ encodes beliefs about token dependence.

Likelihood geometry and sufficient sums. Define precisions and aggregated quantities

$$\lambda_i(\vartheta) := \frac{1}{v_i(\vartheta)} = \frac{1}{\sigma^2}\,\omega_i(\beta, \xi), \qquad \omega_i(\beta, \xi) := L_i^{-\beta}\, s(L_i; \xi)^{-1}, \tag{18}$$

$$\Lambda(\vartheta) \;=\; \sum_{i=1}^G \lambda_i(\vartheta), \qquad T(\vartheta) \;=\; \sum_{i=1}^G \lambda_i(\vartheta)\, g_i. \tag{19}$$

Then the conditional posterior for μ given ϑ is Normal:

$$\mu \,\Big|\, \vartheta, g_i \;\sim\; \mathcal{N}\Big(m(\vartheta),\, \Lambda(\vartheta)^{-1}\Big), \qquad m(\vartheta) = \frac{T(\vartheta)}{\Lambda(\vartheta)} = \frac{\sum_i \lambda_i(\vartheta) g_i}{\sum_i \lambda_i(\vartheta)}. \tag{20}$$

Let $m(\beta, \xi)$ denote $m(\vartheta)$ with σ^2 factored out via equation 18, and define the weighted residual sum of squares

$$\mathrm{RSS}_\omega(\beta, \xi) \;=\; \sum_{i=1}^{G} \omega_i(\beta, \xi) \Big(g_i - m(\beta, \xi)\Big)^2. \tag{21}$$

Combining equation 17–equation 21, the conditional posterior of σ^2 is

$$\sigma^2 \,\Big|\, \beta, \xi, \{g_i\} \;\sim\; \text{Inv-Gamma}\Big(a_0 + \frac{G-1}{2},\; b_0 + \frac{1}{2}\,\mathrm{RSS}_\omega(\beta, \xi)\Big). \tag{22}$$

Proposition 3.4 (Conditional posterior of σ^2)**.** *Under the Gaussian model*

$$g_i \mid \mu, \vartheta \;\sim\; \mathcal{N}\Big(\mu,\; v_i(\vartheta)\Big), \qquad v_i(\vartheta) = \sigma^2 L_i^{\beta} s(L_i; \xi) \;=\; \frac{\sigma^2}{\omega_i(\beta, \xi)},$$

with $\vartheta = (\sigma^2, \beta, \xi)$ and $\omega_i(\beta, \xi) := L_i^{-\beta} s(L_i; \xi)^{-1}$ (cf. equation 18), and priors $p(\mu) \propto 1$, $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$ as in equation 17, the conditional posterior of σ^2 given (β, ξ) and data $\{g_i\}_{i=1}^G$ is

$$\sigma^2 \,\Big|\, \beta, \xi, \{g_i\} \;\sim\; \text{Inv-Gamma}\Big(a_0 + \frac{G-1}{2},\; b_0 + \frac{1}{2}\,\mathrm{RSS}_\omega(\beta, \xi)\Big), \tag{22}$$

where the weighted mean is

$$m(\beta, \xi) \;:=\; \frac{\sum_{i=1}^{G} \omega_i(\beta, \xi)\, g_i}{\sum_{i=1}^{G} \omega_i(\beta, \xi)}$$

and the weighted residual sum of squares is

$$\mathrm{RSS}_\omega(\beta, \xi) \;:=\; \sum_{i=1}^{G} \omega_i(\beta, \xi) \Big(g_i - m(\beta, \xi)\Big)^2. \tag{21}$$

Proof. Step 1: Likelihood in (μ, σ^2) up to (β, ξ)-constants. Conditional on (β, ξ), the variances are $v_i = \sigma^2/\omega_i$. Thus the likelihood is

$$p(\{g_i\} \mid \mu, \sigma^2, \beta, \xi) = \prod_{i=1}^{G} \frac{\sqrt{\omega_i}}{\sqrt{2\pi\sigma^2}} \exp\Big\{-\frac{\omega_i}{2\sigma^2}(g_i - \mu)^2\Big\}.$$

Dropping factors independent of (μ, σ^2) (i.e., $\prod_i \sqrt{\omega_i}$), the kernel is

$$p(\{g_i\} \mid \mu, \sigma^2, \beta, \xi) \;\propto\; (\sigma^2)^{-G/2} \exp\Big\{-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i - \mu)^2\Big\}. \tag{23}$$

Step 2: Multiply by priors. With $p(\mu) \propto 1$ and $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$ whose density kernel is $(\sigma^2)^{-a_0-1} \exp\{-b_0/\sigma^2\}$, the joint posterior kernel is

$$p(\mu, \sigma^2 \mid \beta, \xi, \{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)} \exp\Big\{-\frac{1}{\sigma^2}\Big(b_0 + \frac{1}{2}\sum_{i=1}^{G} \omega_i (g_i - \mu)^2\Big)\Big\}.$$

Step 3: Complete the square in μ. Let $\Lambda_\omega := \sum_{i=1}^G \omega_i$ and $m(\beta,\xi) := \Big(\sum_i \omega_i g_i\Big)/\Lambda_\omega$. Using the weighted-mean identity,

$$\sum_{i=1}^G \omega_i (g_i-\mu)^2 = \sum_{i=1}^G \omega_i \Big(g_i - m(\beta,\xi)\Big)^2 + \Lambda_\omega\,(\mu - m(\beta,\xi))^2,$$

we obtain

$$p(\mu,\sigma^2 \mid \beta,\xi,\{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)} \exp\!\Big\{-\frac{1}{\sigma^2}\Big(b_0+\frac{1}{2}\mathrm{RSS}_\omega(\beta,\xi)\Big)\Big\} \exp\!\Big\{-\frac{\Lambda_\omega}{2\sigma^2}\,(\mu-m(\beta,\xi))^2\Big\}.$$

Step 4: Integrate out μ. Because $p(\mu)\propto 1$ and the μ-kernel is Gaussian, we have

$$\int_{\mathbb{R}} \exp\!\Big\{-\frac{\Lambda_\omega}{2\sigma^2}(\mu-m)^2\Big\}\,d\mu = \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}},$$

which contributes a factor $(\sigma^2)^{1/2}$ (and a constant in (β,ξ)) to the kernel. Therefore,

$$p(\sigma^2 \mid \beta,\xi,\{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+1+\frac{G}{2}\right)} (\sigma^2)^{1/2} \exp\!\Big\{-\frac{1}{\sigma^2}\Big(b_0+\frac{1}{2}\mathrm{RSS}_\omega(\beta,\xi)\Big)\Big\}.$$

Step 5: Identify the inverse-gamma form. Collecting powers of σ^2 gives

$$p(\sigma^2 \mid \beta,\xi,\{g_i\}) \;\propto\; (\sigma^2)^{-\left(a_0+\frac{G-1}{2}+1\right)} \exp\!\Big\{-\frac{1}{\sigma^2}\Big(b_0+\frac{1}{2}\mathrm{RSS}_\omega(\beta,\xi)\Big)\Big\},$$

which is the kernel of Inv-Gamma$\Big(a_0+\frac{G-1}{2},\; b_0+\frac{1}{2}\mathrm{RSS}_\omega(\beta,\xi)\Big)$. □

Remark 3.5. (i) The factorization $\lambda_i(\vartheta) = \sigma^{-2}\omega_i(\beta,\xi)$ implies $m(\beta,\xi)$ does *not* depend on σ^2, since the common factor cancels: $m(\vartheta) = T(\vartheta)/\Lambda(\vartheta) = \Big(\sum \lambda_i g_i\Big)/\Big(\sum \lambda_i\Big) = \Big(\sum \omega_i g_i\Big)/\Big(\sum \omega_i\Big)$.
(ii) The shape update $a_0 \mapsto a_0 + \frac{G-1}{2}$ reflects the loss of one degree of freedom from estimating the (flat-prior) mean μ.
(iii) Constants that depend only on (β,ξ), such as $\prod_i \sqrt{\omega_i}$ and $\Lambda_\omega^{-1/2}$, are immaterial for $p(\sigma^2 \mid \beta,\xi,\{g_i\})$ and are absorbed into the normalizing constant.

3.3 Formal Results and Aggregators

Lemma 3.6 (MVU weights with known v_i)**.** *Let $(g_1,\dots,g_G)$ be scalar random variables such that $\mathbb{E}[g_i]=\mu$ for all i, $\mathrm{Var}(g_i)=v_i\in(0,\infty)$ are known, and the trajectories are cross-uncorrelated: $\mathrm{Cov}(g_i,g_j)=0$ for all $i\neq j$. For any weights $(x_1,\dots,x_G)$ satisfying the unbiasedness constraint $\sum_{i=1}^G x_i = 1/M$, the linear estimator $\widehat{\mu} := \sum_{i=1}^G x_i g_i$ is unbiased for μ/M and satisfies*

$$\mathrm{Var}(\widehat{\mu}) = \sum_{i=1}^G x_i^2 v_i.$$

The unique minimizer of $\mathrm{Var}(\widehat{\mu})$ over all such unbiased linear estimators is

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^G v_j^{-1}}, \qquad \text{(10 revisited)}$$

i.e., inverse-variance weighting with normalization fixed by $\sum_i x_i = 1/M$.

Proof. Under the stated cross-uncorrelatedness, for any weights $(x_1, \ldots, x_G)$ we have

$$\mathrm{Var}\left(\sum_{i=1}^{G} x_i g_i\right) = \sum_{i=1}^{G} x_i^2 \,\mathrm{Var}(g_i) = \sum_{i=1}^{G} x_i^2 v_i.$$

We minimize the strictly convex quadratic function $Q(x) := \sum_{i=1}^{G} x_i^2 v_i$ over the affine subspace $\{x \in \mathbb{R}^G : \sum_{i=1}^{G} x_i = 1/M\}$.

Introduce the Lagrangian

$$\mathcal{L}(x_1, \ldots, x_G, \lambda) = \sum_{i=1}^{G} x_i^2 v_i - \lambda \left(\sum_{i=1}^{G} x_i - \frac{1}{M}\right).$$

The first-order optimality conditions are, for each $i \in \{1, \ldots, G\}$,

$$\frac{\partial \mathcal{L}}{\partial x_i} = 2x_i v_i - \lambda = 0 \quad \Longrightarrow \quad x_i = \frac{\lambda}{2v_i}.$$

Imposing the constraint $\sum_i x_i = 1/M$ yields

$$\frac{1}{M} = \sum_{i=1}^{G} x_i = \frac{\lambda}{2} \sum_{i=1}^{G} v_i^{-1} \quad \Longrightarrow \quad \frac{\lambda}{2} = \frac{1/M}{\sum_{j=1}^{G} v_j^{-1}}.$$

Substituting back gives

$$x_i^\star = \frac{1}{M} \frac{v_i^{-1}}{\sum_{j=1}^{G} v_j^{-1}},$$

which satisfies the constraint and the first-order conditions. Since Q is strictly convex (all $v_i > 0$), the solution is the unique global minimizer. □

Proposition 3.7 (Robustness of plug-in inverse-variance weights)**.** *Assume the setting of Lemma 3.6 and let $\{\widehat{v}_i\}_{i=1}^{G}$ be positive variance estimates such that, for some $c \geq 1$,*

$$\frac{1}{c} v_i \;\leq\; \widehat{v}_i \;\leq\; c\, v_i \qquad \textit{for all } i \in \{1, \ldots, G\}. \tag{24}$$

Define the plug-in inverse-variance weights

$$\widehat{x}_i \;:=\; \frac{1}{M} \frac{\widehat{v}_i^{-1}}{\sum_{j=1}^{G} \widehat{v}_j^{-1}}.$$

Let $V^\star$ denote the minimum variance achieved by the oracle MVU weights of Lemma 3.6, and let $\widehat{V} := \mathrm{Var}(\sum_i \widehat{x}_i g_i)$ denote the variance under the plug-in weights. Then the plug-in rule is variance-robust in the sense that

$$\frac{1}{c^2} V^\star \;\leq\; \widehat{V} \;\leq\; c^2 V^\star.$$

Proof. Let $a_i := v_i^{-1}$ and $\widehat{a}_i := \widehat{v}_i^{-1}$ so that equation 24 is equivalent to $\widehat{a}_i \in [a_i/c,\, ca_i]$. Define $S := \sum_{i=1}^G a_i$ and $\widehat{S} := \sum_{i=1}^G \widehat{a}_i$. A direct computation (see the proof of Lemma 3.6) gives the oracle optimum

$$V^\star = \frac{1}{M^2}\,\frac{1}{S}.$$

For the plug-in weights, cross-trajectory uncorrelatedness gives

$$\widehat{V} = \sum_{i=1}^G \widehat{x}_i^2 v_i = \frac{1}{M^2}\,\frac{\sum_{i=1}^G \widehat{a}_i^2 v_i}{\widehat{S}^2} = \frac{1}{M^2}\,\frac{\sum_{i=1}^G \widehat{a}_i^2/a_i}{\widehat{S}^2}.$$

Since $\widehat{a}_i/a_i \le c$, we have $\widehat{a}_i^2/a_i = \widehat{a}_i(\widehat{a}_i/a_i) \le c\,\widehat{a}_i$ and hence $\sum_i \widehat{a}_i^2/a_i \le c\,\widehat{S}$. Therefore

$$\widehat{V} \le \frac{1}{M^2}\,\frac{c}{\widehat{S}}.$$

Moreover, $\widehat{a}_i \ge a_i/c$ implies $\widehat{S} \ge S/c$, so $\widehat{V} \le \frac{1}{M^2}\frac{c}{S/c} = c^2 V^\star$.

For the lower bound, $\widehat{a}_i/a_i \ge 1/c$ implies $\sum_i \widehat{a}_i^2/a_i \ge (1/c)\widehat{S}$, hence $\widehat{V} \ge \frac{1}{M^2}\frac{1/c}{\widehat{S}}$. Since $\widehat{a}_i \le ca_i$ implies $\widehat{S} \le cS$, we conclude $\widehat{V} \ge \frac{1}{M^2}\frac{1/c}{cS} = \frac{1}{c^2}V^\star$. □

Theorem 3.8 (Posterior-precision aggregation and scale cancellation)**.** *Consider the scalarized Gaussian model equation 15–equation 16 and assume the heteroscedastic variances factorize as*

$$v_i(\vartheta) = \sigma^2\, f_i(\vartheta), \qquad f_i(\vartheta) > 0, \tag{25}$$

so that the conditional precisions satisfy $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, f_i(\vartheta)^{-1}$. Fix ϑ (e.g., $\vartheta = (\beta, \xi)$) and endow σ^2 with any prior such that $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}_{j=1}^G]$ exists and is finite.

Define the posterior-precision scores

$$\widetilde{w}_i(\vartheta) \;:=\; \mathbb{E}\Big[\lambda_i(\vartheta)\,\Big|\,\vartheta, \{g_j\}_{j=1}^G\Big] \;=\; f_i(\vartheta)^{-1}\,\mathbb{E}\Big[\sigma^{-2}\,\Big|\,\vartheta, \{g_j\}_{j=1}^G\Big]. \tag{26}$$

Then the (unique) minimum-variance unbiased linear estimator of μ/M of the form $\widehat{\mu} = \sum_{i=1}^G x_i g_i$ under the constraint $\sum_{i=1}^G x_i = 1/M$ is obtained by normalizing the relative precisions:

$$x_i^{\text{Bayes}}(\vartheta) \;=\; \frac{1}{M}\,\frac{f_i(\vartheta)^{-1}}{\sum_{j=1}^G f_j(\vartheta)^{-1}} \;=\; \frac{1}{M}\,\frac{\widetilde{w}_i(\vartheta)}{\sum_{j=1}^G \widetilde{w}_j(\vartheta)}. \tag{27}$$

In particular, the common scale factor $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$ cancels from the normalized weights.

Proof. Fix ϑ and condition on (σ^2, ϑ). Under equation 25, the conditional variances are $v_i(\vartheta) = \sigma^2 f_i(\vartheta)$ and hence

$$v_i(\vartheta)^{-1} = \lambda_i(\vartheta) = \frac{1}{\sigma^2}\, f_i(\vartheta)^{-1}.$$

Therefore, conditional on (σ^2, ϑ) the variances $\{v_i(\vartheta)\}$ are known and Lemma 3.6 applies, yielding the unique minimizer

$$x_i^\star(\sigma^2, \vartheta) = \frac{1}{M}\,\frac{v_i(\vartheta)^{-1}}{\sum_{j=1}^G v_j(\vartheta)^{-1}} = \frac{1}{M}\,\frac{f_i(\vartheta)^{-1}}{\sum_{j=1}^G f_j(\vartheta)^{-1}},$$

where the factor σ^{-2} cancels from numerator and denominator. This is exactly the first equality in equation 27.

For the second equality, use equation 26 to write $\widetilde{w}_i(\vartheta) = f_i(\vartheta)^{-1}\,\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$; normalizing $\widetilde{w}_i(\vartheta)$ across i cancels the common factor $\mathbb{E}[\sigma^{-2} \mid \vartheta, \{g_j\}]$, giving the same weights as above. □

Remark 3.9 (When ϑ is random). If ϑ itself is endowed with a non-degenerate prior (e.g., a prior on the length exponent β), then two distinct Bayesian constructions become relevant:

1. The *exact* posterior mean of the conditional MVU weights $x_i^\star(\vartheta) \propto f_i(\vartheta)^{-1}$, namely $\mathbb{E}[x_i^\star(\vartheta) \mid \{g_j\}]$.
2. The *posterior-precision approximation* that replaces $f_i(\vartheta)^{-1}$ by its posterior mean $\mathbb{E}[f_i(\vartheta)^{-1} \mid \{g_j\}]$ and then renormalizes across i.

These coincide whenever the posterior over ϑ is sufficiently concentrated or when $f_i(\vartheta)$ is treated as deterministic via Empirical Bayes (as done in §3.8).

Theorem 3.10 (Jeffreys' baseline recovers ΔL with $\alpha = 1$)**.** *Assume independence $s(\cdot;\xi) \equiv 1$, fix $\beta = 1$, and take $a_0, b_0 \to 0$ so that $p(\sigma^2) \propto 1/\sigma^2$. Then*

$$x_i^{\text{Bayes}} \;\propto\; L_i^{-1}. \qquad \text{(12 revisited)}$$

Proof. With $\beta = 1$ and $s \equiv 1$, we have $\lambda_i(\vartheta) = \sigma^{-2} L_i^{-1}$. From equation 22, $\mathbb{E}[\sigma^{-2} \mid \beta{=}1, \{g\}] = (G-1)/\mathrm{RSS}_\omega(\beta{=}1)$, a common factor that cancels in Theorem 3.8, yielding $x_i \propto L_i^{-1}$. □

Proposition 3.11 (Correctness-linked contraction implies $\alpha < 1$)**.** *Suppose the per-token variance contracts with response length via $\kappa(L) \in (0,1]$ with $\kappa'(L) \le 0$ and $\kappa(L) = c\,L^{-\delta}(1 + o(1))$ for some $\delta \in [0,1)$, so that*

$$\mathrm{Var}(g_i \mid L_i = L) = \sigma^2\, L\, \kappa(L)\, s(L;\xi) \;=\; \sigma^2\, L^{1-\delta}\, s(L;\xi)\,(1 + o(1)). \tag{28}$$

Then in the independence benchmark $s \equiv 1$ the implied ΔL exponent is $\alpha = \beta = 1 - \delta < 1$.

Proof. The effective length law in equation 28 gives variance exponent $\beta = 1 - \delta$. Applying Theorem 3.8 with Jeffreys' scale (or any scale prior whose common factor cancels) yields weights $\propto L^{-\beta}$, hence $\alpha = \beta = 1 - \delta < 1$. □

Corollary 3.12 (Serial dependence corrections)**.** *If $s(L;\xi) = \Theta(1)$ (e.g., AR(1) or k-band dependence), then the large-L exponent in Proposition 3.11 is unchanged; dependence enters through the multiplicative factor $s(L;\xi)^{-1}$ in the weights.*

3.4 Informative Prior on the Length Exponent

Retain independence ($s \equiv 1$) but let $\beta \sim \pi_\beta$ with posterior $p(\beta \mid \{g_i\})$. Using equation 27,

$$\widetilde{w}_i \;\propto\; \mathbb{E}\Big[L_i^{-\beta} \,\Big|\, \{g_j\}\Big]. \tag{29}$$

Two useful consequences:

1. *MAP/Laplace approximation.* If $p(\beta \mid \{g_i\})$ is sharply peaked at $\hat{\beta}$, then $x_i^{\text{Bayes}} \propto L_i^{-\hat{\beta}}$.
2. *Gaussian posterior moment.* If $\beta \mid \{g_i\} \sim \mathcal{N}(m_\beta, \tau_\beta^2)$ (ignoring truncation),

$$\mathbb{E}\Big[L_i^{-\beta} \mid \{g_i\}\Big] \;=\; \exp\Big(-m_\beta \ln L_i + \tfrac{1}{2}\tau_\beta^2 (\ln L_i)^2\Big) \;\approx\; L_i^{-m_\beta} \quad (\tau_\beta^2 \text{ small}), \tag{30}$$

i.e., an effective $\alpha \simeq m_\beta$.

3.5 Why $\beta < 1$ When Longer Responses Are More Likely Correct

Intuition. If longer responses have a higher probability of being correct, then, after baselining, their token contributions are less noisy on average. In a heteroscedastic model, the per-token variance decreases with length, so the sum over L tokens grows sub-linearly in L. Consequently, inverse-variance weights scale as $L^{-\alpha}$ with $\alpha < 1$.

Heteroskedastic variance model tied to correctness. Let $g_i = \sum_{t=1}^{L_i} Z_{i,t}$ be the scalarized trajectory gradient (see equation 5 and equation 16). Suppose

$$\mathrm{Var}(Z_{i,t} \mid L_i = L) \;=\; \sigma^2\, \kappa(L), \qquad \kappa'(L) \le 0, \tag{31}$$

with $\kappa(L) \propto p(L)\{1 - p(L)\}$ for a correctness probability $p(L)$ with $p(L) \uparrow 1$ as $L \uparrow \infty$. Then

$$\mathrm{Var}(g_i \mid L_i = L) \;=\; \sigma^2\, L\, \kappa(L)\, s(L; \xi). \tag{32}$$

If $\kappa(L) = c\, L^{-\delta}(1 + o(1))$ with $\delta \in [0, 1)$, we recover equation 28 and thus $\alpha = 1 - \delta < 1$ by Proposition 3.11. The conclusion persists with serial dependence by Corollary 3.12.

3.6 Weights under k-banded Covariance Structure

We work with the scalarized series $Y_{i,t} := \boldsymbol{u}^\top \boldsymbol{Z}_{i,t}$ and $g_i = \sum_{t=1}^{L_i} Y_{i,t}$ as in equation 15. Assume second–order stationarity and the *k-banded, stretched-geometric* auto-covariance:

$$\gamma(0) = \mathrm{Var}(Y_{i,t}) = \tau^2, \qquad \gamma(h) = \mathrm{Cov}(Y_{i,t}, Y_{i,t-h}) = \tau^2\, \rho^{|h|^\eta}\; \mathbf{1}\{|h| \le k\}, \quad \eta \ge 0, \tag{33}$$

with $|\rho| < 1$ for $\eta > 0$. The case $\eta = 1$ reduces to geometric decay $\gamma(h) = \tau^2 \rho^{|h|}$; $\eta \downarrow 0$ yields *no decay across lags* $\gamma(h) = \tau^2 \rho$ (compound-symmetry within the band). Let $m_i := \min\{k, L_i - 1\}$.

Lemma 3.13 (Dependence shape factor for stretched-geometric k-band)**.** *Under equation 33,*

$$\mathrm{Var}(g_i) = \tau^2 \left[L_i + 2 \sum_{h=1}^{m_i} (L_i - h)\, \rho^{h^\eta} \right] \tag{34}$$

$$= \tau^2\, L_i\, s(L_i; \rho, k, \eta), \tag{35}$$

$$s(L; \rho, k, \eta) = 1 + \frac{2}{L} \sum_{h=1}^{m} (L - h) \rho^{h^\eta}. \tag{36}$$

Proof. By Bartlett's identity (Lemma C.1), $\mathrm{Var}(g_i) = \sum_{h=-(L_i-1)}^{L_i-1}(L_i - |h|)\gamma(h)$. Insert equation 33, collect positive and negative lags, and truncate at m_i. □

Proposition 3.14 (Weighting rule under stretched-geometric k-band)**.** *Under the heteroscedastic model equation 16 with $v_i(\vartheta) = \sigma^2 L_i^\beta\, s(L_i;\rho,k,\eta)$, the posterior-precision (Bayes) weights satisfy, up to a common factor,*

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\, s(L_i;\rho,k,\eta)} \;=\; \frac{1}{L_i^\beta\Big(1+\frac{2}{L_i}\sum_{h=1}^{m_i}(L_i-h)\rho^{h^\eta}\Big)}. \tag{37}$$

In particular, under the Jeffreys baseline $\beta = 1$, $x_i \propto \Big[L_i + 2\sum_{h=1}^{m_i}(L_i-h)\rho^{h^\eta}\Big]^{-1}$.

Proof. From equation 16 and Lemma 3.13, $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, L_i^{-\beta}\, s(L_i;\rho,k,\eta)^{-1}$. Apply Theorem 3.8 and note $1/\sigma^2$ cancels upon normalization. □

Lemma 3.15 (Monotonicity in the decay exponent)**.** *For fixed (ρ,k,L) with $\rho \in (0,1)$, $s(L;\rho,k,\eta)$ is non-increasing in η:*

$$0 \le \eta_1 < \eta_2 \;\Rightarrow\; s(L;\rho,k,\eta_1) \;\ge\; s(L;\rho,k,\eta_2).$$

Consequently, the weight magnitude $[L^\beta s(L;\rho,k,\eta)]^{-1}$ is non-decreasing in η.

Proof. For $h \ge 1$ and $\rho \in (0,1)$, the map $\eta \mapsto \rho^{h^\eta}$ is decreasing, hence each summand in equation 35 decreases in η. □

No-decay limit ($\eta = 0$): constant-lag covariance within the band. Setting $\eta = 0$ in equation 33 gives $\gamma(h) = \tau^2\rho$ for $1 \le |h| \le k$. Then

$$s(L;\rho,k,0) \;=\; 1 + \frac{2\rho}{L}\sum_{h=1}^{m}(L-h) \;=\; 1 + 2m\rho - \frac{m(m+1)}{L}\rho, \qquad m = \min\{k, L-1\}. \tag{38}$$

Accordingly,

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\Big(1 + 2m_i\rho - \frac{m_i(m_i+1)}{L_i}\rho\Big)}, \qquad m_i = \min\{k, L_i - 1\}. \tag{39}$$

Compound symmetry. Taking $k \ge L-1$ in equation 38 yields $m = L-1$ and

$$s(L;\rho,k\ge L-1,0) = 1 + \rho(L-1) = (1-\rho) + \rho L, \tag{40}$$

which is the compound-symmetry shape from Lemma 3.35. Thus

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i^\beta\Big(1+\rho(L_i-1)\Big)} \qquad \text{and, for } \beta = 1, \qquad x_i \;\propto\; \frac{1}{(1-\rho)L_i + \rho L_i^2}. \tag{41}$$

Remark 3.16 (Asymptotics). For fixed k and $L \to \infty$, $s(L;\rho,k,\eta) = 1 + 2\sum_{h=1}^{k}\rho^{h^\eta} + O(L^{-1})$, a positive constant depending on (ρ,k,η). Hence the length exponent is unchanged and dependence multiplies the ΔL rule by a constant. Under compound symmetry (no band), $s(L;\rho,\cdot,0) = (1-\rho) + \rho L$, so $v_i = \Theta(L_i^{\beta+1})$ and, under $\beta = 1$, $x_i \sim (\rho L_i^2)^{-1}$ as $L_i \to \infty$.

3.7 Why $\beta > 1$ When Longer Responses Are More Likely Correct (with tail degradation)

Intuition. If longer responses are more likely to be correct up to a point, then after baselining, their token contributions become less noisy as length grows toward an optimum length $L^\star$. Beyond $L^\star$, if the model gets lost, correctness drops and noise increases again. In a heteroskedastic model, this corresponds to a *U-shaped* per-token variance profile $\kappa(L)$: initially decreasing, then increasing. Consequently, the variance of the sum over L tokens grows *sublinearly* before $L^\star$ and *superlinearly* in the tail, yielding inverse-variance weights that scale like $L^{-\beta(L)}$ with $\beta(L) < 1$ for moderate L and $\beta(L) > 1$ in the tail.

Correctness-weighted Heteroscedastic variance model (non-monotone variance profile). Let $g_i = \sum_{t=1}^{L_i} Z_{i,t}$ be the scalarized trajectory gradient (see equation 5 and equation 16). As before, write

$$\mathrm{Var}(Z_{i,t} \mid L_i = L) \;=\; \sigma^2\,\kappa(L), \qquad \mathrm{Var}(g_i \mid L_i = L) \;=\; \sigma^2\,L\,\kappa(L)\,s(L;\xi), \tag{42}$$

where $s(L;\xi)$ encodes serial dependence (Sec. 3.6). We now allow $\kappa(L)$ to be *non-monotone*:

Assumption 3.17 (Tail degradation)**.** *There exist $L^\star \in (1,\infty)$ and exponents $\delta \in [0,1)$, $\theta > 0$, and constants $c_-, c_+ > 0$ such that*

$$\kappa(L) = \begin{cases} c_-\,L^{-\delta}\big(1+o(1)\big), & L \nearrow L^\star \text{ (moderate regime)},\\ c_+\,L^{\;\theta}\big(1+o(1)\big), & L \to \infty \text{ (tail regime)}. \end{cases}$$

That is, κ decreases polynomially up to $L^\star$ and increases polynomially beyond it.

Local length exponent and effective law. Define the *local variance elasticity*

$$\varepsilon(L) \;:=\; \frac{d\log\kappa(L)}{d\log L}.$$

From equation 42, $\log v(L) = \log\sigma^2 + \log L + \log\kappa(L) + \log s(L;\xi)$, so (ignoring the slowly varying $s(L;\xi)$ for the exponent),

$$\beta_{\mathrm{eff}}(L) \;:=\; \frac{d\log v(L)}{d\log L} \;\approx\; 1+\varepsilon(L). \tag{43}$$

Assumption 3.17 implies

$$\beta_{\mathrm{eff}}(L) \approx \begin{cases} 1-\delta, & L \lesssim L^\star,\\ 1+\theta, & L \gg L^\star, \end{cases} \qquad \text{so} \quad v(L) \;\asymp\; \begin{cases} \sigma^2\,L^{\,1-\delta}\,s(L;\xi), & L \lesssim L^\star,\\ \sigma^2\,L^{\,1+\theta}\,s(L;\xi), & L \gg L^\star. \end{cases}$$

Proposition 3.18 (Posterior-precision weights with tail degradation)**.** *Under the heteroskedastic Normal model with variance $v(L)$ as in equation 42 and Assumption 3.17, the Bayes (posterior-precision) weights satisfy, up to a common factor,*

$$x_i^{\mathrm{Bayes}} \;\propto\; \frac{1}{v(L_i)} \;\approx\; \begin{cases} L_i^{-(1-\delta)}\,s(L_i;\xi)^{-1}, & L_i \lesssim L^\star,\\ L_i^{-(1+\theta)}\,s(L_i;\xi)^{-1}, & L_i \gg L^\star. \end{cases} \tag{44}$$

Equivalently, the implied ΔL exponent is $\beta(L) \approx \beta_{\text{eff}}(L)$, i.e., $\beta(L) < 1$ before $L^\star$ and $\beta(L) > 1$ in the tail.

Proof. Theorem 3.8 yields $x_i^{\text{Bayes}} \propto \mathbb{E}[1/v(L_i) \mid \text{data}]$. Under Assumption 3.17, $v(L)$ is regularly varying with index $1-\delta$ for $L \lesssim L^\star$ and $1+\theta$ for $L \to \infty$, hence the stated asymptotics. The multiplicative $s(L;\xi)$ enters as in Sec. 3.6. □

Impact of serial dependence. The factor $s(L;\xi)$ multiplies $v(L)$ and hence divides the weights:

$$x_i^{\text{Bayes}} \;\propto\; \frac{1}{L_i\,\kappa(L_i)\,s(L_i;\xi)}.$$

For k-banded dependence with fixed k (Sec. 3.6), $s(L;\xi) \to s_\infty(\xi)$, a constant; the exponents in equation 3.7 remain $1 \pm (\delta, \theta)$. Under compound symmetry (constant-lag covariance; Sec. 3.9), $s(L;\rho) = (1-\rho) + \rho L$ contributes an extra factor $\sim \rho L$ as $L \to \infty$, so the tail exponent becomes $2+\theta$ and

$$x_i^{\text{Bayes}} \;\propto\; L_i^{-(2+\theta)} \quad \text{(tail, compound symmetry)}.$$

Concrete parametric example (unimodal correctness). Let the correctness probability be unimodal in $\log L$, e.g.,

$$p(L) = \Big(1 + \exp\{-a_0 - a_1 \log L + a_2 (\log L)^2\}\Big)^{-1}, \qquad a_2 > 0,$$

so $p(L) \uparrow$ for small L and $p(L) \downarrow$ for large L. If the variance factor is $\kappa(L) \propto p(L)\{1-p(L)\}$ around the interior (maximal near $p = 1/2$) but is *augmented* by a tail noise term $\nu(L) \asymp L^\theta$ (credit-assignment drift), i.e.,

$$\kappa(L) \;\asymp\; c_- L^{-\delta} \;\text{ for } L \lesssim L^\star, \qquad \kappa(L) \;\asymp\; c_+ L^{\theta} \;\text{ for } L \gg L^\star,$$

then Proposition 3.18 applies with the same exponents.

Remark 3.19. Practical summary.

- For **moderate lengths** where correctness improves, use $\beta \simeq 1-\delta < 1$.
- In the **long-tail** where the model gets lost, use $\beta \simeq 1+\theta > 1$ (and add $+1$ more under compound symmetry).
- The posterior-precision rule automatically enacts this via $x_i \propto [L_i\,\kappa(L_i)\,s(L_i;\xi)]^{-1}$; the common scale cancels.

3.8 Empirical Bayes estimator for length and dependence

We consider trajectories $i = 1, \ldots, G$ with length $L_i \in \mathbb{N}$ and a scalarized trajectory-gradient statistic $g_i \in \mathbb{R}$ (e.g., a fixed-direction projection of the per-trajectory policy-gradient, as in equation 15). Adopt the Normal working model

$$g_i \mid \mu, \sigma^2, \vartheta \sim \mathcal{N}\Big(\mu,\; v_i(\vartheta)\Big), \qquad v_i(\vartheta) = \sigma^2\, L_i^{\beta}\, s(L_i;\xi), \tag{45}$$

where $\mu \in \mathbb{R}$ is the common location, $\sigma^2 > 0$ is a global scale, $\beta \in \mathbb{R}$ (the length exponent) controls the length–variance law, and $s(L;\xi) > 0$ captures within-trajectory dependence via a low-dimensional shape ξ. We denote $\vartheta = (\beta, \xi)$. Define the precisions and normalized weights

$$\omega_i(\beta,\xi) = \frac{1}{L_i^{\beta}\, s(L_i;\xi)}, \qquad w_i(\beta,\xi) = \frac{\omega_i(\beta,\xi)}{\sum_{j=1}^{G} \omega_j(\beta,\xi)}. \tag{46}$$

We take $p(\mu) \propto 1$ and Jeffreys $p(\sigma^2) \propto 1/\sigma^2$; weak priors π_β, π_ξ may be used or set flat for EB/type-II ML.

Weighted aggregator. The Bayes estimator of μ under equation 45 (for fixed ϑ) is the precision-weighted mean

$$m(\beta,\xi) = \sum_{i=1}^{G} w_i(\beta,\xi)\, g_i = \frac{\sum_{i=1}^{G} \omega_i(\beta,\xi)\, g_i}{\sum_{i=1}^{G} \omega_i(\beta,\xi)}. \tag{47}$$

Let

$$e_i(\beta,\xi) = g_i - m(\beta,\xi), \qquad \Lambda_\omega(\beta,\xi) = \sum_{i=1}^{G} \omega_i(\beta,\xi),$$

and

$$\mathrm{RSS}_\omega(\beta,\xi) = \sum_{i=1}^{G} \omega_i(\beta,\xi)\, e_i(\beta,\xi)^2. \tag{48}$$

Proposition 3.20 (Posterior in closed form, conditional on (β,ξ))**.** *Under equation 45 with $p(\mu) \propto 1$ and $p(\sigma^2) \propto \frac{1}{\sigma^2}$,*

$$\mu \mid \sigma^2, \mathbf{g}, \vartheta \sim \mathcal{N}\Big(m(\beta,\xi),\, \sigma^2/\Lambda_\omega(\beta,\xi)\Big), \quad \sigma^2 \mid \mathbf{g}, \vartheta \sim \mathrm{Inv{-}Gamma}\left(\frac{G-1}{2}, \frac{\mathrm{RSS}_\omega(\beta,\xi)}{2}\right),$$

where $\mathbf{g} = (g_1, \ldots, g_G)$ and we use the inverse-gamma parameterization with density $p(x) \propto x^{-(a+1)} \exp(-b/x)$ for $x > 0$, shape $a > 0$ and scale $b > 0$.

Proof. We first write down the likelihood explicitly. Given $\vartheta = (\beta,\xi)$,

$$v_i(\vartheta) = \sigma^2 L_i^{\beta} s(L_i;\xi) = \frac{\sigma^2}{\omega_i(\beta,\xi)}.$$

Thus the conditional density of g_i is

$$p(g_i \mid \mu, \sigma^2, \vartheta) = \frac{1}{\sqrt{2\pi v_i(\vartheta)}} \exp\left(-\frac{(g_i-\mu)^2}{2v_i(\vartheta)}\right) = \frac{\sqrt{\omega_i}}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\omega_i}{2\sigma^2}(g_i-\mu)^2\right),$$

where, for brevity, we write $\omega_i = \omega_i(\beta,\xi)$. By independence across i,

$$p(\mathbf{g} \mid \mu, \sigma^2, \vartheta) = \prod_{i=1}^{G} p(g_i \mid \mu, \sigma^2, \vartheta) = (2\pi\sigma^2)^{-G/2} \left(\prod_{i=1}^{G} \omega_i^{1/2}\right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i-\mu)^2\right).$$

Multiplying by the priors $p(\mu) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$ yields the unnormalized joint posterior

$$p(\mu, \sigma^2 \mid \mathbf{g}, \vartheta) \;\propto\; (\sigma^2)^{-G/2} \frac{1}{\sigma^2} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i - \mu)^2 \right). \tag{49}$$

Next we complete the square in μ. Let

$$T = \sum_{i=1}^{G} \omega_i g_i, \qquad \Lambda_\omega = \sum_{i=1}^{G} \omega_i, \qquad m = \frac{T}{\Lambda_\omega}.$$

Then

$$\sum_{i=1}^{G} \omega_i (g_i - \mu)^2 = \sum_{i=1}^{G} \omega_i \Big[(g_i - m) + (m - \mu) \Big]^2 = \sum_{i=1}^{G} \omega_i (g_i - m)^2 + 2(m - \mu) \sum_{i=1}^{G} \omega_i (g_i - m) + \Lambda_\omega (\mu - m)^2.$$

The middle term vanishes because

$$\sum_{i=1}^{G} \omega_i (g_i - m) = \sum_{i=1}^{G} \omega_i g_i - m \sum_{i=1}^{G} \omega_i = T - \Lambda_\omega m = 0.$$

Hence

$$\sum_{i=1}^{G} \omega_i (g_i - \mu)^2 = \Lambda_\omega (\mu - m)^2 + \sum_{i=1}^{G} \omega_i (g_i - m)^2 = \Lambda_\omega (\mu - m)^2 + \mathrm{RSS}_\omega,$$

with RSS_ω as in equation 48. Substituting into equation 49 gives

$$p(\mu, \sigma^2 \mid \mathbf{g}, \vartheta) \;\propto\; (\sigma^2)^{-(G/2+1)} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right).$$

Conditional posterior of μ. For fixed σ^2, the dependence on μ is exactly that of a Normal kernel with mean m and variance σ^2/Λ_ω. Normalizing in μ therefore yields

$$\mu \mid \sigma^2, \mathbf{g}, \vartheta \sim \mathcal{N}\Big(m(\beta, \xi),\; \sigma^2 / \Lambda_\omega(\beta, \xi) \Big).$$

Marginal posterior of σ^2. Integrate out μ from $p(\mu, \sigma^2 \mid \mathbf{g}, \vartheta)$. The μ-integral is

$$\int_{-\infty}^{\infty} \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) d\mu = \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}}.$$

Thus

$$p(\sigma^2 \mid \mathbf{g}, \vartheta) \;\propto\; (\sigma^2)^{-(G/2+1)} \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right).$$

The multiplicative constant $\sqrt{2\pi/\Lambda_\omega} \prod_i \omega_i^{1/2}$ does not depend on σ^2, so we drop it. We are left with

$$p(\sigma^2 \mid \mathbf{g}, \vartheta) \;\propto\; (\sigma^2)^{-(G/2+1-1/2)} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right) = (\sigma^2)^{-(G+1)/2} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right).$$

Comparing with the inverse-gamma kernel $x^{-(a+1)} \exp(-b/x)$ on $x > 0$, we read off $a = (G-1)/2$ and $b = \mathrm{RSS}_\omega/2$. This proves the claimed inverse-gamma posterior for σ^2. $\square$

Proposition 3.21 (GLS optimality of the aggregator)**.** *Fix* (β, ξ)*. Among all linear unbiased estimators* $\tilde{\mu} = \sum_{i=1}^{G} a_i g_i$ *with* $\sum_{i=1}^{G} a_i = 1$*,* $m(\beta, \xi)$ *in equation 47 attains the minimum variance* $\mathrm{Var}(\tilde{\mu}) = \sigma^2/\Lambda_\omega(\beta, \xi)$*.*

Proof. For fixed ϑ, the model equation 45 implies

$$\mathbb{E}[g_i \mid \mu, \vartheta] = \mu, \qquad \mathrm{Var}(g_i \mid \mu, \vartheta) = v_i(\vartheta) = \frac{\sigma^2}{\omega_i},$$

and the g_i are conditionally independent. Consider a generic linear estimator $\tilde{\mu} = \sum_i a_i g_i$ with coefficients $a_i \in \mathbb{R}$.

Unbiasedness constraint.

$$\mathbb{E}[\tilde{\mu} \mid \mu, \vartheta] = \sum_{i=1}^{G} a_i \mathbb{E}[g_i \mid \mu, \vartheta] = \sum_{i=1}^{G} a_i \mu = \mu \sum_{i=1}^{G} a_i.$$

Thus $\tilde{\mu}$ is unbiased for μ if and only if $\sum_{i=1}^{G} a_i = 1$.

Variance under the constraint. Using independence and $\mathrm{Var}(g_i) = \sigma^2/\omega_i$,

$$\mathrm{Var}(\tilde{\mu} \mid \mu, \vartheta) = \sum_{i=1}^{G} a_i^2 \,\mathrm{Var}(g_i \mid \mu, \vartheta) = \sigma^2 \sum_{i=1}^{G} \frac{a_i^2}{\omega_i}.$$

We must minimize $\sum_i a_i^2/\omega_i$ subject to $\sum_i a_i = 1$. Introduce a Lagrange multiplier λ and consider

$$\mathcal{L}(a_1, \ldots, a_G, \lambda) = \sum_{i=1}^{G} \frac{a_i^2}{\omega_i} - \lambda \left(\sum_{i=1}^{G} a_i - 1 \right).$$

Setting partial derivatives to zero,

$$\frac{\partial \mathcal{L}}{\partial a_i} = \frac{2a_i}{\omega_i} - \lambda = 0 \quad \Rightarrow \quad a_i = \frac{\lambda}{2}\,\omega_i, \qquad i = 1, \ldots, G.$$

Imposing $\sum_i a_i = 1$ gives

$$\sum_{i=1}^{G} a_i = \frac{\lambda}{2} \sum_{i=1}^{G} \omega_i = \frac{\lambda}{2} \Lambda_\omega = 1 \quad \Rightarrow \quad \lambda = \frac{2}{\Lambda_\omega}.$$

Thus

$$a_i = \frac{\omega_i}{\Lambda_\omega} = w_i(\beta, \xi),$$

and the unique minimizer is exactly the weight vector in equation 47. The corresponding variance is

$$\mathrm{Var}(m(\beta, \xi) \mid \mu, \vartheta) = \sigma^2 \sum_{i=1}^{G} \frac{w_i^2}{\omega_i} = \sigma^2 \sum_{i=1}^{G} \frac{\omega_i^2}{\Lambda_\omega^2\, \omega_i} = \sigma^2 \frac{\sum_i \omega_i}{\Lambda_\omega^2} = \frac{\sigma^2}{\Lambda_\omega}.$$

Hence $m(\beta, \xi)$ is the Gauss–Markov / GLS estimator with minimum variance among all linear unbiased estimators. □

Theorem 3.22 (EB marginal objective for (β, ξ))**.** *Under equation 45 with the above priors, the marginal type-II log-posterior for $\vartheta = (\beta, \xi)$ is (up to an additive constant)*

$$\ell(\beta, \xi) = \log \pi_\beta(\beta) + \log \pi_\xi(\xi) + \tfrac{1}{2} \sum_{i=1}^{G} \log \omega_i(\beta, \xi) - \tfrac{1}{2} \log \Lambda_\omega(\beta, \xi) - \tfrac{G-1}{2} \log \mathrm{RSS}_\omega(\beta, \xi). \quad (50)$$

Proof. We integrate out μ and σ^2 from the joint model

$$p(\mathbf{g}, \mu, \sigma^2 \mid \vartheta) = p(\mathbf{g} \mid \mu, \sigma^2, \vartheta)\, p(\mu)\, p(\sigma^2)$$

and then add the priors π_β, π_ξ on (β, ξ).

From the likelihood calculation in the proof of Proposition 3.20, we have

$$p(\mathbf{g} \mid \mu, \sigma^2, \vartheta) = (2\pi\sigma^2)^{-G/2} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{G} \omega_i (g_i - \mu)^2 \right),$$

and $\sum_{i=1}^{G} \omega_i (g_i - \mu)^2 = \Lambda_\omega (\mu - m)^2 + \mathrm{RSS}_\omega$. Using $p(\mu) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$,

$$p(\mathbf{g}, \mu, \sigma^2 \mid \vartheta) \propto (\sigma^2)^{-(G/2+1)} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right).$$

Integrate with respect to μ. As in Proposition 3.20,

$$\int_{-\infty}^{\infty} \exp\left(-\frac{\Lambda_\omega}{2\sigma^2} (\mu - m)^2 \right) d\mu = \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}},$$

hence

$$\begin{aligned} p(\mathbf{g}, \sigma^2 \mid \vartheta) &\propto (\sigma^2)^{-(G/2+1)} \sqrt{\frac{2\pi\sigma^2}{\Lambda_\omega}} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right) \\ &\propto (\sigma^2)^{-(G+1)/2} \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \Lambda_\omega^{-1/2} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right), \end{aligned}$$

where we have absorbed constants depending on 2π into a factor that does not depend on ϑ.

Now integrate out σ^2 using the standard identity for inverse-gamma integrals:

$$\int_0^{\infty} (\sigma^2)^{-(a+1)} \exp\left(-\frac{b}{\sigma^2} \right) d\sigma^2 = 2\, b^{-a}\, \Gamma(a), \qquad a > 0,\ b > 0.$$

Here we take $a = (G-1)/2$, $b = \mathrm{RSS}_\omega/2$. Thus

$$\int_0^{\infty} (\sigma^2)^{-(G+1)/2} \exp\left(-\frac{\mathrm{RSS}_\omega}{2\sigma^2} \right) d\sigma^2 \propto \mathrm{RSS}_\omega^{-(G-1)/2}.$$

It follows that the marginal likelihood of the data given ϑ is

$$p(\mathbf{g} \mid \vartheta) \propto \left(\prod_{i=1}^{G} \omega_i^{1/2} \right) \Lambda_\omega^{-1/2} \mathrm{RSS}_\omega^{-(G-1)/2},$$

up to factors independent of ϑ. Taking logarithms and adding $\log \pi_\beta(\beta) + \log \pi_\xi(\xi)$ yields

$$\ell(\beta, \xi) = \log p(\mathbf{g} \mid \vartheta) + \log \pi_\beta(\beta) + \log \pi_\xi(\xi) \tag{51}$$

$$= \log \pi_\beta(\beta) + \log \pi_\xi(\xi) + \frac{1}{2} \sum_{i=1}^{G} \log \omega_i - \frac{1}{2} \log \Lambda_\omega \tag{52}$$

$$- \frac{G-1}{2} \log \mathrm{RSS}_\omega + C. \tag{53}$$

where C does not depend on (β, ξ). Dropping the additive constant proves equation 50. □

3.8.1 Prompt-group (GRPO) extension: group-specific means

In GRPO-style training, rollouts are naturally organized into *prompt groups*: for each prompt $p \in \{1, \ldots, P\}$ we sample $N_p \geq 2$ trajectories and construct within-group normalized advantages/baselines. To align the EB variance-law estimation with this grouping while preserving the closed-form structure above, we extend equation 45 to allow a *group-specific* location parameter μ_p that is integrated out with the same flat prior.

Let $\mathcal{I}_p$ be the index set of trajectories in group p, with $|\mathcal{I}_p| = N_p$ and total $G := \sum_{p=1}^{P} N_p$. We observe $(g_{p,i}, L_{p,i})$ for $i \in \mathcal{I}_p$, and adopt the grouped working model

$$g_{p,i} \mid \mu_p, \sigma^2, \vartheta \sim \mathcal{N}\Big(\mu_p,\ v_{p,i}(\vartheta)\Big), \qquad v_{p,i}(\vartheta) = \sigma^2 L_{p,i}^{\beta} s(L_{p,i}; \xi), \tag{54}$$

with independent improper priors $p(\mu_p) \propto 1$ and $p(\sigma^2) \propto 1/\sigma^2$. Define group precisions and weights

$$\omega_{p,i}(\beta, \xi) = \frac{1}{L_{p,i}^{\beta} s(L_{p,i}; \xi)}, \qquad \Lambda_{\omega,p} = \sum_{j \in \mathcal{I}_p} \omega_{p,j}, \qquad w_{p,i} = \frac{\omega_{p,i}}{\Lambda_{\omega,p}}, \tag{55}$$

and the within-group precision-weighted aggregator

$$m_p(\beta, \xi) = \sum_{i \in \mathcal{I}_p} w_{p,i}(\beta, \xi)\, g_{p,i}. \tag{56}$$

Let $e_{p,i}(\beta, \xi) = g_{p,i} - m_p(\beta, \xi)$ and define the grouped RSS

$$\mathrm{RSS}_\omega^{\mathrm{grp}}(\beta, \xi) = \sum_{p=1}^{P} \sum_{i \in \mathcal{I}_p} \omega_{p,i}(\beta, \xi)\, e_{p,i}(\beta, \xi)^2. \tag{57}$$

Theorem 3.23 (Grouped EB marginal objective for $\vartheta = (\beta, \xi)$)**.** *Under equation 54 with the above priors, the marginal type-II log-posterior for $\vartheta = (\beta, \xi)$ is (up to an additive constant)*

$$\ell_{\mathrm{grp}}(\beta, \xi) = \log \pi_\beta(\beta) + \log \pi_\xi(\xi) + \tfrac{1}{2} \sum_{p=1}^{P} \sum_{i \in \mathcal{I}_p} \log \omega_{p,i}(\beta, \xi) - \tfrac{1}{2} \sum_{p=1}^{P} \log \Lambda_{\omega,p}(\beta, \xi) - \tfrac{G-P}{2} \log \mathrm{RSS}_\omega^{\mathrm{grp}}(\beta, \xi). \tag{58}$$

When $P = 1$ (a single group), equation 58 reduces to equation 50.

Proof. Conditioned on ϑ, the likelihood factorizes over groups: $p(\{g_{p,i}\} \mid \{\mu_p\}, \sigma^2, \vartheta) = \prod_{p=1}^{P} \prod_{i \in \mathcal{I}_p} \mathcal{N}\big(g_{p,i}; \mu_p, \sigma^2/\omega_{p,i}\big)$. For each group p, completing the square yields $\sum_{i \in \mathcal{I}_p} \omega_{p,i}(g_{p,i} - \mu_p)^2 = \Lambda_{\omega,p}(\mu_p - m_p)^2 + \sum_{i \in \mathcal{I}_p} \omega_{p,i}(g_{p,i} - m_p)^2$. Integrating out μ_p under $p(\mu_p) \propto 1$ gives a factor $\sqrt{2\pi\sigma^2/\Lambda_{\omega,p}}$ per group, hence the grouped marginal (conditional on σ^2) is proportional to $(\sigma^2)^{-G/2} \prod_{p,i} \omega_{p,i}^{1/2} \cdot \prod_p \Lambda_{\omega,p}^{-1/2} \cdot \exp\{-\mathrm{RSS}_\omega^{\mathrm{grp}}/(2\sigma^2)\}$. Multiplying by $p(\sigma^2) \propto 1/\sigma^2$ and integrating out σ^2 produces the exponent $(G - P)/2$ in equation 58, since integrating out P group means reduces the degrees of freedom by P. Taking logarithms and adding $\log \pi_\beta + \log \pi_\xi$ yields equation 58. □

Lemma 3.24 (Weighted RSS derivative identity)**.** *$\forall$ component $\phi \in \{\beta\} \cup \{\xi$ components$\}$,*

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega(\beta, \xi) = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 = \sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}. \tag{59}$$

Proof. Recall $\mathrm{RSS}_\omega(\beta, \xi) = \sum_{i=1}^{G} \omega_i e_i^2$, with $e_i = g_i - m$ and $m = m(\beta, \xi)$. We differentiate RSS_ω with respect to ϕ. Using the product and chain rules,

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \left[\frac{\partial \omega_i}{\partial \phi} e_i^2 + \omega_i \frac{\partial}{\partial \phi}(e_i^2) \right].$$

Since $e_i = g_i - m$ and g_i does not depend on ϕ, $\frac{\partial e_i}{\partial \phi} = -\frac{\partial m}{\partial \phi}$, so

$$\frac{\partial}{\partial \phi}(e_i^2) = 2e_i \frac{\partial e_i}{\partial \phi} = 2e_i \left(-\frac{\partial m}{\partial \phi} \right) = -2e_i \frac{\partial m}{\partial \phi}.$$

Therefore,

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 - 2\frac{\partial m}{\partial \phi} \sum_{i=1}^{G} \omega_i e_i.$$

To simplify the second term, observe that

$$\sum_{i=1}^{G} \omega_i e_i = \sum_{i=1}^{G} \omega_i (g_i - m) = \sum_{i=1}^{G} \omega_i g_i - m \sum_{i=1}^{G} \omega_i = T - m\Lambda_\omega = 0,$$

because $m = T/\Lambda_\omega$. Thus the term involving $\partial m/\partial \phi$ cancels, and we obtain

$$\frac{\partial}{\partial \phi} \mathrm{RSS}_\omega = \sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2.$$

Finally, use $\frac{\partial \omega_i}{\partial \phi} = \omega_i \frac{\partial \log \omega_i}{\partial \phi}$ to get the second equality:

$$\sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi} e_i^2 = \sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}.$$

□

Proposition 3.25 (Gradient of the EB objective)**.** *With* $\Lambda_\omega, \mathrm{RSS}_\omega$ *as above,*

$$\frac{\partial \ell}{\partial \phi} = \frac{1}{2}\left[\sum_{i=1}^{G} \frac{\partial \log \omega_i}{\partial \phi} - \frac{1}{\Lambda_\omega}\sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi}\right] - \frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi} + \frac{\partial \log \pi(\phi)}{\partial \phi}, \tag{60}$$

where $\pi(\phi)$ *denotes the marginal prior factor in* $\pi_\beta \pi_\xi$ *involving* ϕ*.*

Proof. Starting from equation 50, we differentiate term by term with respect to ϕ.

Prior term. By definition,

$$\frac{\partial}{\partial \phi}\Big(\log \pi_\beta(\beta) + \log \pi_\xi(\xi)\Big) = \frac{\partial \log \pi(\phi)}{\partial \phi},$$

where $\pi(\phi)$ denotes the appropriate prior factor.

Sum of log-precisions. For the third term in equation 50,

$$\frac{\partial}{\partial \phi}\left(\frac{1}{2}\sum_{i=1}^{G} \log \omega_i\right) = \frac{1}{2}\sum_{i=1}^{G} \frac{1}{\omega_i}\frac{\partial \omega_i}{\partial \phi} = \frac{1}{2}\sum_{i=1}^{G} \frac{\partial \log \omega_i}{\partial \phi}.$$

Log of total precision. For the fourth term,

$$\frac{\partial}{\partial \phi}\left(-\frac{1}{2}\log \Lambda_\omega\right) = -\frac{1}{2}\frac{1}{\Lambda_\omega}\frac{\partial \Lambda_\omega}{\partial \phi} = -\frac{1}{2}\frac{1}{\Lambda_\omega}\sum_{i=1}^{G} \frac{\partial \omega_i}{\partial \phi}.$$

Residual sum of squares term. For the final term,

$$\frac{\partial}{\partial \phi}\left(-\frac{G-1}{2}\log \mathrm{RSS}_\omega\right) = -\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\frac{\partial \mathrm{RSS}_\omega}{\partial \phi}.$$

By Lemma 3.24, $\frac{\partial \mathrm{RSS}_\omega}{\partial \phi} = \sum_i \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}$, hence

$$-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\frac{\partial \mathrm{RSS}_\omega}{\partial \phi} = -\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G} \omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \phi}.$$

Collecting the contributions from all terms gives equation 60. □

Corollary 3.26 (Gradient in the length direction)**.** *Since* $\log \omega_i = -\beta \log L_i - \log s(L_i;\xi)$*,* $\partial \log \omega_i / \partial \beta = -\log L_i$ *and* $\partial \omega_i / \partial \beta = -\omega_i \log L_i$*. Hence*

$$\frac{\partial \ell}{\partial \beta} = \frac{1}{2}\left[-\sum_{i=1}^{G} \log L_i + \frac{\sum_{i=1}^{G} \omega_i \log L_i}{\Lambda_\omega}\right] + \frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G} \omega_i e_i^2 \log L_i + \frac{\partial \log \pi_\beta(\beta)}{\partial \beta}. \tag{61}$$

Proof. Apply Proposition 3.25 with $\phi = \beta$. We have

$$\frac{\partial \log \omega_i}{\partial \beta} = -\log L_i, \qquad \frac{\partial \omega_i}{\partial \beta} = \omega_i \frac{\partial \log \omega_i}{\partial \beta} = -\omega_i \log L_i.$$

Substitute these into equation 60. The first bracketed term becomes

$$\frac{1}{2}\left[\sum_{i=1}^{G}\frac{\partial \log \omega_i}{\partial \beta} - \frac{1}{\Lambda_\omega}\sum_{i=1}^{G}\frac{\partial \omega_i}{\partial \beta}\right] = \frac{1}{2}\left[\sum_{i=1}^{G}(-\log L_i) - \frac{1}{\Lambda_\omega}\sum_{i=1}^{G}(-\omega_i \log L_i)\right]$$

$$= \frac{1}{2}\left[-\sum_{i=1}^{G}\log L_i + \frac{\sum_{i=1}^{G}\omega_i \log L_i}{\Lambda_\omega}\right].$$

The second term in equation 60 becomes

$$-\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2 \frac{\partial \log \omega_i}{\partial \beta} = -\frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2(-\log L_i)$$

$$= \frac{G-1}{2}\frac{1}{\mathrm{RSS}_\omega}\sum_{i=1}^{G}\omega_i e_i^2 \log L_i.$$

Adding the prior derivative $\partial \log \pi_\beta(\beta)/\partial\beta$ yields equation 61. □

Corollary 3.27 (Grouped gradient in the length direction (prompt groups))**.** *Under the grouped EB objective equation 58, the gradient with respect to the length exponent β is*

$$\frac{\partial \ell_{\mathrm{grp}}}{\partial \beta} = \frac{1}{2}\left[-\sum_{p=1}^{P}\sum_{i\in\mathcal{I}_p}\log L_{p,i} + \sum_{p=1}^{P}\frac{\sum_{i\in\mathcal{I}_p}\omega_{p,i}\log L_{p,i}}{\Lambda_{\omega,p}}\right] + \frac{G-P}{2}\frac{1}{\mathrm{RSS}_\omega^{\mathrm{grp}}}\sum_{p=1}^{P}\sum_{i\in\mathcal{I}_p}\omega_{p,i}e_{p,i}^2\log L_{p,i} + \frac{\partial \log \pi_\beta(\beta)}{\partial \beta}. \tag{62}$$

Proof. Differentiate equation 58 with respect to β, using $\partial_\beta \log \omega_{p,i} = -\log L_{p,i}$ and $\partial_\beta \omega_{p,i} = -\omega_{p,i}\log L_{p,i}$. The second term in equation 58 contributes a sum of groupwise normalizers $\Lambda_{\omega,p}$, and integrating out P group means yields the replacement $G-1 \mapsto G-P$ in the RSS exponent. The resulting expression is equation 62. □

Definition 3.28 (Stretched-geometric k-band dependence)**.** *Given $k \in \mathbb{N}$, $\rho \in (-1,1)$, $\eta \geq 0$, define with $m = \min\{k, L-1\}$*

$$s(L;\rho,k,\eta) = 1 + \frac{2}{L}\sum_{h=1}^{m}(L-h)\,\rho^{h^\eta}. \tag{63}$$

Lemma 3.29 (Derivatives of the k-band shape)**.** *Let $s_i = s(L_i;\rho,k,\eta)$, $m_i = \min\{k, L_i - 1\}$. Then*

$$\frac{\partial \log \omega_i}{\partial \rho} = -\frac{1}{s_i}\frac{\partial s_i}{\partial \rho}, \quad \frac{\partial s_i}{\partial \rho} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i-h)\,h^\eta\,\rho^{h^\eta-1},$$

$$\frac{\partial \log \omega_i}{\partial \eta} = -\frac{1}{s_i}\frac{\partial s_i}{\partial \eta}, \quad \frac{\partial s_i}{\partial \eta} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i-h)\,\rho^{h^\eta}(\log\rho)\,h^\eta\log h.$$

Proof. By Definition 3.28,

$$s_i = 1 + \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i-h)\,\rho^{h^\eta}.$$

For $\partial s_i/\partial\rho$, note that $\partial(1)/\partial\rho = 0$ and we can differentiate the sum termwise:

$$\frac{\partial s_i}{\partial \rho} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i - h)\,\frac{\partial}{\partial \rho}\Big(\rho^{h^\eta}\Big).$$

Using $\rho^{h^\eta} = \exp(h^\eta \log \rho)$, the chain rule gives

$$\frac{\partial}{\partial \rho}\Big(\rho^{h^\eta}\Big) = h^\eta \rho^{h^\eta - 1},$$

so

$$\frac{\partial s_i}{\partial \rho} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i - h)\, h^\eta\, \rho^{h^\eta - 1}.$$

For $\partial s_i/\partial\eta$, again differentiate termwise:

$$\frac{\partial s_i}{\partial \eta} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i - h)\,\frac{\partial}{\partial \eta}\Big(\rho^{h^\eta}\Big).$$

Write $\rho^{h^\eta} = \exp(h^\eta \log \rho)$. Then

$$\frac{\partial}{\partial \eta}\Big(\rho^{h^\eta}\Big) = \rho^{h^\eta}\frac{\partial}{\partial \eta}\Big(h^\eta \log \rho\Big) = \rho^{h^\eta}(\log \rho) h^\eta \log h.$$

Thus

$$\frac{\partial s_i}{\partial \eta} = \frac{2}{L_i}\sum_{h=1}^{m_i}(L_i - h)\,\rho^{h^\eta}(\log \rho) h^\eta \log h.$$

Finally, since $\omega_i = L_i^{-\beta} s(L_i;\xi)^{-1}$,

$$\log \omega_i = -\beta \log L_i - \log s_i,$$

so

$$\frac{\partial \log \omega_i}{\partial \rho} = -\frac{1}{s_i}\frac{\partial s_i}{\partial \rho}, \qquad \frac{\partial \log \omega_i}{\partial \eta} = -\frac{1}{s_i}\frac{\partial s_i}{\partial \eta}.$$

□

Proposition 3.30 (Unbiasedness of $m(\beta,\xi)$)**.** *For fixed (β,ξ), the estimator $m(\beta,\xi)$ in equation 47 is unbiased for μ.*

Proof. By equation 45, for each i, $\mathbb{E}[g_i \mid \mu,\vartheta] = \mu$. Using equation 47,

$$\mathbb{E}[m(\beta,\xi) \mid \mu,\vartheta] = \mathbb{E}\left[\sum_{i=1}^{G} w_i g_i \;\middle|\; \mu,\vartheta\right] = \sum_{i=1}^{G} w_i\, \mathbb{E}[g_i \mid \mu,\vartheta] = \mu \sum_{i=1}^{G} w_i.$$

From equation 46, $\sum_i w_i = 1$. Therefore $\mathbb{E}[m(\beta,\xi) \mid \mu,\vartheta] = \mu$, i.e., $m(\beta,\xi)$ is unbiased. □

Proposition 3.31 (Link to the ΔL rule)**.** *If $s(L;\xi) \equiv 1$ and $\beta = 1$, then $\omega_i = 1/L_i$ and $w_i \propto 1/L_i$, recovering the ΔL baseline weighting.*

Proof. If $s(L;\xi) \equiv 1$ and $\beta = 1$, then for each i,

$$\omega_i = \frac{1}{L_i^{\beta} s(L_i;\xi)} = \frac{1}{L_i \cdot 1} = \frac{1}{L_i}.$$

Hence

$$w_i = \frac{\omega_i}{\sum_{j=1}^{G} \omega_j} = \frac{L_i^{-1}}{\sum_{j=1}^{G} L_j^{-1}} \propto \frac{1}{L_i},$$

which is the standard ΔL choice of weights. □

Assumption 3.32 (Regularity and identifiability)**.** *(i)* $\{(g_i, L_i)\}_{i\geq 1}$ *are i.i.d. or strictly stationary and ergodic with* $\sup_i \mathbb{E}[g_i^4] < \infty$.

(ii) The parameter space $\Theta = \{\beta\} \times \Xi$ *is compact (for example,* $\beta \in [\beta_{\min}, \beta_{\max}]$, $|\rho| \leq \rho_{\max} < 1$, $\eta \in [0, \eta_{\max}]$, *and fixed* k*).*

(iii) The population objective $\ell_\infty(\vartheta) := \lim_{G\to\infty} G^{-1}\,\mathbb{E}[\ell_G(\vartheta)]$ *exists and has a unique maximizer* $\vartheta^\star = (\beta^\star, \xi^\star)$.

(iv) $s(L;\xi)$ *is smooth in* ξ *and bounded away from* 0 *and* $+\infty$ *on* $\operatorname{supp}(L) \times \Xi$.

Here $\ell_G(\vartheta)$ denotes the EB objective equation 50 built from the first G trajectories; we may equivalently work with the rescaled version $G^{-1}\ell_G(\vartheta)$, which only differs by a factor independent of ϑ.

Theorem 3.33 (Consistency & asymptotic normality of the EB estimator)**.** *Let* $\widehat{\vartheta}_G = \arg\max_{\vartheta\in\Theta} \ell_G(\vartheta)$ *be any (measurable) maximizer. Under Assumption 3.32,* $\widehat{\vartheta}_G \xrightarrow{p} \vartheta^\star$ *as* $G \to \infty$.

If, in addition, ℓ_∞ *is twice continuously differentiable at* $\vartheta^\star$ *with negative-definite Hessian* $H = -\nabla^2 \ell_\infty(\vartheta^\star)$*, and the gradient process satisfies a suitable central limit theorem (for example,* $\sqrt{G}\,\nabla \ell_G(\vartheta^\star) \xrightarrow{d} \mathcal{N}(0, J)$ *for some positive semi-definite matrix* J*), then*

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) \;\xrightarrow{d}\; \mathcal{N}\Big(0,\; H^{-1}\, J\, H^{-1}\Big),$$

where J *is the asymptotic covariance of* $\sqrt{G}\,\nabla \ell_G(\vartheta^\star)$.

Proof. Consistency. We can write $\ell_G(\vartheta)$ as a continuous function of empirical averages of bounded (or suitably integrable) functions of (g_i, L_i). Indeed, the terms $\sum_i \log \omega_i$, Λ_ω, and RSS_ω are all sums of such functions, and the mapping from these empirical sums to $\ell_G(\vartheta)$ is continuous in both the empirical sums and in ϑ. By the (vector) law of large numbers and the continuous mapping theorem, for each fixed ϑ, the rescaled objective $G^{-1}\ell_G(\vartheta)$ converges almost surely to a deterministic limit $\ell_\infty(\vartheta)$. Under the compactness of Θ and smoothness/boundedness conditions in Assumption 3.32, standard results in M-estimation (e.g., an argmax continuous mapping theorem) imply that this convergence is uniform on Θ. By assumption, ℓ_∞ has a unique maximizer $\vartheta^\star$. The uniform convergence combined with uniqueness yields $\widehat{\vartheta}_G \xrightarrow{p} \vartheta^\star$ (see, e.g., **?**, Theorem 5.7).

Asymptotic normality. Consider the first-order optimality condition

$$\nabla \ell_G(\widehat{\vartheta}_G) = 0.$$

Assume ℓ_G is twice continuously differentiable in a neighborhood of $\vartheta^\star$ with probability tending to one, and expand $\nabla \ell_G$ in a Taylor series around $\vartheta^\star$:

$$\nabla \ell_G(\widehat{\vartheta}_G) = \nabla \ell_G(\vartheta^\star) + \nabla^2 \ell_G(\tilde{\vartheta}_G)(\widehat{\vartheta}_G - \vartheta^\star),$$

for some intermediate point $\tilde{\vartheta}_G$ on the line segment between $\widehat{\vartheta}_G$ and $\vartheta^\star$. Setting $\nabla \ell_G(\widehat{\vartheta}_G) = 0$ and rearranging gives

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) = -\Big(\nabla^2 \ell_G(\tilde{\vartheta}_G)\Big)^{-1} \sqrt{G}\, \nabla \ell_G(\vartheta^\star).$$

By the consistency already established, $\tilde{\vartheta}_G \xrightarrow{p} \vartheta^\star$. Under the stated differentiability and regularity assumptions, $\nabla^2 \ell_G(\tilde{\vartheta}_G)$ converges in probability to $\nabla^2 \ell_\infty(\vartheta^\star) = -H$, so $\Big(\nabla^2 \ell_G(\tilde{\vartheta}_G)\Big)^{-1} \xrightarrow{p} H^{-1}$.

By assumption, $\sqrt{G}\, \nabla \ell_G(\vartheta^\star) \xrightarrow{d} \mathcal{N}(0, J)$. Applying Slutsky's theorem to the product of the converging matrices and vectors yields

$$\sqrt{G}(\widehat{\vartheta}_G - \vartheta^\star) \xrightarrow{d} -H^{-1}\mathcal{N}(0, J) = \mathcal{N}(0, H^{-1} J H^{-1}),$$

which proves the stated asymptotic normality. □

Remark 3.34 (Smooth but not globally concave). While $\ell(\beta, \xi)$ is generally not globally concave, the gradient $\nabla \ell$ and local Hessian $\nabla^2 \ell$ are explicit via Lemma 3.24, Proposition 3.25, and Lemma 3.29. In practice, damping or exponential moving averages in (β, ξ) and simple box constraints on the parameter space Θ yield stable online updates; see the separate Algorithms section for concrete update rules.

3.9 Weights under non-vanishing constant-lag covariance (compound symmetry)

We work with the scalarized token series $Y_{i,t} := \boldsymbol{u}^\top \boldsymbol{Z}_{i,t}$ as in equation 15. Assume second–order stationarity and

$$\gamma(0) = \mathrm{Var}(Y_{i,t}) = \tau^2, \qquad \gamma(h) = \mathrm{Cov}(Y_{i,t}, Y_{i,t-h}) = \tau^2 \rho \quad \text{for all } h \neq 0, \tag{64}$$

with $\rho \in (-1/(L_i - 1),\, 1)$ to ensure positive definiteness for a given length L_i. (For uniformly valid models across unbounded L, take $\rho \in [0, 1)$.)

Lemma 3.35 (Trajectory variance under compound symmetry)**.** *Let $g_i = \sum_{t=1}^{L_i} Y_{i,t}$. Under equation 64,*

$$\mathrm{Var}(g_i) \;=\; \tau^2 \left[L_i \;+\; 2\rho \sum_{h=1}^{L_i - 1} (L_i - h) \right] \;=\; \tau^2\, L_i \Big(1 + \rho(L_i - 1)\Big) \;=\; \tau^2 \Big((1-\rho) L_i + \rho L_i^2\Big). \tag{65}$$

Equivalently, with the shape factor $s(L; \rho) := \mathrm{Var}(g_i)/(\tau^2 L)$,

$$s(L; \rho) \;=\; 1 + \rho(L - 1) \;=\; (1 - \rho) + \rho L. \tag{66}$$

Proof. By Bartlett's identity (Lemma C.1),

$$\mathrm{Var}(g_i) = \sum_{h=-(L_i-1)}^{L_i-1} (L_i - |h|)\gamma(h) = L_i\gamma(0) + 2\sum_{h=1}^{L_i-1}(L_i - h)\gamma(h).$$

Insert equation 64 to obtain the first equality in equation 65; the second follows from $\sum_{h=1}^{L-1}(L-h) = L(L-1)/2$. The shape factor is equation 66.

(Alternative matrix proof.) The $L_i \times L_i$ covariance matrix of $(Y_{i,1}, \ldots, Y_{i,L_i})$ is $\tau^2\Big((1-\rho)\boldsymbol{I} + \rho\,\mathbf{1}\mathbf{1}^\top\Big)$. Then $\mathrm{Var}(g_i) = \mathbf{1}^\top \mathrm{Cov}\,\mathbf{1} = \tau^2\Big((1-\rho)L_i + \rho L_i^2\Big)$. □

Proposition 3.36 (Weighting rule under constant-lag covariance)**.** *Under the heteroscedastic model equation 16 with $v_i(\vartheta) = \sigma^2 L_i^\beta\, s(L_i;\rho)$ and $s(L;\rho)$ from equation 66, the posterior-precision (Bayes) weights satisfy, up to a common factor,*

$$x_i^{\mathrm{Bayes}} \;\propto\; \frac{1}{L_i^\beta\, s(L_i;\rho)} \;=\; \frac{1}{L_i^\beta\Big(1+\rho(L_i-1)\Big)}. \tag{67}$$

In particular, under the Jeffreys baseline with $\beta = 1$,

$$x_i \;\propto\; \frac{1}{L_i\Big(1+\rho(L_i-1)\Big)} \;=\; \frac{1}{(1-\rho)L_i + \rho L_i^2}. \tag{68}$$

Proof. From equation 16 and Lemma 3.35, $\lambda_i(\vartheta) = 1/v_i(\vartheta) = (1/\sigma^2)\, L_i^{-\beta}\, s(L_i;\rho)^{-1}$. By Theorem 3.8, $x_i^{\mathrm{Bayes}} \propto \mathbb{E}[\lambda_i(\vartheta) \mid \{g\}]$; the factor $1/\sigma^2$ cancels upon normalization, yielding equation 67. The specialization $\beta = 1$ gives equation 68. □

Corollary 3.37 (Asymptotics and effective length law)**.** *If $\rho > 0$ is fixed and $L \to \infty$, then $s(L;\rho) \sim \rho L$ and*

$$v_i(\vartheta) \;\sim\; \sigma^2 L_i^\beta\, \rho L_i \quad\Longrightarrow\quad v_i(\vartheta) = \Theta(L_i^{\beta+1}).$$

Under the Jeffreys baseline $\beta = 1$, $v_i = \Theta(L_i^2)$ and $x_i \propto L_i^{-2}$ asymptotically. Thus persistent positive correlation induces a quadratic penalty in length relative to the i.i.d. case.

Remark 3.38 (Parameter constraints). For a fixed length L, compound symmetry is positive definite for $\rho \in (-1/(L-1), 1)$. To model batches with unbounded L, take $\rho \in [0, 1)$ so that $s(L;\rho)$ remains positive and the covariance is valid for all L.

3.10 Variance-law plug-ins and CAPO variants

From covariance to a one-dimensional target. The preceding sections derive a common structural conclusion: for a scalarized trajectory-gradient signal g at length L, the effect of within-trajectory token covariance enters the statistically efficient weighting rule only through the *induced variance law*

$$v(L) \;:=\; \mathrm{Var}(g \mid L). \tag{69}$$

In the working model equation 45 (or its GRPO/grouped extension equation 54), $v(L)$ takes the parametric form

$$v(L;\vartheta) = \sigma^2 L^\beta s(L;\xi), \qquad \vartheta = (\beta,\xi), \qquad s(L;\xi) > 0. \tag{70}$$

Crucially, for the group-based update used in GRPO-style training, the CAPO reweighting requires only a *plug-in* estimate $\widehat{v}(L)$ (up to a common scale), which is converted into within-group precision weights

$$\widehat{\omega}_{p,i} \;\propto\; \widehat{v}(L_{p,i})^{-1}, \qquad \widehat{w}_{p,i} = \frac{\widehat{\omega}_{p,i}}{\sum_{j\in\mathcal{I}_p}\widehat{\omega}_{p,j}}. \tag{71}$$

The remainder of the CAPO update (baseline construction, advantage normalization, and policy optimization) is identical across plug-ins (§4).

Named plug-ins. We use the term *CAPO variant* to mean "CAPO with a specific choice of $\widehat{v}(L)$". This section defines four submission-ready plug-ins that cover the key dependence regimes analyzed in §2.4–§3.9.

3.10.1 L-CAPO: independence / length-only plug-in (includes ΔL)

Under the i.i.d. token model (§3.1), Bartlett's identity reduces to $\mathrm{Var}(g \mid L) = \tau^2 L$, i.e. $v(L) \propto L$. Equivalently, equation 70 holds with the *identity* dependence shape $s(L;\xi) \equiv 1$ and the correctly specified exponent $\beta = 1$.

Proposition 3.39 (L-CAPO as the independence plug-in)**.** *Assume i.i.d. tokens so that $\gamma(h) = 0$ for $h \neq 0$ and $\mathrm{Var}(g \mid L) = \tau^2 L$. Then the plug-in rule equation 71 with $\widehat{v}(L) = L$ yields within-group weights $\widehat{w}_{p,i} \propto L_{p,i}^{-1}$, i.e. the ΔL normalization ($\alpha = 1$) as the correctly specified independence special case. More generally,* ***L-CAPO*** *denotes the length-only family obtained by fixing $s(L;\xi) \equiv 1$ and estimating only the length exponent β (diagonal EB), so that $\widehat{v}(L) = L^{\widehat{\beta}}$ and $\widehat{w}_{p,i} \propto L_{p,i}^{-\widehat{\beta}}$.*

Proof. Under i.i.d. tokens, $v(L)$ is linear and inverse-variance weights are inverse-length (Lemma 3.6). Plugging $\widehat{v}(L) = L$ into equation 71 yields $\widehat{\omega} \propto 1/L$ and hence $\widehat{w} \propto 1/L$ after within-group normalization. The length-only generalization with $\widehat{v}(L) = L^{\widehat{\beta}}$ follows by the same plug-in mapping. □

3.10.2 LV-CAPO: empirical-Bayes length + dependence plug-in

LV-CAPO is the fully structured empirical-Bayes instantiation developed in §3.8: it fits (β,ξ) by marginal likelihood (either the single-group objective equation 50 or the prompt-group objective equation 58) under a chosen structured dependence family $s(L;\xi)$ (e.g., the k-banded stretched-geometric model of Definition 3.28). The resulting plug-in is

$$\widehat{v}_{\mathrm{LV}}(L) = L^{\widehat{\beta}}\, s(L;\widehat{\xi}), \qquad \widehat{\omega}^{\mathrm{LV}}_{p,i} \propto \widehat{v}_{\mathrm{LV}}(L_{p,i})^{-1}. \tag{72}$$

The dependence correction $s(L;\widehat{\xi})$ makes LV-CAPO sensitive to correlation-induced variance inflation even when $v(L)$ remains asymptotically linear (summable dependence) or becomes superlinear (persistent dependence).

3.10.3 CAPO-Q: quadratic induced-variance plug-in (compound symmetry / random intercept)

Compound symmetry (§3.9) provides a minimal low-dimensional surrogate for persistent within-trajectory correlation: the covariance between any two distinct token positions is approximately constant. Lemma 3.35 implies

$$\mathrm{Var}(g \mid L) \;\propto\; (1-\rho)L + \rho L^2, \tag{73}$$

and under equation 70 this becomes $v(L;\beta,\rho) \propto L^\beta((1-\rho)+\rho L) = (1-\rho)L^\beta + \rho L^{\beta+1}$. This motivates **CAPO-Q**, a quadratic plug-in that fits a two-term variance law and then applies inverse-variance weighting.

Proposition 3.40 (Outcomes-only quadratic plug-in)**.** *Fix β (e.g., $\beta = 1$ for the Jeffreys/independence baseline) and consider the regression model*

$$\mathbb{E}\Big[e_{p,i}^2 \mid L_{p,i}\Big] \;\approx\; a\,L_{p,i}^{\beta} + b\,L_{p,i}^{\beta+1}, \qquad a \ge 0,\; b \ge 0, \tag{74}$$

where $e_{p,i} = g_{p,i} - m_p$ are within-group residuals from equation 56. Let $(\widehat{a},\widehat{b})$ be any nonnegative least-squares fit of equation 74 pooled across groups. Then the quadratic plug-in

$$\widehat{v}_{\mathrm{Q}}(L) = \widehat{a}\,L^{\beta} + \widehat{b}\,L^{\beta+1} \tag{75}$$

induces precision weights $\widehat{\omega}_{p,i}^{\mathrm{Q}} \propto \widehat{v}_{\mathrm{Q}}(L_{p,i})^{-1}$ and within-group weights $\widehat{w}_{p,i}^{\mathrm{Q}}$ via equation 71. If the data are generated under compound symmetry with parameter ρ (Lemma 3.35) and correctly specified β, then (a,b) are proportional to $((1-\rho),\rho)$ and CAPO-Q coincides with the oracle inverse-variance rule (up to scale).

Proof. Lemma 3.35 gives $\mathrm{Var}(g \mid L) = \tau^2((1-\rho)L + \rho L^2)$. Under equation 70, this is $\sigma^2((1-\rho)L^\beta + \rho L^{\beta+1})$ for some scale σ^2. Thus equation 74 is correctly specified with $(a,b) \propto ((1-\rho),\rho)$. Since CAPO weights depend on $v(L)$ only through relative scale within a group, the common multiplicative factor cancels in equation 71, yielding oracle weights when $\widehat{a},\widehat{b}$ recover the shape. □

3.10.4 CAPO-HAC: lag-window (Newey–West) induced-variance plug-in

When token dependence is *short-range* (summable autocovariance), the induced variance can be estimated directly via long-run variance estimators. For a scalarized token increment process $Y_{p,i,t} = \boldsymbol{u}^\top \boldsymbol{Z}_{p,i,t}$, Bartlett's identity gives

$$v(L) = \mathrm{Var}\Big(\sum_{t=1}^{L} Y_t\Big) = L\gamma(0) + 2\sum_{h=1}^{L-1}(L-h)\gamma(h), \qquad \gamma(h) = \mathrm{Cov}(Y_t, Y_{t-h}). \tag{76}$$

CAPO-HAC replaces the parametric $s(L;\xi)$ with a lag-window estimate of the long-run variance. Let $K \ge 1$ be a bandwidth and define pooled lag-h autocovariance estimates (centered within each trajectory)

$$\widehat{\gamma}(h) = \frac{\sum_{p,i}\sum_{t=h+1}^{L_{p,i}}\Big(Y_{p,i,t} - \bar{Y}_{p,i}\Big)\Big(Y_{p,i,t-h} - \bar{Y}_{p,i}\Big)}{\sum_{p,i}(L_{p,i}-h)}, \qquad \bar{Y}_{p,i} = \frac{1}{L_{p,i}}\sum_{t=1}^{L_{p,i}} Y_{p,i,t}. \tag{77}$$

Then the Bartlett-window (Newey–West) estimator of the induced variance at length L is

$$\widehat{v}_{\mathrm{HAC}}(L) = L\widehat{\gamma}(0) + 2 \sum_{h=1}^{\min\{K,L-1\}} (L-h)\Big(1 - \frac{h}{K+1}\Big)\widehat{\gamma}(h), \tag{78}$$

and CAPO-HAC uses $\widehat{\omega}_{p,i}^{\mathrm{HAC}} \propto \widehat{v}_{\mathrm{HAC}}(L_{p,i})^{-1}$ with within-group normalization equation 71. Under standard mixing conditions and $K \to \infty$ sufficiently slowly, equation 78 is a consistent estimator of the long-run variance and therefore yields consistent inverse-variance reweighting (**??**).

Remark 3.41 (Which plug-in when?). L-CAPO is the independence/length-only surrogate (good when autocovariances are negligible or when a conservative hedge is preferred). LV-CAPO is the fully structured EB surrogate (best when the chosen $s(L;\xi)$ family matches the dependence). CAPO-Q is the minimal persistent-correlation surrogate, designed to capture superlinear $v(L)$ growth with two parameters. CAPO-HAC is a nonparametric short-range surrogate that targets $v(L)$ directly without committing to a parametric $s(L;\xi)$.

4 Algorithms: modular CAPO with variance-law plug-ins

4.1 Workflow overview (precise step-by-step; modular plug-in view)

CAPO is implemented as a GRPO-style, *grouped* training loop with a modular *variance-law plug-in* (Section 3.10). At each optimizer step, we (i) collect prompt groups, (ii) invoke a plug-in to estimate an induced variance law $\widehat{v}(L)$ (up to scale) and hence per-trajectory precisions, (iii) normalize these precisions *within each prompt group* to obtain trajectory weights, and (iv) apply the resulting reweighted group objective. LV-CAPO corresponds to the empirical-Bayes plug-in based on equation 58; L-CAPO, CAPO-Q, and CAPO-HAC are alternative plug-ins that change only Step 2/4 below.

1. **Rollout collection (prompt groups).** At optimizer step t, sample P_t prompts and, for each prompt $p \in \{1, \ldots, P_t\}$, sample N trajectories under the current policy π_{θ_t}. This yields a grouped batch

$$\mathcal{B}_t = \Big\{(p, i,\, g_{p,i},\, L_{p,i},\, \{Y_{p,i,\tau}\}_{\tau=1}^{L_{p,i}})\Big\}_{p=1, i\in\mathcal{I}_p}^{P_t},$$

   where $\mathcal{I}_p$ indexes the N rollouts in group p, $L_{p,i} \in \mathbb{N}$ is the response length, $g_{p,i} \in \mathbb{R}$ is a scalarized trajectory-gradient statistic (e.g., a fixed-direction projection of the per-trajectory policy-gradient, equation 15), and $Y_{p,i,\tau} \in \mathbb{R}$ are token-level scalar increments available to token-aware plug-ins (LV-CAPO and CAPO-HAC). Outcomes-only plug-ins (L-CAPO, CAPO-Q) depend only on $(g_{p,i}, L_{p,i})$.

2. **Variance-law plug-in: compute unnormalized precisions.** Choose a plug-in P that returns a variance-law estimate $\widehat{v}_{\mathsf{P}}(L)$ (up to scale) and set

$$\widehat{\omega}_{p,i} \;\propto\; \widehat{v}_{\mathsf{P}}(L_{p,i})^{-1}.$$

   Concretely:

- **L-CAPO** fixes $s(L) \equiv 1$ and uses $\widehat{v}(L) = L^{\widehat{\beta}}$ (with $\widehat{\beta}$ either fixed to 1 for ΔL or estimated online; Algorithm 4).
- **LV-CAPO** uses the structured family $\widehat{v}(L) = L^{\widehat{\beta}} s(L; \widehat{\xi})$ with $(\widehat{\beta}, \widehat{\xi})$ estimated by grouped EB; Algorithm 2.
- **CAPO-Q** fits a quadratic variance law $\widehat{v}(L) = \widehat{a}\, L^{\beta} + \widehat{b}\, L^{\beta+1}$ motivated by compound symmetry; Algorithm 5.
- **CAPO-HAC** estimates the long-run variance by a Bartlett/Newey–West lag window and sets $\widehat{v}(L) = \widehat{v}_{\mathrm{HAC}}(L)$; Algorithm 6.

3. **Within-group normalization and grouped summaries.** Convert unnormalized precisions into within-group weights

$$\widehat{w}_{p,i} = \frac{\widehat{\omega}_{p,i}}{\sum_{j \in \mathcal{I}_p} \widehat{\omega}_{p,j}}, \qquad m_{p,t} = \sum_{i \in \mathcal{I}_p} \widehat{w}_{p,i}\, g_{p,i}, \qquad e_{p,i} = g_{p,i} - m_{p,t}.$$

If the plug-in requires an EB objective (LV-CAPO/L-CAPO), form grouped summaries

$$\Lambda_{\omega,p} = \sum_{i \in \mathcal{I}_p} \widehat{\omega}_{p,i}, \qquad \mathrm{RSS}_{\omega}^{\mathrm{grp}} = \sum_{p=1}^{P_t} \sum_{i \in \mathcal{I}_p} \widehat{\omega}_{p,i}\, e_{p,i}^2,$$

matching equation 55–equation 57.

4. **Plug-in parameter update (optional).** Update plug-in parameters using the current batch:
   - **LV-CAPO**: update (β, ξ) by ascent on equation 58 using equation 62 and the corresponding ξ-gradient (Proposition 3.25 with groupwise means; optionally warm-start via Algorithm 3).
   - **L-CAPO**: update β only (set $s \equiv 1$) using the same grouped EB structure; ΔL is the fixed $\beta = 1$ special case.
   - **CAPO-Q**: update (a, b) by nonnegative least squares on residual squares $e_{p,i}^2$ versus $L_{p,i}^{\beta}$ and $L_{p,i}^{\beta+1}$ (Proposition 3.40).
   - **CAPO-HAC**: update $\widehat{\gamma}(h)$ and $\widehat{v}_{\mathrm{HAC}}(L)$ using lag-window pooling over token increments (Section 3.10.4).
5. **Policy update.** Use the within-group weights $\{\widehat{w}_{p,i}\}$ to form a weighted GRPO objective (baseline $m_{p,t}$, residual advantages $e_{p,i}$) and take an optimizer step on θ.
6. **Stabilization and diagnostics.** Apply EMA to plug-in parameters, enforce box constraints (e.g., $|\rho| \le \rho_{\max} < 1$ and $\eta \ge 0$ for k-band families), and clip extreme weights. Monitor weight entropy/effective sample size and length-decile stratification.

4.2 Algorithm 1: Modular CAPO training loop (GRPO prompt groups)

Algorithm 1 makes the modularity explicit: a plug-in produces per-trajectory precisions (equivalently, induced-variance estimates), and the outer loop performs the standard GRPO update using within-group normalized weights.

Algorithm 1: CAPO (modular): GRPO update with a variance-law plug-in

Input : Plug-in $\mathsf{P} \in \{\text{L-CAPO}, \text{LV-CAPO}, \text{CAPO-Q}, \text{CAPO-HAC}\}$ with state ψ; group size N; optimizer and policy π_θ.

Output : Updated policy parameters θ and updated plug-in state ψ (if applicable).

for $t = 1, 2, \dots$ **do**

  Sample P_t prompts; for each prompt p sample N trajectories to form $\mathcal{B}_t = \{(p, i, g_{p,i}, L_{p,i}, \{Y_{p,i,\tau}\})\}$.

  $(\widehat{\omega}_{p,i})_{(p,i)\in\mathcal{B}_t},\ \psi \leftarrow \mathsf{P}.\textsc{FitAndPredict}(\mathcal{B}_t, \psi)$ `// plug-in-specific; returns precisions up to scale`

  foreach *group* p **do**

    $\widehat{w}_{p,i} \leftarrow \widehat{\omega}_{p,i} / \sum_{j\in\mathcal{I}_p} \widehat{\omega}_{p,j}$;

    $m_p \leftarrow \sum_{i\in\mathcal{I}_p} \widehat{w}_{p,i} g_{p,i}$;

    $e_{p,i} \leftarrow g_{p,i} - m_p$;

  Form the weighted GRPO loss using $(e_{p,i}, \widehat{w}_{p,i})$ and take an optimizer step on θ.

  Apply EMA/projection/clipping to plug-in parameters and weights as needed.

4.3 Algorithm 2: LV–CAPO in the CAPO training loop

LV-CAPO is the empirical-Bayes variance-law plug-in that jointly fits a length exponent β and a structured dependence shape ξ under the working model of Section 3.8 and its prompt-group extension (Section 3.8.1). In the modular CAPO loop (Algorithm 1), LV-CAPO is implemented as a FitAndPredict routine that:

1. evaluates the current precision weights $\omega_{p,i}(\beta_{t-1}, \xi_{t-1}) = [L_{p,i}^{\beta_{t-1}} s(L_{p,i}; \xi_{t-1})]^{-1}$ on the grouped batch $\mathcal{B}_t$;
2. computes within-group weighted means m_p and grouped summaries $(\Lambda_{\omega,p}, \mathrm{RSS}_\omega^{\mathrm{grp}})$ as in equation 55–equation 57;
3. updates β (and optionally ξ) by ascent on the grouped EB objective equation 58, using the grouped length-gradient equation 62 and either an ACF-moment update (Algorithm 3) or a direct EB gradient update for ξ;
4. returns updated precisions $\omega_{p,i}(\beta_t, \xi_t)$ to be normalized *within each group* by the outer CAPO loop.

4.4 Algorithm 3: ACF-moment estimator for dependence

Algorithm 3 provides a moment-based plug-in estimator for the dependence shape $\xi = (\rho, \eta, \dots)$ in the k-banded stretched-geometric family $s(L; \rho, k, \eta)$ (Definition 3.28). It operates at the *token* level $Y_{i,\tau}$ and:

1. computes per-trajectory empirical autocovariances $\hat{\gamma}_i(h)$ up to lag k, centered by the within-trajectory mean $\bar{Y}_i$;

Algorithm 2: LV–CAPO plug-in: online grouped-EB estimation and precision prediction

Input : Grouped batch $\mathcal{B}_t = \{(p, i, g_{p,i}, L_{p,i}, \{Y_{p,i,\tau}\})\}$; previous EB parameters (β_{t-1}, ξ_{t-1}); learning rates η_β, η_ξ; EMA coefficients γ_β, γ_ξ; constraints/projection proj_Ξ for ξ (e.g., $|\rho| \le \rho_{\max} < 1$, $\eta \ge 0$, fixed k).

Output : Unnormalized precisions $(\omega_{p,i}^{(t)})_{(p,i)\in\mathcal{B}_t}$ and updated parameters (β_t, ξ_t).

`// Evaluate current precisions and within-group normalizers`

foreach *group* p **do**
  foreach $i \in \mathcal{I}_p$ **do**
    $\omega_{p,i} \leftarrow \left(L_{p,i}^{\beta_{t-1}} s(L_{p,i}; \xi_{t-1})\right)^{-1}$
  $\Lambda_{\omega,p} \leftarrow \sum_{j\in\mathcal{I}_p} \omega_{p,j}$
  foreach $i \in \mathcal{I}_p$ **do**
    $w_{p,i} \leftarrow \omega_{p,i}/\Lambda_{\omega,p}$
  $m_p \leftarrow \sum_{i\in\mathcal{I}_p} w_{p,i} g_{p,i}$
  foreach $i \in \mathcal{I}_p$ **do**
    $e_{p,i} \leftarrow g_{p,i} - m_p$

$\mathrm{RSS}_\omega^{\mathrm{grp}} \leftarrow \sum_p \sum_{i\in\mathcal{I}_p} \omega_{p,i} e_{p,i}^2$

$G \leftarrow \sum_p |\mathcal{I}_p|$

$P \leftarrow \left|\{p : \mathcal{I}_p \neq \emptyset\}\right|$

`// Update` β `by grouped EB ascent (Corollary 3.27)`

$g_\beta \leftarrow \frac{1}{2}\left[-\sum_p \sum_{i\in\mathcal{I}_p} \log L_{p,i} + \sum_p \frac{\sum_{i\in\mathcal{I}_p} \omega_{p,i} \log L_{p,i}}{\Lambda_{\omega,p}}\right] + \frac{G-P}{2} \frac{1}{\mathrm{RSS}_\omega^{\mathrm{grp}}} \sum_p \sum_{i\in\mathcal{I}_p} \omega_{p,i} e_{p,i}^2 \log L_{p,i} + \partial_\beta \log \pi_\beta(\beta_{t-1})$

$\tilde{\beta}_t \leftarrow \beta_{t-1} + \eta_\beta g_\beta$

$\beta_t \leftarrow (1-\gamma_\beta)\beta_{t-1} + \gamma_\beta \,\mathrm{clip}(\tilde{\beta}_t; [\beta_{\min}, \beta_{\max}])$

`// Update` ξ `(token-aware) either by ACF-moment or EB gradient`

if *ACF-moment update* **then**
  $\tilde{\xi}_t \leftarrow \mathrm{ACF_Moment_Fit}(\{Y_{p,i,\tau}\}, k)$ `// Algorithm 3`
else
  Compute $g_\xi \leftarrow \nabla_\xi \ell_{\mathrm{grp}}(\beta_{t-1}, \xi_{t-1})$ (grouped analogue of Proposition 3.25)
  $\tilde{\xi}_t \leftarrow \xi_{t-1} + \eta_\xi g_\xi$

$\xi_t \leftarrow (1-\gamma_\xi)\xi_{t-1} + \gamma_\xi \,\mathrm{proj}_\Xi(\tilde{\xi}_t)$

`// Return updated precisions for the outer CAPO loop`

foreach *group* p **do**
  foreach $i \in \mathcal{I}_p$ **do**
    $\omega_{p,i}^{(t)} \leftarrow \left(L_{p,i}^{\beta_t} s(L_{p,i}; \xi_t)\right)^{-1}$

return $(\omega_{p,i}^{(t)}), (\beta_t, \xi_t)$

2. forms a pooled autocovariance $\bar{\gamma}(h)$ by weighting each trajectory i by the number of contributing pairs $L_i - h$;
3. converts to a pooled autocorrelation $\bar{r}(h) = \bar{\gamma}(h)/\bar{\gamma}(0)$;
4. fits (ρ, η) by least squares against the theoretical correlation ρ^{h^η} implied by the stretched-geometric model.

The resulting $\tilde{\xi} = (\tilde{\rho}, \tilde{\eta}, \dots)$ can be used directly in the variance law $v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i; \xi)$, or as a warm start for subsequent EB gradient updates in Algorithm 2 or Algorithm 8.

Algorithm 3: ACF-moment estimator for dependence shape (used inside Algorithm 2)

Input : Token-level increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ for each trajectory i; maximum lag k.
Output : Shape estimate $\tilde{\xi} = (\tilde{\rho}, \tilde{\eta}, \dots)$ for the family $s(L; \rho, k, \eta)$ in Definition 3.28.

foreach *trajectory* i **do**
  $\bar{Y}_i \leftarrow \frac{1}{L_i} \sum_{\tau=1}^{L_i} Y_{i,\tau}$;
  for $h = 0, 1, \dots, k$ **do**
    $\hat{\gamma}_i(h) \leftarrow \frac{1}{L_i - h} \sum_{\tau=1}^{L_i - h} (Y_{i,\tau} - \bar{Y}_i)(Y_{i,\tau+h} - \bar{Y}_i)$.
$\bar{\gamma}(h) \leftarrow \frac{\sum_i (L_i - h)\hat{\gamma}_i(h)}{\sum_i (L_i - h)}$ for $h = 0, \dots, k$; $\bar{r}(h) \leftarrow \bar{\gamma}(h)/\bar{\gamma}(0)$ for $h \geq 1$.
$\tilde{\xi} \leftarrow \arg\min_{\rho,\eta} \sum_{h=1}^{k} \left(\bar{r}(h) - \rho^{h^\eta}\right)^2$
`// Optionally downweight large` h `to reduce noise`

4.5 Algorithm 4: L-CAPO length-only plug-in (includes ΔL)

L-CAPO is the *length-only* variance-law plug-in: it fixes the dependence shape to the identity $s(L) \equiv 1$ and models $v(L) \propto L^\beta$. In the modular CAPO loop (Algorithm 1), L-CAPO returns unnormalized precisions $\omega_{p,i} \propto L_{p,i}^{-\beta}$ to be normalized within each prompt group.

- **ΔL as a special case.** Under token independence, $\beta = 1$ is correctly specified (Proposition 3.39), yielding $\omega_{p,i} \propto 1/L_{p,i}$.
- **Learned-β L-CAPO.** In practice, we can update β online via a lightweight moment regression on squared residuals or via a one-dimensional grouped-EB step with $s \equiv 1$.

4.6 Algorithm 5: CAPO-Q quadratic variance-law plug-in

CAPO-Q is an outcomes-only plug-in motivated by compound symmetry / random-intercept covariance (Section 3.10.3). It fits a two-term variance law

$$\widehat{v}_{\mathrm{Q}}(L) = \widehat{a}\, L^\beta + \widehat{b}\, L^{\beta+1},$$

and returns precisions $\omega_{p,i} \propto \widehat{v}_{\mathrm{Q}}(L_{p,i})^{-1}$ (to be normalized within-group by Algorithm 1). When the compound-symmetry surrogate is correct, CAPO-Q coincides with the oracle inverse-variance rule (Proposition 3.40).

Algorithm 4: L-CAPO plug-in: length-only precisions (optionally fit β)

Input : Grouped batch $\mathcal{B}_t = \{(p, i, g_{p,i}, L_{p,i})\}$; either fixed β (e.g., $\beta = 1$ for ΔL) or previous estimate β_{t-1}; small $\varepsilon > 0$.

Output : Precisions $(\omega_{p,i})$ and (optionally updated) β_t.

if β *is fixed* **then**
  foreach *group* p **do**
    foreach $i \in \mathcal{I}_p$ **do**
      $\omega_{p,i} \leftarrow L_{p,i}^{-\beta}$
  return $(\omega_{p,i}), \beta$

`// Compute crude within-group residuals (uniform baseline) and regress log-squared residuals on log-length`

foreach *group* p **do**
  $m_p \leftarrow \frac{1}{|\mathcal{I}_p|} \sum_{i \in \mathcal{I}_p} g_{p,i}$
  foreach $i \in \mathcal{I}_p$ **do**
    $e_{p,i} \leftarrow g_{p,i} - m_p; \quad z_{p,i} \leftarrow \log(e_{p,i}^2 + \varepsilon); \quad \ell_{p,i} \leftarrow \log L_{p,i}$

Fit $z_{p,i} = c - \beta\, \ell_{p,i} + \epsilon_{p,i}$ by least squares pooled over (p, i); set $\beta_t \leftarrow -$ (estimated slope)

foreach *group* p **do**
  foreach $i \in \mathcal{I}_p$ **do**
    $\omega_{p,i} \leftarrow L_{p,i}^{-\beta_t}$

return $(\omega_{p,i}), \beta_t$

Algorithm 5: CAPO-Q plug-in: fit a quadratic induced-variance law and return precisions

Input : Grouped batch $\mathcal{B}_t = \{(p, i, g_{p,i}, L_{p,i})\}$; fixed β (typically $\beta = 1$) or an external estimate; nonnegativity constraint $a, b \geq 0$.

Output : Precisions $(\omega_{p,i})$ and fitted coefficients $(\widehat{a}, \widehat{b})$.

`// Compute within-group residuals`

foreach *group* p **do**
  $m_p \leftarrow \frac{1}{|\mathcal{I}_p|} \sum_{i \in \mathcal{I}_p} g_{p,i}$
  foreach $i \in \mathcal{I}_p$ **do**
    $e_{p,i} \leftarrow g_{p,i} - m_p; \quad y_{p,i} \leftarrow e_{p,i}^2$

`// Nonnegative least squares fit:` $y_{p,i} \approx a L_{p,i}^{\beta} + b L_{p,i}^{\beta+1}$

$(\widehat{a}, \widehat{b}) \leftarrow \arg\min_{a,b \geq 0} \sum_{p,i} \left(y_{p,i} - a L_{p,i}^{\beta} - b L_{p,i}^{\beta+1} \right)^2$

foreach *group* p **do**
  foreach $i \in \mathcal{I}_p$ **do**
    $\widehat{v} \leftarrow \widehat{a}\, L_{p,i}^{\beta} + \widehat{b}\, L_{p,i}^{\beta+1}; \quad \omega_{p,i} \leftarrow 1/\widehat{v}$

return $(\omega_{p,i}), (\widehat{a}, \widehat{b})$

4.7 Algorithm 6: CAPO-HAC lag-window long-run variance plug-in

CAPO-HAC is a token-aware plug-in that estimates the induced variance $v(L) = \mathrm{Var}(\sum_{t=1}^{L} Y_t)$ directly via a Bartlett/Newey–West lag window (Section 3.10.4). It pools autocovariance estimates across trajectories to reduce noise and returns $\omega_{p,i} \propto \widehat{v}_{\mathrm{HAC}}(L_{p,i})^{-1}$.

Algorithm 6: CAPO-HAC plug-in: Newey–West induced-variance estimation and precisions

Input : Grouped batch $\mathcal{B}_t = \{(p, i, L_{p,i}, \{Y_{p,i,\tau}\})\}$; bandwidth $K \geq 1$.
Output : Precisions $(\omega_{p,i})$ and pooled autocovariances $(\widehat{\gamma}(h))_{h=0}^{K}$.

```
// Estimate pooled autocovariances up to lag K
for h = 0, 1, ..., K do
    N_h ← 0;   S_h ← 0
foreach trajectory (p, i) do
    Ȳ_{p,i} ← (1/L_{p,i}) Σ_{τ=1}^{L_{p,i}} Y_{p,i,τ}
    for h = 0, 1, ..., min{K, L_{p,i} − 1} do
        for τ = h + 1, ..., L_{p,i} do
            S_h ← S_h + (Y_{p,i,τ} − Ȳ_{p,i})(Y_{p,i,τ−h} − Ȳ_{p,i})
            N_h ← N_h + 1
for h = 0, 1, ..., K do
    γ̂(h) ← S_h / N_h
// Map γ̂(·) to an induced-variance estimate at each length and return
   precisions
foreach group p do
    foreach i ∈ I_p do
        L ← L_{p,i};   v̂ ← L γ̂(0)
        for h = 1, ..., min{K, L − 1} do
            v̂ ← v̂ + 2(L − h)(1 − h/(K+1)) γ̂(h)
        ω_{p,i} ← 1/v̂
return (ω_{p,i}), (γ̂(h))_{h=0}^{K}
```

4.8 Algorithm 7: k-banded covariance weights

Algorithm 7 specializes the general EB weighting rule to the k-banded stretched-geometric dependence model of Definition 3.28. Given a batch $\mathcal{B}_t$, a current length exponent β_t, and dependence parameters (ρ_t, η_t), it:

1. optionally refreshes (ρ_t, η_t) using the ACF-moment estimator (Algorithm 3);
2. computes, for each trajectory i, the shape factor

$$s_i = s(L_i; \rho_t, k, \eta_t) = 1 + \frac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h) \rho_t^{h^{\eta_t}}, \qquad m_i = \min\{k, L_i - 1\};$$

3. forms the precision $\omega_i^{(t)} = \left(L_i^{\beta_t} s_i\right)^{-1}$ and normalizes to obtain $w_i^{(t)}$;

4. returns the dependence-corrected weights $\{w_i^{(t)}\}$ and the shape factors $\{s_i\}$.

This gives an explicit implementation of the weighting rule in Proposition 3.14.

Algorithm 7: k-banded covariance weights (stretched-geometric dependence)

Input : Batch $\mathcal{B}_t = \{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\}_{i=1}^{G_t}$;
current length exponent β_t;
current k-banded dependence parameters (ρ_t, η_t);
maximum lag k for the k-band family in Definition 3.28.

Output : Shape factors $s_i = s(L_i; \rho_t, k, \eta_t)$ and trajectory weights $w_i^{(t)} \propto [L_i^{\beta_t}\, s(L_i; \rho_t, k, \eta_t)]^{-1}$.

```
// Optional: refresh (ρ_t, η_t) from token-level ACFs
if update dependence parameters then
    ξ̃_t = (ρ̃_t, η̃_t) ← ACF_Moment_Fit({Y_{i,τ}}_{i∈B_t,τ}, k)        // Alg. 3
    (ρ_t, η_t) ← proj_Ξ(ρ̃_t, η̃_t)
// Compute k-banded shape factors s_i and precisions ω_i
foreach i ∈ B_t do
    m_i ← min{k, L_i − 1}
    s_i ← 1 + (2/L_i) Σ_{h=1}^{m_i} (L_i − h) ρ_t^{h^{η_t}}        // Definition 3.28
    ω_i^(t) ← (L_i^{β_t} s_i)^{−1}
// Normalize to obtain Bayes/precision weights
Z_t ← Σ_{j∈B_t} ω_j^(t)
foreach i ∈ B_t do
    w_i^(t) ← ω_i^(t) / Z_t
// Return k-banded dependence-corrected weights
return {s_i}_{i∈B_t}, {w_i^(t)}_{i∈B_t}
```

4.9 Algorithm 8: Joint EB update for $\beta, \rho,$ & η

Algorithm 8 combines the EB updates for the length exponent β and the k-banded dependence parameters (ρ, η) into a single joint gradient-ascent step on the EB objective $\ell(\beta, \rho, \eta)$, under the stretched-geometric dependence model $s(L; \rho, k, \eta)$. For a given batch $\mathcal{B}_t$, it:

1. computes the E-step statistics $\omega_i, w_i, m_t, e_i, \Lambda_\omega, \mathrm{RSS}_\omega$ under current hyperparameters $(\beta_{t-1}, \rho_{t-1}, \eta_{t-1})$;

2. uses Corollary 3.26 to obtain $g_\beta = \partial_\beta \ell$ and Lemma 3.29 together with Proposition 3.25 to obtain $g_\rho = \partial_\rho \ell$ and $g_\eta = \partial_\eta \ell$;

3. performs projected gradient-ascent steps on β, ρ, η, followed by EMA smoothing with γ_β, γ_ξ;

4. recomputes dependence-corrected weights $w_i^{(t)} \propto [L_i^{\beta_t} s(L_i; \rho_t, k, \eta_t)]^{-1}$ for use in the CAPO update.

This algorithm is the most direct implementation of the EB theory in §3.8 when the practitioner is comfortable optimizing the full parameter vector (β, ρ, η).

Complexity and implementation notes. Per outer step t, computing the EB statistics $\{\omega_i^{(t)}, w_i^{(t)}, m_t, e_i^{(t)}, \Lambda_\omega^{(t)}, \mathrm{RSS}_\omega^{(t)}\}$ has cost $O(G_t)$ for a batch of G_t trajectories. The ACF-moment estimator with maximum lag k has cost $O\Big(\sum_{i \in \mathcal{B}_t} L_i\, k\Big)$, which is typically dominated by the rollout generation cost. In practice, small EMA coefficients $\gamma_\beta, \gamma_\xi \in [0.05, 0.3]$, a bound $|\rho| \leq 0.9$, and clipping of the largest $w_i^{(t)}$ (e.g., at the 99.5th percentile) are sufficient to prevent any single trajectory from dominating the update.

Numerical stability, identifiability, and practical defaults. In RLVR regimes with long-context CoT, the raw length factors $L_i^{-\beta}$ can underflow for large L_i and moderate β. A numerically stable implementation should therefore compute weights in log-space: $\log \omega_i = -\beta \log L_i - \log s(L_i; \xi)$ and then use a log-sum-exp reduction to obtain $\Lambda_\omega = \sum_i \omega_i$. Similarly, when the dependence model includes exponential terms (e.g., $e^{-L/\xi}$), it is preferable to evaluate $\log s(L; \xi)$ directly.

From an identifiability perspective, (β, ξ) are only weakly identifiable in small batches when the empirical length distribution is narrow. In such cases, we recommend (i) restricting β to a conservative interval (e.g., $[0, 1]$), (ii) using the outcomes-only EB objective as a warm-start for the full LV-CAPO updates, and (iii) employing mild regularization or informative priors as in §3.4. These steps stabilize estimation without altering the unbiasedness constraint in equation 6.

5 Experiments

We evaluate CAPO on a controlled RLVR setting designed to isolate the effect of advantage aggregation. All results are produced with a single open-weight model (Qwen2.5–1.5B–Instruct) and a single task distribution (CountDown).

5.1 Setup

Task. CountDown presents a target integer and a small multiset of integers; the model must return an arithmetic expression that evaluates to the target. We use the standard exact-match verifier, yielding a binary outcome reward.

Model. We finetune Qwen2.5–1.5B–Instruct with group-based on-policy RL. For each prompt, we sample $N = 8$ trajectories per update, matching the aggregation dimension assumed by GRPO-style estimators.

Algorithm 8: Joint EB update for length β and k-banded dependence (ρ, η)

Input : Batch $\mathcal{B}_t = \{(g_i, L_i, \{Y_{i,\tau}\}_{\tau=1}^{L_i})\}_{i=1}^{G_t}$;
current parameters $\beta_{t-1}, \rho_{t-1}, \eta_{t-1}$;
learning rates $\eta_\beta, \eta_\rho, \eta_\eta > 0$;
EMA $\gamma_\beta, \gamma_\xi \in (0, 1]$;
constraints $\beta \in [\beta_{\min}, \beta_{\max}],\ |\rho| \le \rho_{\max} < 1,\ \eta \ge 0$;
max lag k for $s(L; \rho, k, \eta)$ (Def. 3.28);
priors $\pi_\beta, \pi_\rho, \pi_\eta$.
Output : Updated β_t, ρ_t, η_t and weights $w_i^{(t)} \propto [L_i^{\beta_t} s(L_i; \rho_t, k, \eta_t)]^{-1}$.

`// Optional warm-start for` (ρ, η) `from ACF moments (Alg. 3)`

if *warm-start* **then**
  $(\rho_{t-1}, \eta_{t-1}) \leftarrow \mathrm{proj}_{\Xi}\big(\mathrm{ACF_Moment_Fit}(\{Y_{i,\tau}\}, k)\big)$

`// E-step: EB statistics under current` $(\beta_{t-1}, \rho_{t-1}, \eta_{t-1})$

foreach $i \in \mathcal{B}_t$ **do**
  $m_i \leftarrow \min\{k,\ L_i - 1\}$;
  $s_i \leftarrow 1 + \dfrac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho_{t-1}^{h^{\eta_{t-1}}}$;
  $\omega_i \leftarrow \left(L_i^{\beta_{t-1}}\, s_i\right)^{-1}$
$\Lambda_\omega \leftarrow \sum_{i \in \mathcal{B}_t} \omega_i$;
foreach $i \in \mathcal{B}_t$ **do**
  $w_i \leftarrow \omega_i / \Lambda_\omega$
$m_t \leftarrow \sum_{i \in \mathcal{B}_t} w_i g_i$;
foreach $i \in \mathcal{B}_t$ **do**
  $e_i \leftarrow g_i - m_t$
$\mathrm{RSS}_\omega \leftarrow \sum_{i \in \mathcal{B}_t} \omega_i e_i^2$

`// M-step: gradients of EB objective` $\ell(\beta, \rho, \eta)$

Compute $g_\beta = \partial_\beta \ell$ from Cor. 3.26 (using current $\{\omega_i, e_i, L_i\}$)
Compute $g_\rho = \partial_\rho \ell$ and $g_\eta = \partial_\eta \ell$ from Prop. 3.25 and Lem. 3.29 (using current $\{\omega_i, e_i, s_i\}$)

`// Gradient-ascent updates with projection and EMA`

$\tilde{\beta}_t \leftarrow \beta_{t-1} + \eta_\beta\, g_\beta$;$\tilde{\rho}_t \leftarrow \rho_{t-1} + \eta_\rho\, g_\rho$;$\tilde{\eta}_t \leftarrow \eta_{t-1} + \eta_\eta\, g_\eta$;
$\beta_t \leftarrow (1 - \gamma_\beta)\beta_{t-1} + \gamma_\beta\, \mathrm{clip}(\tilde{\beta}_t; [\beta_{\min}, \beta_{\max}])$;
$\rho_t \leftarrow (1 - \gamma_\xi)\rho_{t-1} + \gamma_\xi\, \mathrm{clip}(\tilde{\rho}_t; [-\rho_{\max}, \rho_{\max}])$;
$\eta_t \leftarrow (1 - \gamma_\xi)\eta_{t-1} + \gamma_\xi\, \max\{\tilde{\eta}_t,\ 0\}$;

`// Recompute` k`-banded weights at` $(\beta_t, \rho_t, \eta_t)$ `for CAPO`

foreach $i \in \mathcal{B}_t$ **do**
  $m_i \leftarrow \min\{k,\ L_i - 1\}$;
  $s_i \leftarrow 1 + \dfrac{2}{L_i} \sum_{h=1}^{m_i} (L_i - h)\, \rho_t^{h^{\eta_t}}$;
  $\omega_i \leftarrow \left(L_i^{\beta_t}\, s_i\right)^{-1}$
$\Lambda_\omega \leftarrow \sum_{j \in \mathcal{B}_t} \omega_j$;
foreach $i \in \mathcal{B}_t$ **do**
  $w_i^{(t)} \leftarrow \omega_i / \Lambda_\omega$
return $\beta_t, \rho_t, \eta_t, \{w_i^{(t)}\}_{i \in \mathcal{B}_t}$

Hyperparameter	Value
Base policy model	Qwen2.5–1.5B–Instruct
RL stack	VERL
Task / reward	CountDown verifier (binary success)
Trajectories per prompt (N)	8
Max response tokens	$\{512, 1024, 2048\}$
Optimizer	—
Learning rate	—
PPO/GRPO cliprange	—
Rollout temperature	—
Prompts per update (M)	—
Evaluation frequency	—
Random seeds	—

Table 2 Key optimization and rollout hyperparameters used in the CountDown RLVR experiments. We hold these fixed across methods within a given context budget; only the advantage aggregation / normalization rule varies.

Context budgets. We report two response-length regimes: (i) a *short* regime with maximum response length 2048 tokens and (ii) a *long* regime with maximum response length 8192 tokens. Unless otherwise noted, all other hyperparameters are kept identical across regimes.

Implementation. We implement all methods within the same training stack (VERL), holding fixed the rollout procedure, optimizer, and training schedule. This ensures that differences can be attributed to the advantage aggregation / normalization rule rather than to unrelated systems choices.

Group-based objective and estimator. All baselines and CAPO variants use the same group-based on-policy objective. Concretely, for each prompt we sample N trajectories, compute token-level policy-gradient terms using the same clipping and advantage conventions, and differ only in how the per-trajectory advantages (and hence token-level losses) are reweighted as a function of trajectory length and dependence. This isolates the effect of the aggregation rule discussed in §2–§3.

Hyperparameters and reproducibility. We use identical optimizer, rollout, and training-schedule hyperparameters across all methods within a given context budget. For transparency and reproducibility, we report the key optimization hyperparameters in Table 2. Unless otherwise stated, all reported curves are averaged over multiple random seeds; the stability metrics in Table 4 are computed from the per-seed validation trajectories prior to averaging.

5.2 Baselines and method variants

We compare CAPO against a set of aggregation baselines that progressively add or remove normalizations:

- **DAPO normalization** (token-mean): GRPO with token-averaged policy loss.
- **GRPO normalization** (seq-mean-token-mean): GRPO with per-sequence normalization of the loss, which empirically reduces length bias but can increase variance.
- **Dr. GRPO normalization:** GRPO with distributional-reweighting correction (Dr. GRPO) combined with the GRPO normalization.
- ΔL **normalization** ($\alpha \in \{0.75, 1.0\}$): length-weighted normalization via GRPO++-style trajectory weighting.

CAPO variants (variance-law plug-ins). CAPO is a modular GRPO-style framework: each variant differs only by the estimator of the induced variance law $v(L) = \mathrm{Var}(g \mid L)$ (Section 3.10) used to produce within-group precision weights (Section 4). We implement four plug-ins:

- **L-CAPO**: length-only/independence plug-in ($s \equiv 1$), including ΔL as the fixed-β=1 special case (Proposition 3.39).
- **LV-CAPO**: structured empirical-Bayes plug-in that jointly fits a length exponent and a dependence shape (Algorithm 2).
- **CAPO-Q**: quadratic variance-law plug-in motivated by compound symmetry / random intercepts (Algorithm 5).
- **CAPO-HAC**: lag-window/Newey–West long-run variance plug-in that targets $v(L)$ directly under short-range dependence (Algorithm 6).

In the main CountDown runs (Tables 3–4), we focus on L-CAPO and LV-CAPO, which bracket the independence and structured-dependence extremes. CAPO-Q and CAPO-HAC are evaluated in our controlled synthetic covariance benchmark (Appendix D.1.2), where the dependence regime is known by construction and the plug-in behavior is directly interpretable.

5.3 Metrics

Accuracy. For each validation prompt, we sample $N = 8$ trajectories and report (i) *Avg@8*, the average verifier success over trajectories, and (ii) *Best@8*, the success rate of the best trajectory within each group (equivalently, pass@8 under a binary verifier).

Stability and efficiency. We summarize learning-curve stability and sample efficiency with:

- **Monotonicity:** the Pearson correlation between evaluation step and validation Avg@8.
- **Negative drift rate (NDR):** the fraction of evaluation intervals with a decrease in validation Avg@8.
- **Tokens-to-target (TTT):** the cumulative number of training tokens processed until validation Avg@8 first exceeds a fixed target threshold.

Method	CountDown (1.5B, 2048)	CountDown (1.5B, 8192)
GRPO Norm	—	—
DAPO Norm	—	—
Dr. GRPO Norm	—	—
ΔL ($\alpha = 1$)	—	—
ΔL ($\alpha = 0.75$)	—	—
LV–CAPO (best)	—	—

Table 3 Main accuracy on CountDown with Qwen2.5–1.5B–Instruct. We report Avg@8 under a short (2048) and long (8192) response budget.

Method	Monotonicity ↑	NDR ↓	TTT ↓
ΔL (best)	—	—	—
LV–CAPO (best)	—	—	—

Table 4 Stability and efficiency diagnostics computed from validation curves. Higher monotonicity is better; lower NDR and lower TTT are better.

5.4 Main results

Table 3 reports end performance under both context budgets. Figure 3 shows representative learning dynamics.

5.5 Stability and efficiency

Table 4 aggregates stability and efficiency diagnostics. Figure 4 visualizes the stability trade-off across methods.

5.6 Length equity

To make length effects explicit, we stratify validation by response-length deciles and report verifier success within each stratum.

5.7 Additional diagnostics and failure modes

Weight degeneracy. A central practical concern in any variance-based weighting scheme is *degeneracy* of the normalized weights w_i: if one trajectory receives nearly all mass, the effective number of trajectories contributing to the update collapses. To monitor this, we recommend reporting the effective sample size $\mathrm{ESS} := (\sum_i w_i)^2/(\sum_i w_i^2)$ and/or the entropy $-\sum_i w_i \log w_i$ of the normalized weights. CAPO includes explicit clipping and ε-regularization steps (Algorithms 4–2) to prevent pathological collapse.

Length-conditioned bias. Length-based normalization can introduce systematic differences in how short and long trajectories are reinforced. The length-decile analysis in Figure 5 is intended to expose such effects. For publication-quality reporting, we recommend additionally stratifying by reward outcome (success/failure) and by prompt difficulty to distinguish genuine credit assignment from spurious length preferences.

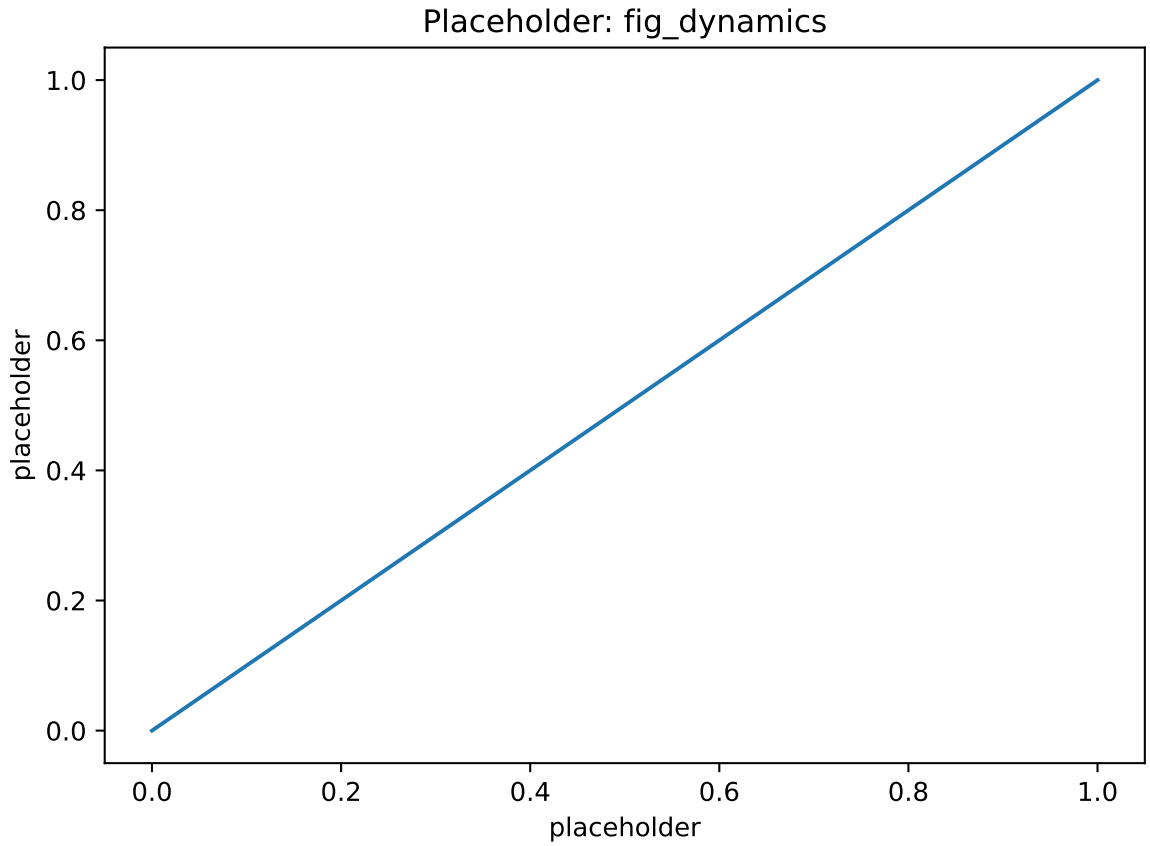

Figure 3 Training dynamics (validation Avg@8 vs. global step) for a representative subset of methods.

Model misspecification. The variance law $v(L) \propto L^{\beta} s(L; \xi)$ is a working model. When this model is misspecified, CAPO remains unbiased (because equation 6 enforces unbiasedness), but suboptimal weighting can increase variance and exacerbate instability. The robustness results in §3.9 and the covariance examples in Appendix D provide guidance for diagnosing when simple dependence surrogates (e.g., exponential or compound-symmetry) are adequate.

5.8 Synthetic covariance benchmark

The online CountDown experiments exercise CAPO in a full RLVR pipeline, but they do not isolate the specific statistical question that CAPO targets: *how a batch of heterogeneous-length trajectories should be reweighted when within-trajectory dependence induces structured heteroscedasticity.* To answer this question under maximum experimental control, we include a base-model-free simulator that generates trajectory lengths L_i and scalar returns $g_i \sim \mathcal{N}(0, v(L_i))$ under configurable covariance structures *without querying a base model* (Appendix D.1.2).

Setup. We simulate $G = P \times N$ rollouts organised into P prompt groups of N trajectories, matching the on-policy workflow. All methods operate *within each prompt group*, differing

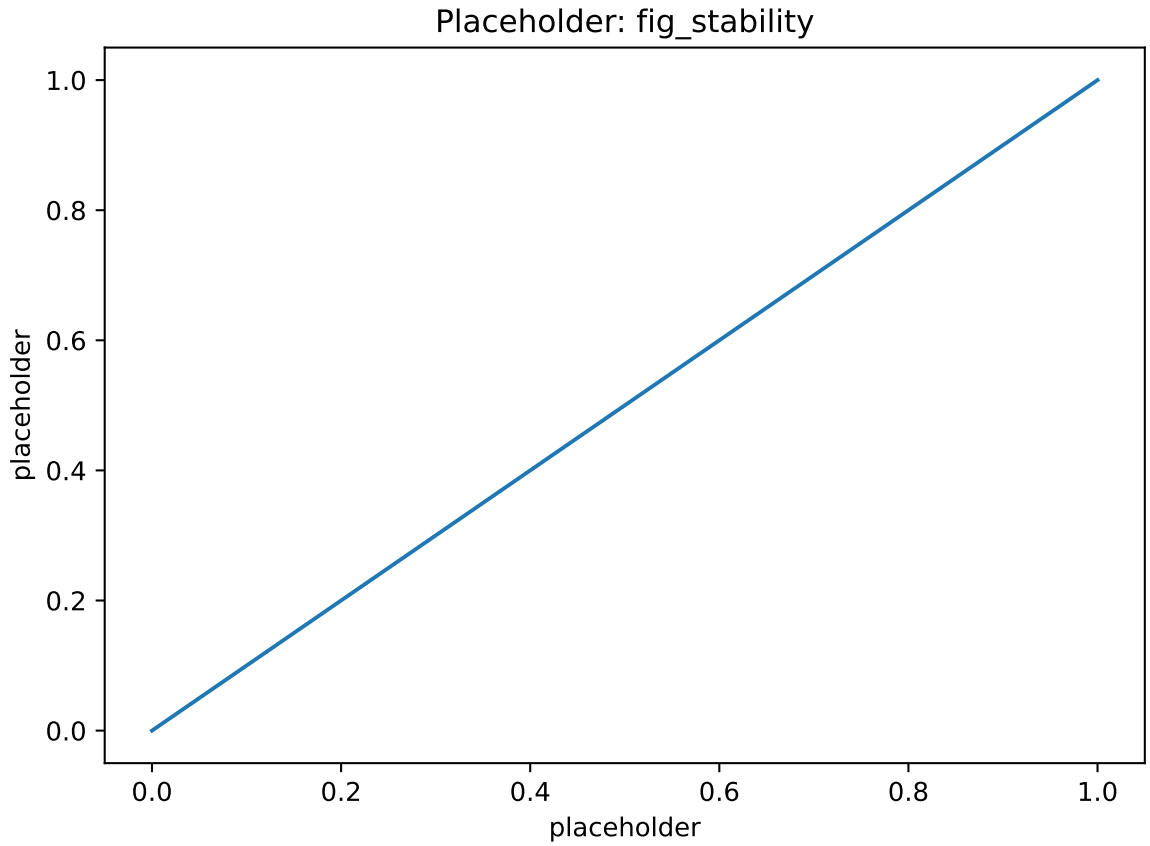

Figure 4 Stability summary. Each point corresponds to a method, plotted by monotonicity (higher is better) and negative drift rate (lower is better).

only in the weighting rule applied to the N rollouts: GRPO uses uniform weights (z-score normalisation), ΔL uses $w_i \propto L_i^{-\alpha}$ with a fixed α, and CAPO uses precision weights $w_i \propto 1/v(L_i)$ derived from modelling how variance depends on length and within-trajectory token covariance. We compare oracle instantiations (true $v(L)$ known) to isolate the effect of the weighting rule itself, rather than estimation error.

Covariance regimes. Six canonical regimes span short-range and long-range dependence: (i) IID token noise ($v(L) = L$), (ii)–(iii) geometric (AR(1)) dependence at correlation levels $\rho \in \{0.7, 0.9\}$, (iv) k-banded stretched-geometric dependence ($\rho = 0.7, k = 16, \eta = 1.5$), and (v)–(vi) compound-symmetry (CS) regimes with off-diagonal correlation $\rho \in \{0.01, 0.05\}$, where variance grows superlinearly in L. These regimes directly mirror the families analysed in §3.6–§3.9.

Diagnostics. We report two complementary views of how each method transforms an identical batch $\{(g_i, L_i)\}$:

(a) Length amplification (Table 5, Figure 6). The normalised mean-squared advantage profile $\mathbb{E}[A^2 \mid L]/\mathbb{E}[A^2 \mid L_{\min}]$ measures whether an advantage construction amplifies or suppresses

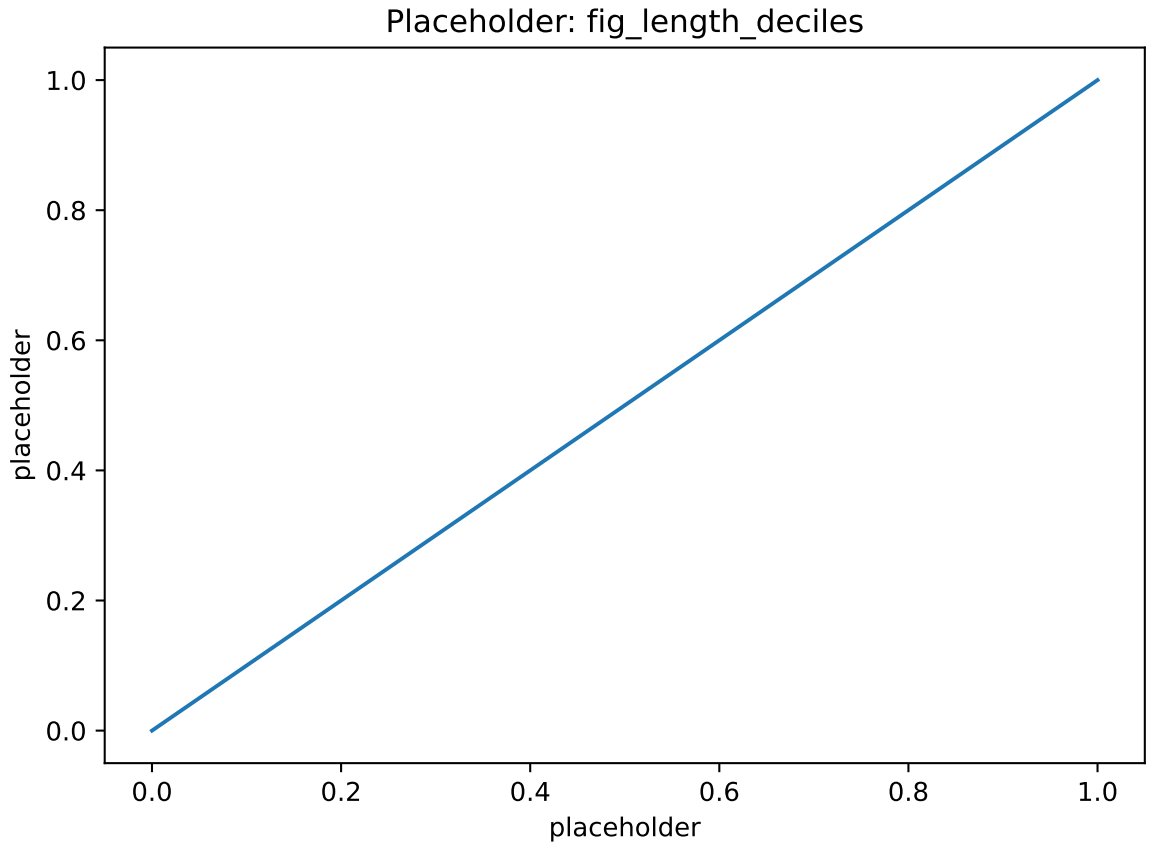

Figure 5 Verifier success by response-length decile. This analysis highlights whether a method disproportionately favors short (or long) trajectories even when end-to-end accuracy is similar.

long trajectories. GRPO amplifies long (noisy) trajectories by 3–7× across regimes. ΔL ($\alpha = 1.0$) and CAPO both *suppress* long trajectories (ratios $\ll 1$), but for different reasons: ΔL applies a fixed correction $L^{-\alpha}$ that is optimal only under IID, while CAPO's precision weighting $w_i \propto 1/v(L_i)$ adapts to the true variance law. Under compound symmetry, where $v(L)$ grows superlinearly, ΔL's fixed-α rule under-corrects: L-CAPO (which assumes $v(L) = L^\beta$) shows amplification ratios rising above 1 at CS $\rho = 0.05$, whereas dependence-aware LV-CAPO achieves the lowest amplification across all long-range regimes.

Where CAPO-Q and CAPO-HAC fit in this benchmark. The synthetic benchmark is particularly well-suited for benchmarking plug-ins because the generative covariance regime is controlled. In addition to L-CAPO (length-only) and LV-CAPO (structured EB), we implement two simplified plug-ins: (i) **CAPO-Q**, which uses the quadratic CS variance law $v(L) \propto (1-\rho)L + \rho L^2$ (Section 3.10.3); and (ii) **CAPO-HAC**, which estimates $v(L)$ by a lag-window/Newey–West long-run variance estimator (Section 3.10.4). When the simulator uses compound symmetry, CAPO-Q coincides with oracle inverse-variance weighting (Proposition 3.40); when the simulator uses short-range dependence (IID/AR(1)/k-band), CAPO-HAC targets the same long-run variance appearing in Bartlett's identity and is consistent under standard mixing conditions. For readability, the main plots focus on GRPO, ΔL, L-CAPO,

Method	IID	AR(1) ($\rho = 0.7$)	AR(1) ($\rho = 0.9$)	k-band ($\rho = 0.7$, $k = 16$, $\eta = 1.5$)	CS (ρ
GRPO (z-score)	3.19	3.42	3.76	3.30	5.
ΔL ($\alpha = 0.75$)	0.21	0.22	0.25	0.22	0.
ΔL ($\alpha = 1.0$)	**0.09**	**0.09**	**0.10**	**0.09**	**0.**
L-CAPO (oracle $\beta = 1$)	0.28	0.31	0.37	0.29	0.
LV-CAPO (oracle)	0.28	0.28	0.25	0.28	0.

Table 5 Length amplification ratio $\mathbb{E}[A^2 \mid L_{\max}]/\mathbb{E}[A^2 \mid L_{\min}]$. Ratios $\gg 1$ indicate that long trajectories dominate the gradient (GRPO); ratios $\ll 1$ indicate precision-optimal suppression. Column-wise best (lowest ratio) is bolded. Under short-range dependence, ΔL (α=1) achieves the lowest ratio; under compound symmetry, LV-CAPO adapts to the superlinear variance law and dominates.

and LV-CAPO; CAPO-Q and CAPO-HAC behave as predicted by their matched-regime theory and serve as practical ablations between the fully nonparametric and fully parametric ends of the CAPO family.

(b) Signal allocation (Figure 7). The fraction of total squared signal $\sum_i A_i^2$ contributed by each length bin shows where update energy concentrates. GRPO over-allocates to the longest bin; CAPO redistributes signal towards shorter (higher-precision) trajectories, concentrating gradient energy where the signal-to-noise ratio is highest.

Relative efficiency. Beyond the distributional view, Figure 8 reports the baseline-MSE relative efficiency $\mathrm{RelEff}(M) = \mathrm{Var}(\hat{m}_{\mathrm{unif}})/\mathrm{Var}(\hat{m}_M)$, estimated via Monte Carlo. This is the primary metric: values above 1 mean the method achieves a *strictly lower-variance* baseline estimate than uniform averaging. LV-CAPO achieves the highest relative efficiency across all regimes, with gains of 2–6× under dependence, because its inverse-variance weighting is the minimum-variance unbiased (MVU) linear combination (Lemma 3.6). ΔL provides partial correction (1–2×), while GRPO is the 1.0× reference by construction.

Summary. These synthetic experiments confirm three properties of CAPO's per-group precision weighting: (i) it eliminates the length amplification that destabilises GRPO in long-horizon settings; (ii) it concentrates update signal on high-precision trajectories where the gradient carries the most information per unit noise; and (iii) it achieves the statistically optimal variance reduction for baseline estimation, with the largest gains appearing under the strong within-trajectory dependence structures that arise in real RLVR training.

Full simulator details, variance law derivations, and reproducibility instructions appear in Appendix D.1.2. The reference implementation is provided as `capo.experiments.analysis.synthetic_benchmark`.

6 Conclusion

This paper developed *Covariance-Aware Policy Optimization (CAPO)*, a variance-aware aggregation principle for policy-gradient updates under the length heterogeneity and within-

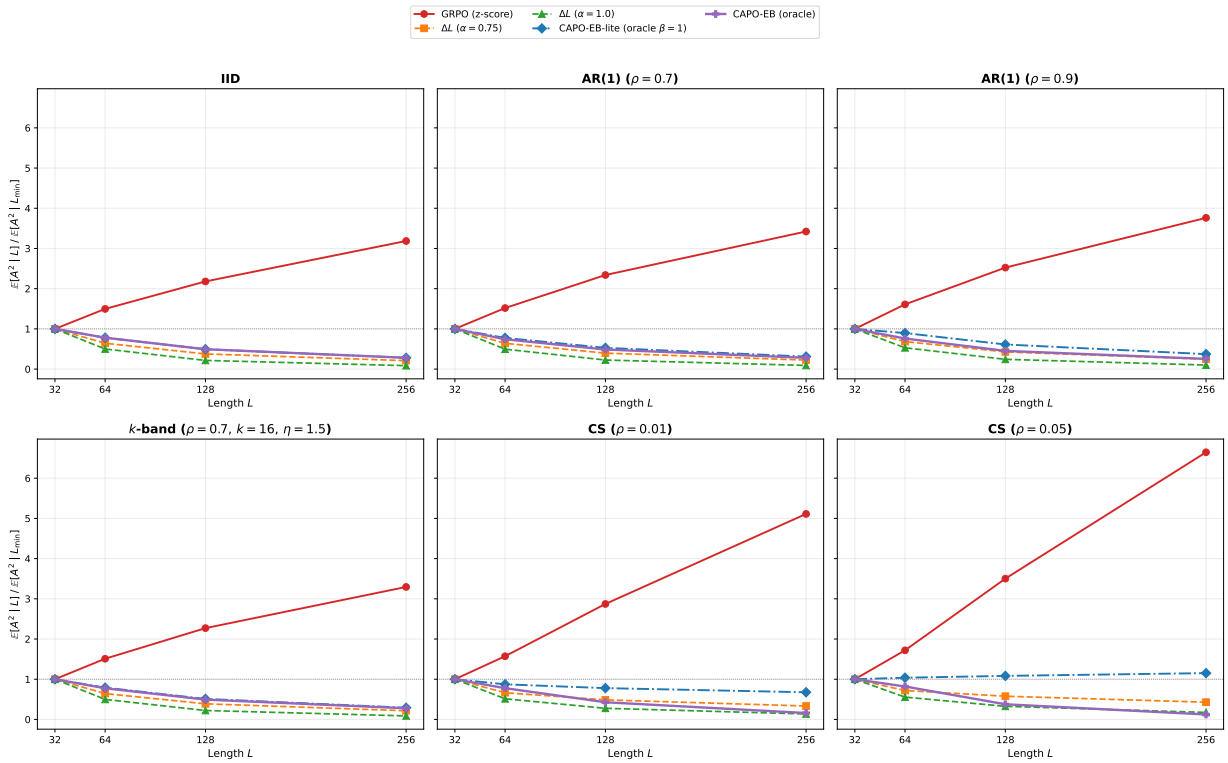

Figure 6 Synthetic benchmark: normalised mean-squared advantage $\mathbb{E}[A^2 \mid L]/\mathbb{E}[A^2 \mid L_{\min}]$ versus trajectory length under six covariance regimes. Curves above 1 indicate length amplification (long trajectories dominate the gradient); curves below 1 indicate precision-optimal suppression of noisy trajectories. GRPO amplifies long trajectories by 3–7×; LV-CAPO suppresses them by concentrating weight on high-precision (short) trajectories.

trajectory dependence that arise in reinforcement learning with verifiable rewards (RLVR). Starting from a minimum-variance unbiased estimation (MVUE) formulation of trajectory reweighting, we showed that inverse-variance weights arise as the unique optimal linear unbiased rule (Lemma 3.6) and that classical length-normalizations such as ΔL correspond to special cases when trajectory variance grows proportionally to length. We then introduced a flexible heteroscedastic Gaussian working model with structured dependence through a *shape factor* $s(L;\xi)$, established a posterior-precision view of optimal weighting with an invariant Jeffreys baseline, and derived principled sublinear length exponents $\beta < 1$ under correctness-linked variance reduction. To make these ideas operational, we proposed empirical-Bayes (EB) estimators for (β, ξ), provided asymptotic guarantees under weak dependence, and gave implementable CAPO variants that interpolate between outcomes-only weighting and full token-level dependence corrections.

Beyond the specific RLVR setting, the CAPO perspective clarifies a general design principle for stochastic-gradient aggregation with variable-length or variable-cost samples: *the correct normalization is determined by how the sample-level covariance grows with length and dependence, rather than by length alone.* In this sense, CAPO connects the practical engineering of stable LLM RL training to classical ideas from weighted least squares, generalized least squares, and empirical Bayes.

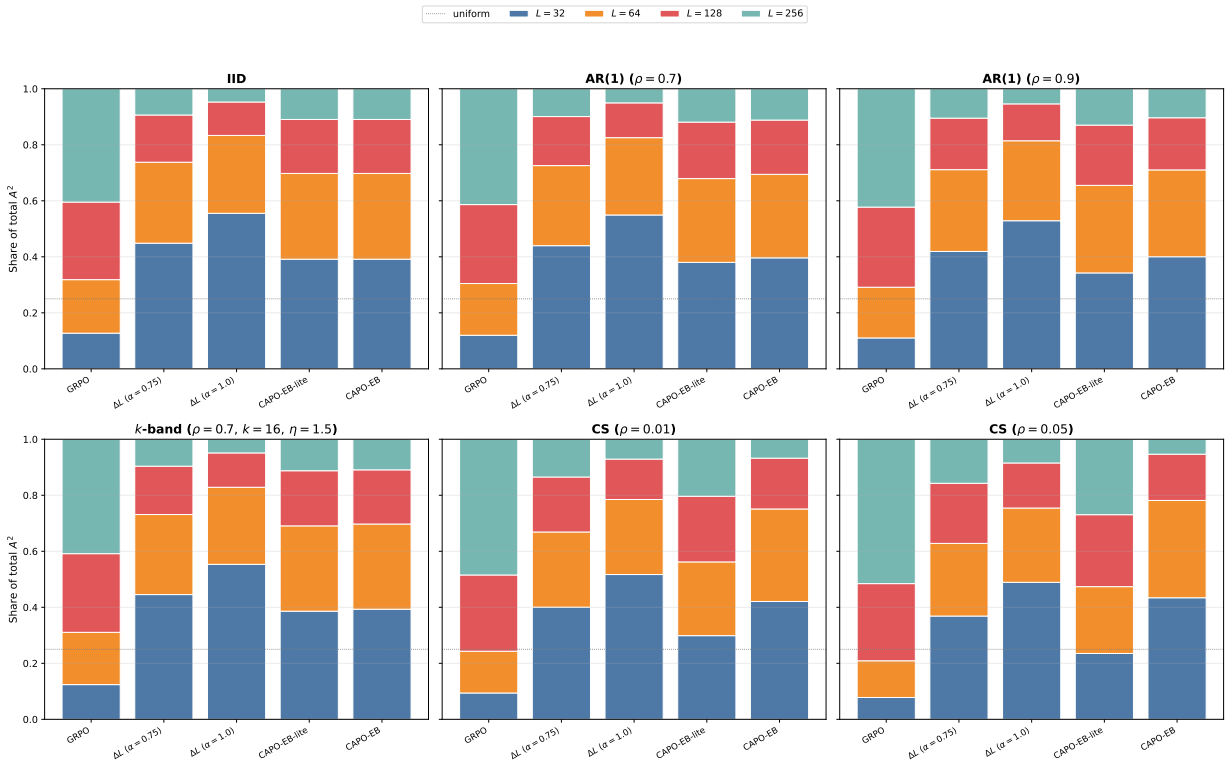

Figure 7 Signal allocation across length bins. Each bar shows the fraction of total $\sum_i A_i^2$ contributed by a given length bin. GRPO over-allocates to the longest bin; CAPO shifts signal towards shorter trajectories where the signal-to-noise ratio is highest.

Limitations and future directions. CAPO is intentionally conservative in its modeling assumptions: we treat the within-trajectory covariance through low-dimensional summaries (bandwidth, decay, or compound-symmetry surrogates) and adopt cross-trajectory uncorrelatedness as a working approximation (Assumption 2.1). Two directions appear especially important for future work. First, prompt-level random effects and group-based baselines can induce non-negligible dependence across trajectories within a batch; extending CAPO to explicit block-structured cross-trajectory covariance is a natural next step. Second, policy-gradient contributions may exhibit heavy tails or non-Gaussianity under rare but extreme rewards; robustness to tail behavior and stable estimation of variance under such distributions merit dedicated study. We view these extensions as complementary to the present contribution: CAPO provides a rigorous baseline framework upon which richer covariance models and robust estimators can be layered without sacrificing unbiasedness.

7 Acknowledgment

We thank the authors of the open-source RL and LLM tooling ecosystems that make controlled RLVR experimentation possible. We also thank colleagues and reviewers for discussions that improved the clarity of the estimation-theoretic framing and the dependence modeling in this manuscript.

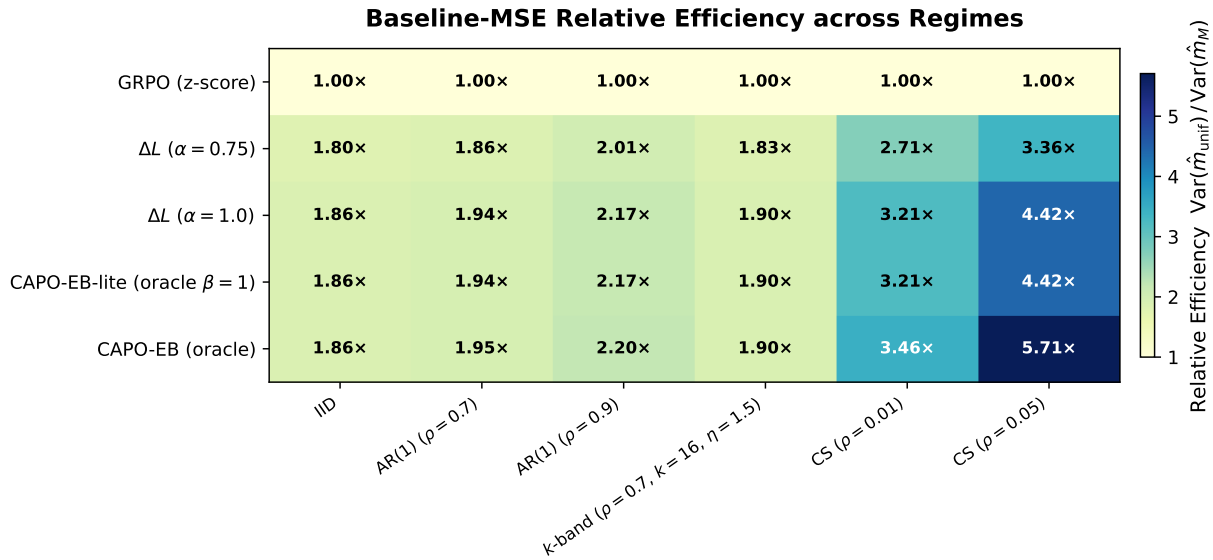

Figure 8 Baseline-MSE relative efficiency $\mathrm{Var}(\hat{m}_{\mathrm{unif}})/\mathrm{Var}(\hat{m}_M)$ across methods and covariance regimes. Values above 1.0 indicate strictly lower-variance baseline estimation than uniform averaging. LV-CAPO achieves 2–6× relative efficiency under dependence, the theoretical optimum for inverse-variance weighting.

A Proof of the Minimum-Variance Unbiased (MVU) Weights

Lemma A.1 (Inverse-variance MVU weights)**.** *Assume* $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) = \mathbf{0}$ *for* $i \neq j$ *and let* $v_i := \Psi\big(\mathrm{Var}(\boldsymbol{g}_i)\big) \in (0, \infty)$ *for a fixed positive, linear, monotone functional* Ψ *on* $\mathbb{S}_+^p$ *(e.g.,* $\Psi(\boldsymbol{A}) = \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}$ *with* $\|\boldsymbol{u}\|{=}1$*, or* $\Psi(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A})$*). Among all linear unbiased estimators* $\widehat{\boldsymbol{g}} = \sum_{i=1}^G x_i \boldsymbol{g}_i$ *satisfying* $\sum_{i=1}^G x_i = 1/M$*, the weights*

$$x_i^\star \;=\; \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^G v_j^{-1}} \tag{10}$$

uniquely minimize the dispersion $\Psi\big(\mathrm{Var}(\widehat{\boldsymbol{g}})\big)$*.*

Proof. By linearity of covariance and the cross-covariance assumption,

$$\mathrm{Var}(\widehat{\boldsymbol{g}}) = \mathrm{Var}\Big(\sum_{i=1}^G x_i \boldsymbol{g}_i\Big) = \sum_{i=1}^G x_i^2\,\mathrm{Var}(\boldsymbol{g}_i).$$

Applying Ψ (positive, linear, monotone on $\mathbb{S}_+^p$),

$$\Psi\big(\mathrm{Var}(\widehat{\boldsymbol{g}})\big) = \sum_{i=1}^G x_i^2\,\Psi\big(\mathrm{Var}(\boldsymbol{g}_i)\big) = \sum_{i=1}^G v_i\,x_i^2.$$

Thus the problem reduces to the strictly convex quadratic program

$$\min_{\boldsymbol{x}} \sum_{i=1}^G v_i x_i^2 \quad \text{s.t.} \quad \sum_{i=1}^G x_i = \frac{1}{M},$$

with $v_i > 0$. The Lagrangian is $\mathcal{L}(\boldsymbol{x}, \lambda) = \sum_i v_i x_i^2 - \lambda\Big(\sum_i x_i - \frac{1}{M}\Big)$. First-order optimality yields $2v_i x_i - \lambda = 0$, i.e.

$$x_i = \frac{\lambda}{2v_i}.$$

Imposing the constraint,

$$\sum_{i=1}^{G} x_i = \frac{\lambda}{2}\sum_{i=1}^{G}\frac{1}{v_i} = \frac{1}{M} \quad \Longrightarrow \quad \lambda = \frac{2}{M\sum_{j=1}^{G} v_j^{-1}}.$$

Substituting back gives

$$x_i^\star = \frac{1}{M}\,\frac{v_i^{-1}}{\sum_{j=1}^{G} v_j^{-1}},$$

which is the unique minimizer by strict convexity. □

Remark A.2. Taking $\Psi(\boldsymbol{A}) = \boldsymbol{u}^\top \boldsymbol{A}\boldsymbol{u}$ recovers the familiar "inverse-variance" rule with $v_i = \boldsymbol{u}^\top \mathrm{Var}(\boldsymbol{g}_i)\boldsymbol{u}$. Taking $\Psi = \mathrm{tr}$ yields the same weights with $v_i = \mathrm{tr}\,\mathrm{Var}(\boldsymbol{g}_i)$. If $\mathrm{Cov}(\boldsymbol{g}_i, \boldsymbol{g}_j) \neq \boldsymbol{0}$ for some $i \neq j$, the optimal weights depend on the full covariance matrix and are not generally diagonal inverse-variance.

B ΔL under Token Independence

This appendix provides an explicit, self-contained derivation of the ΔL warm-up case (token independence) and a notation bridge to **?**. The main text in §3 states the result succinctly; here we record the derivation and the exact conditions under which the inverse-length rule is MVU.

Under an i.i.d. token model with $\mathrm{Var}(\boldsymbol{Z}_{i,t}) = \boldsymbol{\Sigma}_0$ and $\boldsymbol{\Gamma}_i(k) = \boldsymbol{0}$ for $k \neq 0$,

$$\mathrm{Var}(\boldsymbol{g}_i) = L_i\,\boldsymbol{\Sigma}_0 \quad \Longrightarrow \quad v_i \propto L_i. \tag{79}$$

Substituting into the MVU weights equation 10 yields $x_i \propto L_i^{-1}$, i.e., the ΔL rule with exponent $\alpha = 1$. More generally, the practical one-parameter family

$$x_i \;\propto\; L_i^{-\alpha}, \qquad \alpha \in [0, 1], \tag{80}$$

interpolates between constant weighting ($\alpha = 0$) and inverse-length weighting ($\alpha = 1$), providing a baseline that CAPO formalizes via posterior precisions in §3.

C Bartlett summation identity for trajectory covariance

Lemma C.1 (Bartlett summation identity for trajectory covariance)**.** *Assume second–order stationarity along the token index for trajectory i: for all t and all lags $k \in \mathbb{Z}$,*

$$\mathbb{E}\,\boldsymbol{Z}_{i,t} = \boldsymbol{m}_i \quad \textit{(constant)}, \qquad \mathrm{Cov}(\boldsymbol{Z}_{i,t}, \boldsymbol{Z}_{i,t-k}) = \boldsymbol{\Gamma}_i(k).$$

Let $\boldsymbol{g}_i = \sum_{t=1}^{L_i} \boldsymbol{Z}_{i,t}$. *Then*

$$\mathrm{Var}(\boldsymbol{g}_i) \;=\; \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k). \tag{8}$$

Proof. Write $\widetilde{\boldsymbol{Z}}_{i,t} := \boldsymbol{Z}_{i,t} - \boldsymbol{m}_i$. Then

$$\mathrm{Var}(\boldsymbol{g}_i) = \mathrm{Cov}\Big(\sum_{t=1}^{L_i} \widetilde{\boldsymbol{Z}}_{i,t}, \sum_{s=1}^{L_i} \widetilde{\boldsymbol{Z}}_{i,s}\Big) = \sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \mathrm{Cov}\big(\widetilde{\boldsymbol{Z}}_{i,t}, \widetilde{\boldsymbol{Z}}_{i,s}\big).$$

By stationarity, $\mathrm{Cov}(\widetilde{\boldsymbol{Z}}_{i,t}, \widetilde{\boldsymbol{Z}}_{i,s}) = \boldsymbol{\Gamma}_i(t-s)$. Hence

$$\mathrm{Var}(\boldsymbol{g}_i) = \sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \boldsymbol{\Gamma}_i(t-s).$$

Group terms by the lag $k = t - s$. For a fixed k with $|k| \le L_i - 1$, the number of index pairs (t, s) in $\{1, \dots, L_i\}^2$ satisfying $t - s = k$ is $L_i - |k|$ (for $k \ge 0$: $s = 1, \dots, L_i - k$; for $k < 0$: $t = 1, \dots, L_i - |k|$). Therefore

$$\sum_{t=1}^{L_i}\sum_{s=1}^{L_i} \boldsymbol{\Gamma}_i(t-s) = \sum_{k=-(L_i-1)}^{L_i-1} (L_i - |k|)\, \boldsymbol{\Gamma}_i(k),$$

which is equation 8. □

D Illustrative examples

In this section we give two concrete examples illustrating the LV–CAPO algorithm: (i) a simple hand-computable toy example that makes the weighting mechanism transparent, and (ii) a simulated rollout experiment that can be used to compare CAPO with a GRPO-style baseline and empirically demonstrate the variance and sample-efficiency gains predicted by the theory.

D.1 Toy example: three trajectories with different lengths

Consider a single batch of $G = 3$ trajectories with lengths

$$L_1 = 16, \qquad L_2 = 64, \qquad L_3 = 256.$$

Suppose their scalarized returns (e.g., episode rewards) are

$$g_1 = 0.8, \qquad g_2 = 1.0, \qquad g_3 = 1.1.$$

We will assume, for this toy example, that the dependence factor is constant, $s(L; \xi) \equiv 1$, so that

$$\omega_i(\beta, \xi) = \omega_i(\beta) = L_i^{-\beta}$$

and we focus on how the length exponent β changes the weights and the aggregated estimate of μ.

Case 1: GRPO-style ΔL weighting ($\beta = 1$). Setting $\beta = 1$ and $s \equiv 1$ corresponds to the ΔL rule discussed in Proposition 3.31. The (unnormalized) precisions are

$$\omega_1 = L_1^{-1} = 16^{-1} = 0.0625, \quad \omega_2 = 64^{-1} \approx 0.015625, \quad \omega_3 = 256^{-1} \approx 0.00390625.$$

The normalization constant is

$$\Lambda_\omega = \omega_1 + \omega_2 + \omega_3 = 0.0625 + 0.015625 + 0.00390625 = 0.08203125.$$

Hence the normalized weights are

$$w_1 = \frac{\omega_1}{\Lambda_\omega} \approx 0.7619, \quad w_2 \approx 0.1905, \quad w_3 \approx 0.0476.$$

The aggregated estimate is

$$m(\beta = 1) = \sum_{i=1}^{3} w_i g_i \approx 0.7619 \cdot 0.8 + 0.1905 \cdot 1.0 + 0.0476 \cdot 1.1 \approx 0.846.$$

Here the shortest trajectory gets the largest weight, because GRPO implicitly assumes that longer episodes are noisier (variance growing proportionally with length).

Case 2: EB-favored $\beta < 1$ when longer rollouts are less noisy. Suppose the empirical EB fit (e.g., via Algorithm 4) finds that the residual variance decreases approximately like $L^{-0.5}$, corresponding to an optimal $\beta^\star \approx 0.5$. Then

$$\omega_1 = L_1^{-1/2} = 16^{-1/2} = 0.25, \quad \omega_2 = 64^{-1/2} = 0.125, \quad \omega_3 = 256^{-1/2} = 0.0625,$$

and

$$\Lambda_\omega = 0.25 + 0.125 + 0.0625 = 0.4375.$$

The normalized weights become

$$w_1 = \frac{0.25}{0.4375} \approx 0.5714, \quad w_2 \approx 0.2857, \quad w_3 \approx 0.1429.$$

The aggregated estimate is now

$$m(\beta = 0.5) \approx 0.5714 \cdot 0.8 + 0.2857 \cdot 1.0 + 0.1429 \cdot 1.1 \approx 0.892.$$

Relative to the GRPO-style $\beta = 1$ case, EB has shifted mass from the shortest trajectory towards the longer, more stable trajectories, because the data indicated that per-token noise decreases with length. The toy example makes the effect of β on the weights and on the overall estimator $m(\beta, \xi)$ transparent, while keeping all computations fully explicit.

Case 3: EB-favored $\beta > 1$ when very long rollouts are degraded. Conversely, if the empirical analysis (e.g., based on RSS_ω across length buckets) suggests that very long rollouts are degraded due to reward sparsity or drift, the EB objective $\ell(\beta, \xi)$ will favor $\beta > 1$, further downweighting extremely long episodes relative to the $\beta = 1$ baseline. This case can be illustrated by repeating the above calculations with, say, $\beta = 1.5$, in which case L_3 receives much smaller weight than in the GRPO configuration.

These hand-computable examples are convenient for visual demonstrations (e.g., plotting the weights $w_i(\beta)$ as a function of β) and for sanity-checking code paths that implement Algorithm 2.

D.1.1 Simulated rollouts: LV–CAPO versus GRPO

We now outline a fully specified simulated experiment that can be used to compare CAPO with a GRPO-style baseline on a controlled synthetic task. The goal is to isolate the effect of length- and covariance-aware weighting on the variance and sample-efficiency of the training signal.

Generative model. We construct a synthetic rollout generator with the following components:

- **Length distribution.** At each outer step t, draw G_t trajectory lengths i.i.d. from a heavy-tailed discrete distribution to mimic realistic RLHF setups, for example

  $$L_i \sim \text{Uniform}\{32, 64, 128, 256\} \quad \text{or} \quad L_i \sim \text{Discrete-Lognormal}(\mu_L, \sigma_L^2) \text{ clipped to } [16, 512].$$

- **Token-level increments with dependence.** For each trajectory i, generate scalar increments $\{Y_{i,\tau}\}_{\tau=1}^{L_i}$ according to a Gaussian AR(1) process with length-dependent variance:

  $$\begin{aligned} Y_{i,1} &\sim \mathcal{N}(0, \sigma_0^2), \\ Y_{i,\tau} \mid Y_{i,\tau-1} &\sim \mathcal{N}\Big(\rho\, Y_{i,\tau-1},\, \sigma_\epsilon^2 L_i^{-\gamma}\Big), \qquad \tau = 2, \dots, L_i, \end{aligned}$$

  where $\rho \in (0, 1)$ controls temporal dependence and $\gamma > 0$ controls how per-token noise decays with length. For large L_i and fixed ρ, γ, this construction induces an approximate variance for the trajectory average that behaves like $L_i^{-\gamma}$ up to a multiplicative factor depending on ρ.

- **Scalarized return.** Define the scalarized trajectory return as

  $$g_i = \mu^\star + \frac{1}{L_i} \sum_{\tau=1}^{L_i} Y_{i,\tau} + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_{\text{tail}}^2),$$

  where $\mu^\star$ is the true global mean (e.g., the "teacher" target) and ε_i is an additional independent noise term capturing unmodeled effects. By construction, conditional on L_i and $\{Y_{i,\tau}\}$, we have $\mathbb{E}[g_i \mid L_i] = \mu^\star$ and $\text{Var}(g_i \mid L_i) \approx \sigma^2 L_i^{-\gamma}$ for a suitable σ^2, so the true length–variance exponent is $\beta^\star \approx -\gamma$ (under the family $v_i(\vartheta) = \sigma^2 L_i^\beta s(L_i; \xi)$, we have $\beta^\star = -\gamma$ and $s \equiv 1$ in the simplest case).

Compared estimators (LV–CAPO vs. GRPO). On top of this generator, we compare two weighting strategies for estimating $\mu^\star$ (or, in an RL context, for aggregating policy gradients):

- **GRPO-style baseline.** Use the ΔL rule (Proposition 3.31) regardless of the actual noise law:

  $$\omega_i^{\text{GRPO}} = L_i^{-1}, \qquad w_i^{\text{GRPO}} = \frac{\omega_i^{\text{GRPO}}}{\sum_j \omega_j^{\text{GRPO}}}, \qquad m_{\text{GRPO}} = \sum_i w_i^{\text{GRPO}} g_i.$$

  In a full GRPO implementation, these same weights would be used to aggregate per-trajectory gradients or log-prob increments for the policy update.

- **LV–CAPO.** Use the EB objective $\ell(\beta, \xi)$ in equation 50 to fit β (and optionally ξ), and then weight trajectories according to

$$\omega_i^{\text{CAPO}} = L_i^{-\beta_t} s(L_i; \xi_t)^{-1}, \qquad w_i^{\text{CAPO}} = \frac{\omega_i^{\text{CAPO}}}{\sum_j \omega_j^{\text{CAPO}}}, \qquad m_{\text{CAPO}} = \sum_i w_i^{\text{CAPO}} g_i.$$

  In terms of the modular view (Algorithm 1), we instantiate CAPO with the following variance-law plug-ins (Section 3.10):

  - *L–CAPO:* Set $s \equiv 1$ and either fix $\beta = 1$ (ΔL) or estimate β_t using Algorithm 4 (no token-level ACFs required).
  - *LV–CAPO:* Use $s(L; \rho, k, \eta)$ from Definition 3.28, update β_t via EB ascent, and update $\xi_t = (\rho_t, \eta_t)$ via Algorithm 2 (optionally warm-started by Algorithm 3).
  - *CAPO-Q:* Fit a two-term variance law $\widehat{v}(L) = \widehat{a}L^{\beta} + \widehat{b}L^{\beta+1}$ and weight by $\omega_i \propto 1/\widehat{v}(L_i)$ (Algorithm 5).
  - *CAPO-HAC:* Estimate the induced variance directly via a Bartlett/Newey–West lag window and weight by $\omega_i \propto 1/\widehat{v}_{\text{HAC}}(L_i)$ (Algorithm 6).

Simulation protocol. A simple protocol that is straightforward to implement and plot is:

1. Fix hyperparameters: true $\mu^\star$, ρ, γ, σ_0^2, σ_ϵ^2, σ_{tail}^2, length distribution for L_i, batch size G_t, and number of outer steps T.
2. For each outer step $t = 1, \ldots, T$:
   (a) Generate a batch $\mathcal{B}_t$ using the generative model above.
   (b) **GRPO branch:** Compute $(w_i^{\text{GRPO}}, m_{\text{GRPO}}^{(t)})$ using $\omega_i^{\text{GRPO}} = L_i^{-1}$.
   (c) **CAPO branch:** Choose a plug-in (L-CAPO, LV-CAPO, CAPO-Q, or CAPO-HAC), run one iteration of its FitAndPredict routine (Algorithms 4, 2, 5, 6) to compute precisions ω_i and within-group weights w_i^{CAPO}, and form $m_{\text{CAPO}}^{(t)}$.
3. Repeat the entire T-step process over R independent trials (e.g., $R = 500$ or $R = 1000$), re-simulating fresh data each time, and record the sequences $\{m_{\text{GRPO}}^{(t,r)}\}_{t,r}$ and $\{m_{\text{CAPO}}^{(t,r)}\}_{t,r}$.

Evaluation metrics. On this synthetic task the ground truth $\mu^\star$ is known, which allows clean comparisons:

- **Mean-squared error (MSE) over time.** For each method $M \in \{\text{GRPO}, \text{CAPO}\}$, define

$$\text{MSE}_M(t) = \frac{1}{R} \sum_{r=1}^{R} \left(m_M^{(t,r)} - \mu^\star \right)^2.$$

  Plotting $\text{MSE}_{\text{GRPO}}(t)$ and $\text{MSE}_{\text{CAPO}}(t)$ on the same axis typically shows that CAPO reaches a given error threshold (e.g., 10^{-2}) in many fewer outer steps, reflecting its variance-optimal weighting.

- **Variance and bias decomposition.** Because both estimators are unbiased in the model of Section 3.8, the MSE is dominated by variance, $\mathrm{MSE}_M(t) \approx \mathrm{Var}(m_M^{(t)})$. One can empirically confirm this by computing

$$\widehat{\mathrm{Bias}}_M(t) = \frac{1}{R}\sum_{r=1}^{R} m_M^{(t,r)} - \mu^\star,$$

  and checking that $|\widehat{\mathrm{Bias}}_M(t)|$ is negligible compared to $\sqrt{\mathrm{MSE}_M(t)}$.

- **Effective sample size (ESS).** For each method M at step t, define

$$\mathrm{ESS}_M(t) = \frac{1}{\sum_{i\in\mathcal{B}_t}(w_i^M)^2},$$

  where w_i^M are the normalized weights (either GRPO or CAPO). CAPO typically yields a larger ESS in regimes where GRPO over- or under-weights certain length ranges, indicating more efficient usage of the same batch of trajectories.

Integrating with a policy update. The above simulation evaluates CAPO and GRPO purely as estimators of $\mu^\star$ (i.e., of the scalarized training signal). To embed this into a full policy-learning loop, one can replace the role of g_i by per-trajectory gradient estimates (or log-prob increments) and use the same weights w_i^{GRPO} vs. w_i^{CAPO} to aggregate them. Because the EB estimator $m(\beta,\xi)$ has strictly smaller variance than any fixed-weight scheme of the form $w_i \propto L_i^{-\beta_0}$ when the working model is approximately correct, the same simulation framework can then be used to compare downstream learning curves (e.g., average return or teacher score versus wall-clock steps) and empirically demonstrate the superiority of CAPO over GRPO on this controlled benchmark.

D.1.2 Synthetic covariance benchmark: CAPO variants, GRPO++, and advantage scaling

This section formalizes the synthetic benchmark summarized in §5.8. The objective is to isolate, in a fully controlled setting, how different aggregation/normalization rules transform an identical batch of scalar rollout signals $\{(g_i, L_i)\}_{i=1}^B$ into per-trajectory advantage magnitudes. Unlike the full CountDown pipeline, the benchmark does not require querying a base language model; instead, it directly simulates heteroscedastic returns induced by specified within-trajectory covariance regimes.

Synthetic batch structure. We simulate P prompt groups, each with N rollouts (responses), so that $B := PN$ trajectories are present in a batch. Trajectory i has a discrete length $L_i \in \{32, 64, 128, 256\}$, sampled i.i.d. within and across groups unless otherwise noted. The GRPO-family estimators use the prompt-group index to center/standardize within each group, while CAPO variants estimate EB parameters (β,ξ) from the full batch but normalise weights and compute baselines *within* each prompt group — the same per-group structure used by GRPO and ΔL.

Covariance regimes via an induced variance law. We model each trajectory-level scalar return as a (scaled) sum of token-level increments, $g_i = \sum_{t=1}^{L_i} Y_{i,t}$, where $\{Y_{i,t}\}_{t\geq 1}$ is a zero-mean stationary process with unit marginal variance and autocovariance $\gamma(h) := \mathrm{Cov}(Y_{i,t}, Y_{i,t-h})$. Under stationarity, the induced variance law is

$$\mathrm{Var}(g_i \mid L_i = L) \;=\; L \;+\; 2\sum_{h=1}^{L-1}(L-h)\,\gamma(h) \;=\; L\,s(L;\xi), \tag{81}$$

where $s(L;\xi) := \mathrm{Var}(g_i \mid L_i = L)/L$ is the dependence shape factor used throughout the paper (cf. Lemma 3.13 and §3.9). In all regimes below we keep the base length exponent at $\beta = 1$ so that variation across regimes is attributable entirely to dependence.

We instantiate six regimes spanning short-range and long-range dependence by specifying $\gamma(\cdot)$ and hence $s(\cdot)$:

- **IID:** $\gamma(h) \equiv 0$ for $h \geq 1$, so $\mathrm{Var}(g \mid L) = L$ and $s(L) = 1$.
- **AR(1) / geometric dependence (two levels):** $\gamma(h) = \rho^h$ for $h \geq 1$, yielding

$$\mathrm{Var}(g \mid L) \;=\; L \;+\; 2\sum_{h=1}^{L-1}(L-h)\rho^h, \qquad \rho \in \{0.7, 0.9\}. \tag{82}$$

- k**-banded stretched-geometric dependence:** $\gamma(h) = \rho^{h^\eta}\mathbf{1}\{h \leq k\}$, giving

$$\mathrm{Var}(g \mid L) \;=\; L \;+\; 2\sum_{h=1}^{\min\{k,L-1\}}(L-h)\rho^{h^\eta}, \qquad (\rho, k, \eta) = (0.7, 16, 1.5). \tag{83}$$

- **Compound symmetry (two levels):** $\gamma(h) = \rho_{\mathrm{cs}}$ for $h \geq 1$, so

$$\mathrm{Var}(g \mid L) \;=\; L \;+\; \rho_{\mathrm{cs}}\,L(L-1), \qquad \rho_{\mathrm{cs}} \in \{0.01, 0.05\}. \tag{84}$$

  This regime represents long-range dependence in which off-diagonal correlations do not decay and the trajectory variance grows superlinearly in L.

The figure and tables in §5.8 are generated by the reference implementation `capo.experiments.analysis.sy` `covariance_benchmark` and saved as compile-ready artifacts under `docs/report/artifacts/paper/`.

Compared advantage constructions. Given a batch $\{(g_i, L_i)\}$, we compute scalar advantages using the following rules. To ensure an apples-to-apples comparison, every method uses the *same* structural form: within each prompt group $\mathcal{I}_p$, define

$$w_i \;=\; \frac{\omega_i}{\sum_{j\in\mathcal{I}_p}\omega_j}, \qquad m_p \;=\; \sum_{j\in\mathcal{I}_p} w_j\,g_j, \qquad A_i \;=\; w_i\,(g_i - m_p). \tag{85}$$

The methods differ *only* in the precision weights ω_i:

(i) GRPO. Uniform weights: $\omega_i = 1$. This gives the unweighted group mean $m_p = \bar{g}_p$ and equal per-trajectory influence $w_i = 1/N$.

(ii) ΔL (length scaling). For $\alpha \in (0, \infty)$, set

$$\omega_i^{\Delta L(\alpha)} \;=\; L_i^{-\alpha}. \tag{86}$$

This is structurally identical to L-CAPO with a fixed exponent $\beta = \alpha$ and $s \equiv 1$.

(iii) L–CAPO (length-only weighting). Estimate $\hat{\beta}$ from the batch via Algorithm 4 and set

$$\omega_i^{\text{lite}} \;=\; L_i^{-\hat{\beta}}. \tag{87}$$

(iv) LV–CAPO (dependence-aware weighting). Estimate $(\hat{\beta}, \hat{\xi})$ via Algorithm 2 and set

$$\omega_i^{\text{full}} \;=\; \Big(L_i^{\hat{\beta}}\, s(L_i; \hat{\xi})\Big)^{-1}. \tag{88}$$

In the oracle diagnostic, $\hat{\beta}$ and $\hat{\xi}$ are replaced by the true parameters of the generative model.

Evaluation metrics. For each method and each covariance regime, we compute the mean-squared advantage conditional on length, $\mathbb{E}[A_i^2 \mid L_i = L]$, and summarize length sensitivity by the amplification ratio

$$\text{Amp} \;:=\; \frac{\mathbb{E}[A_i^2 \mid L_i = L_{\max}]}{\mathbb{E}[A_i^2 \mid L_i = L_{\min}]}, \tag{89}$$

where $L_{\min}$ and $L_{\max}$ denote the smallest and largest length bins in the simulation. Values larger than 1 indicate amplification of long trajectories; values below 1 indicate suppression.

To visualize where update energy concentrates, we additionally report the *signal share* contributed by each length bin:

$$\text{Share}(L) \;:=\; \frac{\mathbb{E}\Big[\sum_{i:\, L_i = L} A_i^2\Big]}{\mathbb{E}\Big[\sum_{i=1}^{B} A_i^2\Big]}, \tag{90}$$

which corresponds to the expected fraction of the total squared learning signal arising from trajectories of length L.

Reproducibility. The benchmark used to generate Figures 6–7 and Table 5 is implemented in the repository as `capo.experiments.analysis.synthetic_covariance_benchmark`. The script is self-contained and can be executed on CPU to regenerate the corresponding LaTeX table and PDF figures.

E Annotated Literature Review

This appendix provides an annotated, one-paragraph summary for each cited reference in the main manuscript. Each annotation explains why the work is cited and how it connects to the technical development of CAPO.

E.1 RLVR, RLHF, and LLM Training Systems

?. **?** is cited as an early, influential exemplar of reinforcement learning with verifiable rewards (RLVR) for mathematical reasoning, demonstrating that outcome-supervised reinforcement learning can materially improve long-form chain-of-thought performance. We reference this work to motivate the empirical regime in which CAPO operates: training pipelines where rollouts vary substantially in length and where update stability depends critically on how per-trajectory learning signals are normalized and aggregated. In particular, the DeepSeekMath setting highlights the practical consequences of length heterogeneity (variance growth and instability), which CAPO treats through an explicit variance law and covariance-aware weighting.

?. **?** is cited to document the base model family used in our empirical evaluation and to ground experimental claims in a concrete, reproducible model specification. This reference provides architectural and training-context details for Qwen2.5 that are necessary to interpret downstream RLVR behavior (e.g., typical response-length distributions, scaling behavior, and stability considerations) without conflating CAPO's aggregation effects with undocumented model idiosyncrasies.

?. **?** is cited as a closely related RLVR optimization pipeline that explicitly discusses length-driven instability and proposes a practical normalization strategy (DAPO) that operates at the batch level. We reference this work both as a strong baseline and as motivation for treating length effects as a statistical estimation problem: DAPO stabilizes training by modifying how per-token losses are normalized, whereas CAPO targets the same failure mode by directly estimating trajectory-level variance and applying an inverse-variance reweighting rule. The comparison clarifies when batch-level length scaling is beneficial and when it can be improved by dependence-aware variance modeling.

?. **?** is a primary methodological antecedent for CAPO. It provides a systematic account of ΔL-type length-dependent normalization and empirically demonstrates that suppressing long trajectories can improve stability in RLVR. We cite this work to position CAPO as a principled extension: CAPO recovers the ΔL family as a special case (via MVU/Jeffreys arguments) and then generalizes it by (i) learning the length exponent from data and (ii) incorporating explicit dependence corrections through $s(L;\xi)$, thereby addressing regimes where length alone is an insufficient proxy for gradient dispersion.

?. **?** is cited for its analysis of GRPO-style estimators and for proposing a "Dr. GRPO" variant that removes or modifies standard normalization components to mitigate undesirable statistical side-effects. We reference this work as evidence that small, seemingly innocuous normalization choices (e.g., scaling by within-group standard deviation) can materially change estimator behavior. CAPO's contribution is to replace ad hoc normalization knobs with an explicit variance model and an estimator-theoretically justified weighting rule, while still allowing practitioners to recover Dr. GRPO-like behavior as a boundary case under appropriate assumptions.

?. ? is cited as a canonical reference for modern RLHF pipelines in which policy-gradient updates (typically PPO variants) are combined with reward or preference models. We reference this work to situate RLVR as a specialization of RLHF and to emphasize that variance control and stability in the policy-gradient update remain central challenges when scaling to large language models. CAPO inherits the same basic optimization substrate but focuses on a distinct, RLVR-specific pathology: extreme length heterogeneity that induces structured heteroscedasticity at the trajectory level.

?. ? is cited as a prominent alternative to RL-based fine-tuning—direct preference optimization—that avoids explicit policy-gradient rollouts. We reference it to delimit the scope of CAPO: CAPO is designed for rollout-based objectives (RLHF/RLVR) where per-trajectory stochasticity is intrinsic, and where normalization and aggregation decisions directly control estimator variance. Contrasting with DPO helps clarify that CAPO's statistical machinery is targeted at policy-gradient aggregation rather than preference-model regression or supervised ranking.

E.2 Policy-Gradient and Advantage-Estimation Foundations

?. ? (REINFORCE) is cited as the foundational policy-gradient method that makes explicit the variance challenges inherent to likelihood-ratio gradient estimators. We reference it to justify why variance reduction is not an optional engineering detail but a core statistical requirement, especially when rollouts are long and gradients accumulate correlated noise. CAPO can be viewed as a variance-reduction mechanism that operates at the aggregation layer of REINFORCE/PPO-style updates by optimizing a linear unbiased estimator under heteroscedastic, dependent noise.

?. ? is cited as a standard reference for the principles of policy-gradient estimation, baselines, and the bias–variance trade-offs that arise in practice. We reference it to anchor terminology (advantages, baselines, and return aggregation) and to motivate why variance control must respect unbiasedness constraints when the goal is to preserve correct optimization direction. CAPO's formulation can be interpreted as importing classical estimation-theoretic ideas into this reinforcement-learning setting.

?. ? is cited for generalized advantage estimation (GAE), which is central to modern actor–critic training and provides a canonical template for trading bias against variance in advantage construction. We reference this work to distinguish two levels of variance control: (i) how token- or step-level advantages are computed (e.g., via GAE) and (ii) how trajectory-level gradients are aggregated when response lengths vary. CAPO addresses the latter and is therefore compatible with GAE-style advantage estimators; the cited connection helps clarify how CAPO can be integrated into PPO-type pipelines without altering the underlying credit-assignment mechanism.

?. ? is cited as the dominant practical policy-gradient update used in RLHF and RLVR systems. We reference PPO to emphasize that, even with clipping and other stabilizers, the stochastic-gradient estimator in PPO remains sensitive to how trajectories are normalized and combined—particularly in long-horizon language-model rollouts. CAPO is intended as

a drop-in aggregation layer that can be used within PPO/GRPO-style updates to reduce gradient dispersion induced by length heterogeneity and within-trajectory dependence.

E.3 Statistical Foundations: Dependence Modeling and Empirical Bayes

?. **?** is cited as a classical reference for estimation with autocorrelated errors and for the relationship between time-series dependence and efficient generalized least squares (GLS) procedures. We reference this work to motivate the dependence-aware extensions of CAPO: when token-level gradient increments exhibit serial correlation, the trajectory-level variance is not determined by length alone, and GLS-style corrections become statistically natural. The cited perspective supports our use of structured covariance models and helps interpret CAPO as a practical, structured-GLS approximation for RLVR gradient aggregation.

?. **?** is cited to situate CAPO's empirical-Bayes (EB) approach within the broader EB tradition. We reference it because CAPO estimates hyperparameters (e.g., length exponent and dependence shape) by marginal likelihood and then plugs the estimates into a precision-weighted decision rule—an EB pattern that is well-studied in large-scale inference. This connection clarifies why CAPO's "learned normalization" can be treated as statistically principled shrinkage/weighting rather than as a purely heuristic tuning mechanism.

?. **?** is cited for Jeffreys' prior and the invariance principles that motivate it. We reference this work because CAPO's Bayesian warm-up derivation uses a Jeffreys scale prior to obtain an aggregation rule that is invariant to reparameterizations of the variance scale, and this derivation provides a principled route to recovering ΔL-type behavior as a special case. The citation therefore grounds our choice of baseline prior in established Bayesian theory rather than in convenience.

?. **?** is cited as a standard reference for the asymptotic theory of M-estimators and consistency arguments. We reference it to justify technical steps in the convergence analysis of CAPO's EB hyperparameter estimates, where we appeal to general results ensuring that maximizers of empirical criteria converge to population maximizers under regularity conditions. This citation supports the rigor of the EB theory by linking our proofs to well-established asymptotic tools.